



Università degli Studi di Milano-Bicocca
Dipartimento di Fisica “Giuseppe Occhialini”
Corso di Laurea in Fisica

Pulse Shaping for Enhanced Superconducting Qubit Readout

Supervisor:

Prof. Andrea Giachero

Co-supervisor:

Dott. Pietro Campana

Candidate:

Alice Arena

Student Number:

912874

ACADEMIC YEAR 2025/2026

Abstract

This thesis explores the dispersive measurement of superconducting qubits, with particular attention to optimizing the shape of the applied signal. The work is developed on three levels: the study of the formalism required to describe the qubit-resonator system, the development of a semiclassical model for field dynamics to analytically compare various readout signals (pulses), and, finally, a complete experimental implementation, from device calibration to the measurement of the residual photon population in the resonator on real superconducting quantum systems.

A transmon qubit is created by replacing the inductor of a superconducting LC circuit with a Josephson junction, a nonlinear element that makes the circuit's energy spectrum anharmonic. This allows the two lowest levels, the states $|0\rangle$ and $|1\rangle$, to be isolated as the computational basis, thus creating an artificial atom. To read its state without modifying it, the qubit is capacitively coupled to a microwave resonator; if the mismatch between the frequencies of the qubit and the resonator is much greater than their coupling constant, the interaction enters the so-called dispersive regime, in which no actual exchange of excitations occurs, but the resonator's frequency is shifted by an amount χ depending on whether the qubit is in the ground state or the excited state. By sending a readout pulse to the resonator and measuring the amplitude and phase of the transmitted or reflected signal, i.e., its quadrature components I , the real part, and Q , the imaginary part, in the so-called IQ plane, it is therefore possible to determine the state of the qubit. At the end of the measurement, however, a certain number of photons remain trapped in the resonator: through the same dispersive coupling that makes the readout possible, these residual photons continuously shift the qubit's frequency, causing measurement-induced dephasing that limits the fidelity of subsequent quantum operations. It is therefore important to have techniques capable of rapidly emptying the resonator, returning it to the ground state $|0\rangle$, also called the vacuum state, as quickly as possible, rather than relying solely on the passive decay of the cavity. This also allows for faster operation.

The first chapter builds the theoretical tools needed to describe this system. After recalling the representation of a qubit on the Bloch sphere, it shows how the quantization of an LC circuit produces a harmonic oscillator and how the Josephson junction introduces the anharmonic term that transforms this oscillator into a transmon. Next, the Jaynes-Cummings Hamiltonian, which describes the coupling between the qubit and the resonator, and the effective dispersive Hamiltonian, which is the basis for qubit readout, are introduced. Furthermore, the driving of the resonator and the qubit in the rotating reference frame is discussed. The chapter concludes by introducing the density matrix formalism, necessary to describe a qubit interacting with an external environment, together with the characteristic relaxation time T_1 and decoherence time T_2 used to quantify that interaction.

The second chapter develops a semiclassical model of the dynamics of the readout resonator. Starting from input-output theory, the quantum Langevin equation governing the evolution of the resonator field operator is derived, including an external drive and losses through the transmission line; in the semiclassical limit, the equation reduces to two differential equations for the conditional amplitudes $\alpha_g(t)$ and $\alpha_e(t)$, one for each state of the qubit. The objectives that an effective readout pulse must satisfy are then defined: a rapid rise of the intracavity field, a good separation between the two conditioned trajectories in the IQ plane, a measurement time short compared to the qubit relaxation times, a controlled number of photons during the measurement and, above all, a low residual photon population at the end of the pulse. Two pulse shapes, obtained by analytically solving these equations, are compared. The first is the rectangular pulse, whose solution shows a sudden increase (*overshoot*) in the photon population followed by

a passive decay with rate κ once the drive is turned off. The second is called the CLEAR (*Cavity Level Excitation and Reset*) pulse, composed of five segments: two short *kick* pulses before the steady-state readout segment and two short depletion pulses after it. The amplitudes of the four kicks are obtained by solving a 2×2 linear system that forces the conditioned trajectories to reach the desired steady-state amplitude at the beginning of the readout and to vanish at the end of the sequence. Numerical simulations confirm that the CLEAR pulse eliminates the overshoot observed for the rectangular pulse and reduces the residual photon population to a value compatible with 0, compared with a residual population of approximately 0.43 photons left by the rectangular pulse in the same time window. The chapter finally analyzes the trade-off between kick duration, the required drive amplitude and the χ/κ ratio, showing that a faster suppression of the residual population requires a proportionally more intense drive.

The third chapter describes the experimental implementation, on a real transmon device, of the methods developed in the previous chapters. Microwave theory principles relevant to the control and readout chain are briefly reviewed, and the cryogenic environment is described. Control and acquisition are performed via an RFSoc (*Radio-Frequency System on Chip*), a board that combines on a single integrated circuit a programmable logic (*Field-Programmable Gate Array*, or FPGA), for the execution of real-time processing instructions, digital-to-analog converters, which generate the microwave pulses and analog-to-digital converters, which digitize the return signal. The entire chain is controlled through the open-source Qibo software ecosystem.

The calibration of the qubit-resonator system was conducted through a structured sequence of experiments, in which each measurement provides an essential parameter for the next. Specifically, the following were performed: low-power resonator spectroscopy, qubit spectroscopy, a Rabi calibration of the pulse amplitudes to induce the π and $\pi/2$ rotations on the qubit, a measurement of the coherence times T_1 , T_2 , and the Hahn-echo T_2 , resonator one-tone spectroscopy (punchout) used to identify the readout operating point, a direct measurement of the dispersive shift χ , a Ramsey calibration to refine the qubit transition frequency, and finally, a calibration of the AC-Stark shift. The latter uses the dispersive coupling to convert the readout pulse amplitude, defined in arbitrary DAC units, into an average intracavity photon number via a linear relationship.

With these parameters, the CLEAR pulse is constructed by computing the amplitudes of the four kicks relative to the steady-state readout amplitude. To measure the population, a pair of $\pi/2$ pulses is applied immediately after the readout pulse under test; this produces a Ramsey fringe whose shape at short times is distorted by the residual AC-Stark shift induced by the photons still present in the cavity; by fitting this distortion to an analytical model, the residual photon population n_0 is extracted, corrected for the loss of excited population due to relaxation during the readout pulse itself. The comparison between the rectangular pulse and the CLEAR pulse, conducted over a wide range of normalized readout powers, shows that the residual population of the rectangular pulse grows linearly with power, reaching approximately 5 photons at the highest power explored, while that of the CLEAR pulse remains below approximately 0.5 photons over the entire range.

Overall, the thesis provides a comprehensive treatment of dispersive readout of superconducting qubits that combines the theoretical framework, semiclassical modeling, and a complete experimental implementation on real hardware. It demonstrates both analytically and experimentally the advantage of the CLEAR pulse over a conventional readout pulse in reducing residual photons in the resonator, a key requirement for the rapid and repeated readout of superconducting qubits.

Contents

Introduction	1
1 Foundations of superconducting qubits	3
1.1 From quantum mechanics to quantum technologies	3
1.1.1 The qubit	3
1.2 Superconducting circuits and circuit quantum electrodynamics	5
1.2.1 The LC circuit as a quantum harmonic oscillator	5
1.2.2 Anharmonicity and the transmon qubit	6
1.2.3 The microwave resonator	7
1.2.4 Dispersive regime	8
1.2.5 Resonator control	9
1.3 Open Quantum Systems	10
1.3.1 Density Matrix Formalism	10
1.3.2 Relaxation, dephasing and error sources	11
2 Semiclassical dynamics of the resonator	13
2.1 The resonator as an open quantum system	13
2.1.1 Resonator linewidth, lifetime and quality factor	13
2.1.2 Internal loss and external coupling	14
2.2 The Langevin equation for the driven resonator	14
2.2.1 Coupling between the resonator and the transmission line	14
2.2.2 Derivation of the driven Langevin equation	16
2.2.3 State-dependent resonator dynamics	16
2.2.4 Photon number and IQ plane representation	17
2.3 Readout pulses	17
2.3.1 Objectives of a readout pulse	18
2.3.2 The rectangular pulse	19
2.3.3 The CLEAR pulse	19
2.3.4 Overview of alternative pulse-shaping strategies	20
2.4 Semiclassical simulation of resonator reset pulses	21
2.4.1 Aim of the simulations	21
2.4.2 Simulation method and parameters	22
2.4.3 Results	23
3 Experimental implementation of CLEAR resonator reset	31
3.1 Experimental setup	31
3.1.1 Microwave signal propagation and S-parameters	31
3.1.2 Cryogenic environment and microwave lines	33
3.1.3 Room-temperature electronics	33
3.1.4 The Qibo software stack	35
3.2 Calibration	36
3.2.1 Low-power resonator spectroscopy	37
3.2.2 Wide-range qubit power spectroscopy	37
3.2.3 Resonator punchout	38
3.2.4 Qubit spectroscopy	39
3.2.5 Rabi calibration of the π and $\pi/2$ pulse	40

3.2.6	Ramsey calibration of the qubit transition frequency	41
3.2.7	Coherence times: T_1 , T_2 , and Hahn-echo T_2	42
3.2.8	Dispersive shift	44
3.2.9	AC-Stark-shift calibration of the intracavity photon number	45
3.2.10	Final calibrated parameters	48
3.3	CLEAR pulse implementation	48
3.3.1	From calibrated parameters to hardware pulse amplitudes	49
3.3.2	Pulse-sequence implementation in Qibolab	49
3.3.3	Extracting the residual photon population from a Ramsey trace	51
3.3.4	Probe-frequency midpoint calibration	52
3.3.5	Residual population versus drive power	53
3.3.6	Cross-check: residual population versus relaxation delay	54
	Conclusions	57

Introduction

Quantum computers promise to solve problems that are, in practice, out of reach for classical machines: simulating molecules and materials at the quantum level, searching unsorted data or factoring large numbers with a speed-up no classical algorithm can match, and processing information in ways that could reshape cryptography and secure communication. Over the past decade this promise has stopped being purely theoretical: companies such as Google and IBM have built devices large enough to perform computations that would be impractical to reproduce classically, and steady progress continues across several physical platforms. And yet, none of this has translated into practically useful quantum algorithms running error-free: today’s processors are still too noisy, and we are, in the words commonly used to describe this stage of the field, in the Noisy Intermediate-Scale Quantum (NISQ) era.

Closing the gap between an impressive demonstration and a genuinely useful quantum computer is not simply a matter of adding more qubits. Every qubit carries with it a whole set of imperfect operations, i.e. initialization, single- and two-qubit gates, readout, and each of these introduces errors that accumulate as a computation grows in size and depth. For this reason, alongside the effort to scale up the number of qubits, a large and very active part of current research is dedicated to making each qubit, and every operation performed on it, more accurate and faster. This thesis belongs to this second line of research, and focuses on one operation in particular: reading out the qubit state, which every experiment with superconducting qubits has to perform, typically many times over.

A superconducting qubit is read out indirectly, by coupling it to a microwave resonator and probing the resonator instead of the qubit itself; this is what makes the measurement possible without immediately destroying the fragile quantum state it is meant to reveal. The price for this indirection is that, once the measurement pulse is switched off, a handful of photons are typically left behind in the resonator, and it takes time for them to leak away. While they linger, these residual photons keep perturbing the qubit’s frequency through the very same coupling that made the readout possible in the first place, effectively causing the qubit to lose track of its own phase. If a second operation is attempted before the resonator has emptied, this residual disturbance directly degrades its fidelity. The straightforward fix, which is simply waiting for the cavity to empty on its own, works, but it is slow.

An alternative, and the one this thesis is about, is to not wait passively but to actively drive the resonator back to the vacuum state. This idea is not new: an established technique, known as the CLEAR pulse (*Cavity Level Excitation and Reset*), was proposed by McClure et al. [1] and has since been adopted, in different variants, in several superconducting-qubit laboratories. Before starting this project, however, the technique had never been implemented within our own group, on our own hardware and software stack, an RFSoc-based control system driven through the open-source Qibo ecosystem. The motivation behind this thesis is precisely this gap: an optimal-control technique can be well understood on paper and reported to work elsewhere, but demonstrating it end to end, from the underlying theory to a working calibration on a real device, is a different and necessary step before it can be trusted and reused.

Concretely, this thesis develops the CLEAR pulse on three levels: it builds the theoretical framework needed to describe a qubit coupled to a readout resonator and its interaction with an external environment (Chapter 1); it constructs a semiclassical model of the resonator dynamics that makes it possible to derive the CLEAR pulse analytically and to compare it, under idealized conditions, against a conventional rectangular readout pulse (Chapter 2); and it implements the entire pipeline experimentally on a real transmon device, from the calibration of the qubit-resonator system to a dedicated measurement of the photon population left in the resonator

after each pulse (Chapter 3). The result, presented and discussed in Secs. 3.3.5 - 3.3.6, is a direct, quantitative confirmation, on our own hardware, of the advantage that motivated this work in the first place.

Beyond this specific comparison, the possible payoff of this thesis is meant to outlast it. The calibration routines, the pulse-generation code, and the photon-population estimator developed here constitute a validated, ready-to-use building block for active resonator reset that can be reused directly in any future experiment that needs a fast, low-dephasing qubit readout.

Chapter 1

Foundations of superconducting qubits

1.1 From quantum mechanics to quantum technologies

Quantum physics was born in 1900, with the introduction of the quantum hypothesis by Max Planck. Throughout the twentieth century, the theoretical framework of quantum mechanics was progressively consolidated, enabling the development of technologies such as the transistor and laser. Beginning in the 1980s, the possibility of directly exploiting quantum mechanical phenomena, such as superposition and entanglement, to process and transmit information began to be considered. From these ideas, quantum computing and quantum information science emerged. This ongoing period is widely known as the Second Quantum Revolution [2]. Devices developed in this era are expected to surpass classical computers in certain tasks, achieving what is known as *quantum advantage*.

These technologies are used in four key areas:

- quantum computing, to solve computational problems intractable for classical computers;
- quantum communication, which ensures ultra-secure information transmission through quantum cryptography;
- sensors and metrology, capable of measuring physical quantities such as time and magnetic fields with unprecedented precision;
- quantum simulation, which is essential for modeling complex materials and molecular interactions in fields ranging from pharmaceuticals to particle physics.

This chapter introduces the theoretical concepts required for the description of superconducting-qubit readout developed in the following chapters. We first recall the basic notion of a qubit and its physical implementation in superconducting circuits. We then introduce the transmon qubit and the microwave resonator, focusing on their interaction in the dispersive regime, which enables qubit-state discrimination. Finally, we discuss the effect of driving and dissipation, providing the basis for the semiclassical description of resonator dynamics developed in Chapter 2.

1.1.1 The qubit

The qubit represents the fundamental unit of quantum information. Unlike a classical bit, which can only assume one of two discrete states (0 or 1), a qubit is a two-level quantum system that can exist in a superposition of both states simultaneously.

The state of a qubit is described by a state vector $|\psi\rangle$ belonging to a two-dimensional complex Hilbert space. A general pure state can be written as a linear combination of the computational basis states $|0\rangle$ and $|1\rangle$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle. \quad (1.1)$$

Here, α and β are complex numbers called probability amplitudes. The physical state must satisfy the normalization condition

$$|\alpha|^2 + |\beta|^2 = 1. \quad (1.2)$$

When the qubit is measured in the computational basis, $|\alpha|^2$ and $|\beta|^2$ represent the probabilities of obtaining the outcomes 0 and 1, respectively.

The state of a qubit can be visualized geometrically using the Bloch sphere. A general pure state may be parametrized as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle, \quad (1.3)$$

where θ and ϕ determine the position of the state on the sphere. In this representation, every pure state corresponds to a point on the surface of a unit sphere. The north and south poles represent the states $|0\rangle$ and $|1\rangle$, respectively, while points on the equator correspond to equally weighted superpositions, for which $|\alpha| = |\beta|$. Their specific position on the equator is determined by the relative phase between the two probability amplitudes.

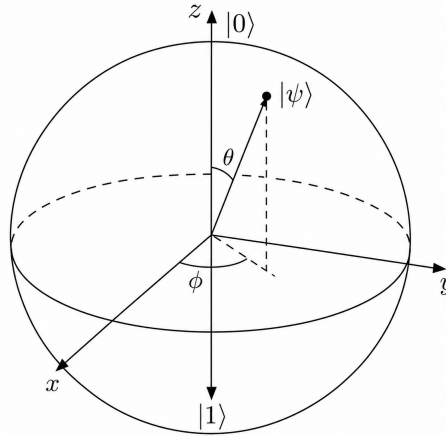


Figure 1.1: Bloch sphere representation of a pure qubit state.

Measurement converts quantum information into classical information. For a projective measurement in the computational basis, the state is projected onto either $|0\rangle$ or $|1\rangle$.

For the qubit state defined in Eq. (1.1), the probability of obtaining the outcome 0 is

$$P(0) = |\alpha|^2, \quad (1.4)$$

whereas the probability of obtaining the outcome 1 is

$$P(1) = |\beta|^2. \quad (1.5)$$

After the measurement, the qubit is left in the state associated with the observed outcome.

1.2 Superconducting circuits and circuit quantum electrodynamics

Superconducting qubits are among the most advanced and widely adopted physical platforms for quantum information processing. These devices are artificial atoms constructed from superconducting electrical circuits operated at cryogenic temperatures, where variables such as charge and magnetic flux display quantum behavior. The logical states $|0\rangle$ and $|1\rangle$ are encoded in two energy levels of the circuit and are manipulated using microwave pulses.

The fundamental building block of such devices is the LC circuit, which behaves as a quantum harmonic oscillator once quantized. However, its equally spaced energy spectrum prevents individual transitions from being selectively addressed, making it unsuitable as a qubit on its own.

This limitation can be overcome by introducing a Josephson junction, a nonlinear element that makes the energy spectrum anharmonic [3], [4]. The resulting circuit can therefore be operated as an effective two-level system. In particular, this chapter focuses on the transmon qubit [5], one of the most widely used superconducting-qubit architectures.

We first present the quantization of the LC circuit and then examine how the Josephson junction modifies its dynamics and gives rise to the transmon. We subsequently couple the qubit to a microwave resonator and introduce the dispersive regime, which provides the physical mechanism underlying qubit readout.

The following discussion is mainly based on the review by Krantz et al. [6].

1.2.1 The LC circuit as a quantum harmonic oscillator

To introduce the circuit-quantization framework employed throughout this chapter, the LC circuit is first considered, consisting of a capacitor C and an inductor L . The capacitor stores electrical energy, which is analogous to kinetic energy, while the inductor stores magnetic energy, analogous to potential energy. We denote by $\Phi(t)$ the magnetic flux through the inductor,

$$\Phi(t) = \int_{-\infty}^t V(t') dt',$$

which acts as the generalized position coordinate. The circuit obeys

$$V = L \frac{dI}{dt}, \quad I = C \frac{dV}{dt}.$$

The kinetic and potential energies of the circuit are given by

$$\tau_C = \frac{1}{2} C \dot{\Phi}^2, \quad \mu_L = \frac{1}{2L} \Phi^2,$$

respectively. Consequently, the Lagrangian of the system can be expressed as

$$\mathcal{L} = \tau_C - \mu_L = \frac{1}{2} C \dot{\Phi}^2 - \frac{1}{2L} \Phi^2. \quad (1.6)$$

Eq. (1.6) exhibits the same mathematical form as the Lagrangian of a mechanical harmonic oscillator. During the oscillation, energy is periodically exchanged between the electric field stored in the capacitor and the magnetic field stored in the inductor.

We can derive the Hamiltonian by computing the Legendre transformation,

$$H = \dot{\Phi} \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} - \mathcal{L} = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} = \frac{1}{2} C V^2 + \frac{1}{2} L I^2,$$

where $Q = C \dot{\Phi}$ is the conjugate charge, which acts as the momentum coordinate.

To move from a classical to a quantum description, we promote the flux and charge to operators

$$\hat{\Phi} = \Phi, \quad \hat{Q} = -i\hbar \partial / \partial \Phi.$$

$\hat{\Phi}$ and $\hat{Q}$ are linked by the following commutation relation:

$$[\hat{\Phi}, \hat{Q}] = i\hbar.$$

This commutation relation expresses the fact that flux and charge are conjugate quantum observables and cannot simultaneously have arbitrarily well-defined values. Subsequently, we define new operators $\hat{\phi}$ and $\hat{n}$ as follows

$$\hat{\phi} = \frac{2\pi}{\Phi_0} \hat{\Phi}, \quad \hat{n} = \frac{\hat{Q}}{2e},$$

where $\Phi_0 = h/2e$ is the superconducting flux quantum, and e is the electron charge. In these variables, the Hamiltonian takes the following form:

$$\hat{H} = 4E_C \hat{n}^2 + \frac{1}{2} E_L \hat{\phi}^2, \quad (1.7)$$

where $E_C = e^2/(2C)$ indicates the charging energy and $E_L = (\Phi_0/2\pi)^2/L$ represents the inductive energy. The operators $\hat{\phi}$ and $\hat{n}$ satisfy

$$[\hat{\phi}, \hat{n}] = i.$$

Eq. (1.7) describes the Hamiltonian of a harmonic oscillator with frequency

$$\omega_{LC} = \frac{1}{\hbar} \sqrt{8E_L E_C} = \frac{1}{\sqrt{LC}},$$

whose quantized energy levels are $E_k = \hbar\omega_{LC}(k + \frac{1}{2})$. Introducing the creation and annihilation operators $\hat{b}^\dagger$ and $\hat{b}$ via

$$\hat{n} = n_{\text{zpf}} i(\hat{b} - \hat{b}^\dagger), \quad \hat{\phi} = \phi_{\text{zpf}}(\hat{b} + \hat{b}^\dagger),$$

where the zero-point fluctuation amplitudes are given by $n_{\text{zpf}} = (E_L/32E_C)^{1/4}$ and $\phi_{\text{zpf}} = (2E_C/E_L)^{1/4}$, the Hamiltonian can be rearranged in the canonical harmonic-oscillator form

$$\hat{H} = \hbar\omega_{LC} \left(\hat{b}^\dagger \hat{b} + \frac{1}{2} \right).$$

Consequently, even in the ground state $|0\rangle$, the oscillator exhibits finite fluctuations of the conjugate variables $\hat{n}$ and $\hat{\phi}$ of characteristic magnitude n_{zpf} and ϕ_{zpf} .

1.2.2 Anharmonicity and the transmon qubit

For the system to function as a qubit, it must exhibit anharmonicity, which restricts transitions to the two lowest energy levels. Otherwise, a drive at a frequency resonant with ω_{LC} may excite higher energy levels. Anharmonicity is provided by replacing the inductor with a Josephson junction, which obeys the current-phase relations

$$I = I_c \sin \phi,$$

$$V = \frac{\hbar}{2e} \frac{d\phi}{dt},$$

where I_c is the critical current. The junction therefore adds a nonlinear term $-E_J \cos \phi$ to the Hamiltonian, with the Josephson energy given by $E_J = \hbar I_c/(2e)$. Including the junction and a shunt capacitance C_S leads to the Hamiltonian

$$H = 4E_C n^2 - E_J \cos \phi,$$

with $E_C = e^2/(2C_R)$, where $C_R = C_S + C_J$ also accounts for the intrinsic contribution C_J of the junction itself. In the regime $E_J \gg E_C$ (transmon) the cosine can be expanded,

$$E_J \cos \phi \approx E_J \left(1 - \frac{\phi^2}{2} + \frac{\phi^4}{24} \right),$$

producing an anharmonic oscillator. Keeping terms up to fourth order and substituting the ladder operators $\hat{b}$ and $\hat{b}^\dagger$, the quadratic part becomes

$$\hat{H}_0 = \sqrt{8E_J E_C} \left(\hat{b}^\dagger \hat{b} + \frac{1}{2} \right). \quad (1.8)$$

The fourth order contribution to the Hamiltonian is

$$\hat{H}_4 = -\frac{E_C}{12} \left(\hat{b} + \hat{b}^\dagger \right)^4. \quad (1.9)$$

According to the rotating-wave approximation, only excitation-number-conserving terms are retained, leading to

$$\left(\hat{b} + \hat{b}^\dagger \right)^4 \simeq 6\hat{b}^{\dagger 2} \hat{b}^2 + 12\hat{b}^\dagger \hat{b} + 3. \quad (1.10)$$

Neglecting constant energy shifts, the Hamiltonian therefore becomes

$$\hat{H} \simeq \left(\sqrt{8E_J E_C} - E_C \right) \hat{b}^\dagger \hat{b} - \frac{E_C}{2} \hat{b}^{\dagger 2} \hat{b}^2. \quad (1.11)$$

Defining $\hbar\omega_q = \sqrt{8E_J E_C} - E_C$, the anharmonicity

$$\alpha \equiv \omega_q^{1 \rightarrow 2} - \omega_q^{0 \rightarrow 1} \approx -\frac{E_C}{\hbar}$$

is small compared to ω_q , but sufficient to isolate two levels.

Therefore, the transmon is a weakly anharmonic oscillator. Approximating the system as a two-level qubit with Pauli operator $\hat{\sigma}_z$ leads to the simplified Hamiltonian

$$\hat{H} = \hbar \frac{\omega_q}{2} \hat{\sigma}_z,$$

where we adopt the convention

$$\hat{\sigma}_z = |e\rangle\langle e| - |g\rangle\langle g|,$$

so that the excited state corresponds to the +1 eigenvalue and

$$E_e - E_g = \hbar\omega_q.$$

This convention is used throughout the rest of this work.

A practical implementation of a transmon qubit is the Xmon geometry [7], in which the cross-shaped structure provides the high shunt capacitance required for operation in the transmon regime. Fig. 1.2 shows an example of a superconducting circuit containing two Xmon qubits, each coupled to a readout resonator. The coupling claw is a gap between the qubit cross and the resonator that serves as a coupling capacitance, allowing the exchange of microwave excitations between the two systems.

The device includes both a fixed-frequency qubit, QB-0, and a flux-tunable qubit, QB-1. In the latter case, the Josephson element is implemented as a SQUID, consisting of a superconducting ring interrupted by two Josephson junctions. The effective Josephson energy, and thus the qubit's transition frequency, can be controlled by applying an external magnetic flux through the ring. The resonators are connected to a common transmission line used to send and receive the microwave readout signal.

1.2.3 The microwave resonator

A microwave resonator is the element that mediates between the quantum system and the classical measurement apparatus. The resonator serves two primary functions. First, it stores photons, and therefore enhances the interaction between the electromagnetic field and the qubit. Second, because photons can leak out through the measurement line, the resonator also acts as the interface through which information about the qubit state is extracted.

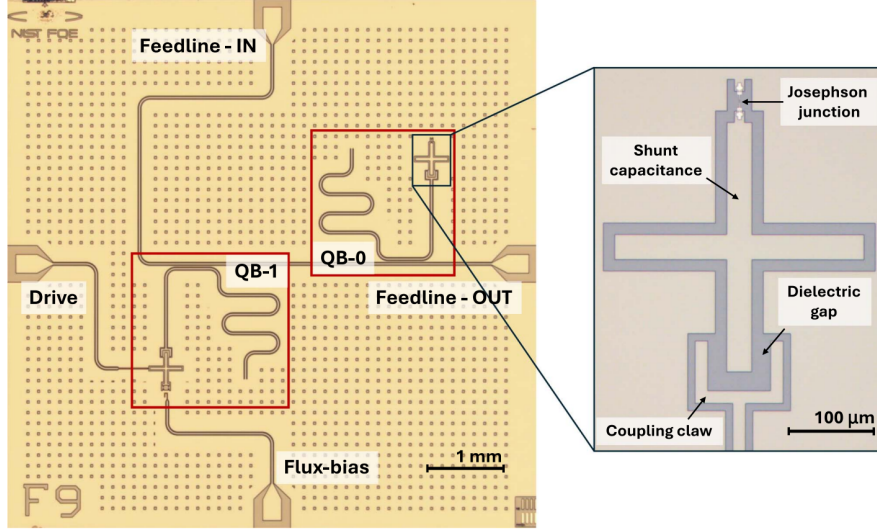


Figure 1.2: Scheme of a superconducting circuit containing a fixed-frequency Xmon qubit, QB-0, and a flux-tunable Xmon qubit, QB-1, each coupled to an individual readout resonator. Reproduced from Ref. [8].

Consider a qubit of frequency ω_q coupled to a resonator of frequency ω_r . The total Hamiltonian can be written

$$\hat{H} = \hat{H}_q + \hat{H}_r + \hat{H}_{\text{int}},$$

where $\hat{H}_{q,r}$ are the uncoupled qubit and resonator Hamiltonians, and $\hat{H}_{\text{int}}$ represents the interaction. Capacitive coupling arises from a mutual capacitance C_g between the qubit and resonator. Denoting the total capacitances of the qubit and the resonator by C_q and C_r , respectively, the interaction Hamiltonian takes the form

$$\hat{H}_{\text{int}} = 4e^2 \frac{C_g}{C_q C_r} \hat{n}_q \hat{n}_r.$$

In this expression, $\hat{n}_q$ and $\hat{n}_r$ represent the charge operators for the qubit and the resonator. Within the two-level approximation adopted here, the transmon is restricted to its computational subspace spanned by $|g\rangle$ and $|e\rangle$. Defining the raising and lowering operators $\hat{\sigma}_+ = |e\rangle\langle g|$ and $\hat{\sigma}_- = |g\rangle\langle e|$, together with the resonator creation and annihilation operators $\hat{a}^\dagger$ and $\hat{a}$, the coupled system is described by

$$\hat{H} = \hbar \frac{\omega_q}{2} \hat{\sigma}_z + \hbar \omega_r \hat{a}^\dagger \hat{a} + \hbar g (\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger), \quad (1.12)$$

where g is the qubit-resonator coupling strength. This is the Jaynes-Cummings Hamiltonian. We define the qubit-resonator detuning as

$$\Delta = \omega_q - \omega_r.$$

The limit in which the qubit and resonator are far detuned is discussed in the following section.

1.2.4 Dispersive regime

Direct exchange of excitations between the qubit and the resonator is not required for qubit readout. What is needed instead is a configuration in which the resonator response depends on the qubit state, while resonant transfer of excitations between the two systems is effectively suppressed. This is realized in the dispersive regime, where the qubit and resonator frequencies are sufficiently detuned.

When

$$g \ll |\Delta|,$$

the coherent exchange described by the Jaynes-Cummings interaction is strongly inhibited, and the qubit-resonator coupling term can be perturbatively removed to the second order in g/Δ by applying a Schrieffer-Wolff transformation, leading to the effective dispersive Hamiltonian (see Ref. [9, Eq. (44)]),

$$\hat{H}_{\text{disp}} = \hbar(\omega_r + \chi\hat{\sigma}_z)\hat{a}^\dagger\hat{a} + \frac{\hbar\tilde{\omega}_q}{2}\hat{\sigma}_z, \quad (1.13)$$

where $\chi = \frac{g^2}{\Delta}$ is the dispersive frequency shift and $\tilde{\omega}_q = \omega_q + \frac{g^2}{\Delta}$ is the Lamb-shifted qubit frequency. The resonator is thus conditionally shifted by $\pm\chi$ depending on whether the qubit is in $|e\rangle$ or $|g\rangle$, so that probing its frequency reveals the qubit state.

1.2.5 Resonator control

In order to perform a measurement, the resonator must be actively driven by a microwave pulse. The envelope of this pulse determines how the resonator is populated and subsequently emptied, and therefore plays a central role in the readout dynamics considered in this thesis. We first introduce the corresponding Hamiltonian and then move to a frame rotating at the drive frequency, where the dynamical description becomes simpler.

To describe a drive at frequency ω_d and pulse $\epsilon(t)$, assumed to be real, in the laboratory frame, a convenient Hamiltonian is

$$\hat{H}_{\text{drive}} = \hbar\epsilon(t)(\hat{a}^\dagger e^{-i\omega_d t} + \hat{a}e^{i\omega_d t}), \quad (1.14)$$

(see Ref. [10, Eq. (20)]).

Starting from the dispersive resonator Hamiltonian that includes the drive in the laboratory frame,

$$\hat{H} = \hbar(\omega_r + \chi\hat{\sigma}_z)\hat{a}^\dagger\hat{a} + \hbar\epsilon(t)(\hat{a}^\dagger e^{-i\omega_d t} + \hat{a}e^{i\omega_d t}),$$

we move to a frame rotating at the frequency ω_d by using the unitary transformation

$$\hat{U}(t) = \exp(i\omega_d \hat{a}^\dagger \hat{a} t).$$

Since the transformation is time-dependent, the Hamiltonian in the rotating frame is

$$\hat{H}_{\text{rot}} = \hat{U}\hat{H}\hat{U}^\dagger + i\hbar\dot{\hat{U}}\hat{U}^\dagger.$$

The useful operator identities are

$$\hat{U}\hat{a}\hat{U}^\dagger = \hat{a}e^{-i\omega_d t}, \quad \hat{U}\hat{a}^\dagger\hat{U}^\dagger = \hat{a}^\dagger e^{i\omega_d t}, \quad \hat{U}\hat{a}^\dagger\hat{a}\hat{U}^\dagger = \hat{a}^\dagger\hat{a}.$$

The additional term due to the time dependence of $U(t)$ is

$$i\hbar\dot{\hat{U}}\hat{U}^\dagger = -\hbar\omega_d\hat{a}^\dagger\hat{a}.$$

Therefore, the free resonator part transforms as

$$\hbar(\omega_r + \chi\hat{\sigma}_z)\hat{a}^\dagger\hat{a} \longrightarrow \hbar(\omega_r + \chi\hat{\sigma}_z)\hat{a}^\dagger\hat{a} - \hbar\omega_d\hat{a}^\dagger\hat{a}. \quad (1.15)$$

Introducing the detuning

$$\delta_r = \omega_r - \omega_d,$$

Eq. (1.15) reduces to

$$\hbar(\delta_r + \chi\hat{\sigma}_z)\hat{a}^\dagger\hat{a}.$$

The drive term becomes time-independent:

$$\hat{U}[\hbar\epsilon(t)(\hat{a}^\dagger e^{-i\omega_d t} + \hat{a}e^{i\omega_d t})]\hat{U}^\dagger = \hbar\epsilon(t)(\hat{a}^\dagger + \hat{a}).$$

Thus, in the rotating frame, the resonator Hamiltonian reads

$$\hat{H}_s = \hbar(\delta_r + \chi\hat{\sigma}_z)\hat{a}^\dagger\hat{a} + \hbar\epsilon(t)(\hat{a}^\dagger + \hat{a}). \quad (1.16)$$

Eq. (1.16) provides the starting point for the description of dispersive resonator readout. The effective resonator frequency depends on the qubit state through the term $\chi\hat{\sigma}_z$. In particular, since

$$\hat{\sigma}_z|e\rangle = +|e\rangle, \quad \hat{\sigma}_z|g\rangle = -|g\rangle,$$

the two possible effective resonator frequencies in the rotating frame are

$$\delta_r + \chi \quad \text{and} \quad \delta_r - \chi.$$

Consequently, the same readout pulse $\epsilon(t)$ produces two different resonator responses depending on whether the qubit is in $|e\rangle$ or $|g\rangle$. This state-dependent response is the basis of dispersive qubit readout and will be studied in detail in Chapter 2.

A similar rotating-frame treatment can be applied to a microwave drive acting directly on the qubit. After restricting the transmon dynamics to its two lowest energy levels, a drive with envelope $\Omega(t)$, frequency ω_d , and phase ϕ can be described in the laboratory frame by

$$\hat{H}_{q,\text{drive}} = \frac{\hbar\Omega(t)}{2} \left[\hat{\sigma}_+ e^{-i(\omega_d t + \phi)} + \hat{\sigma}_- e^{i(\omega_d t + \phi)} \right].$$

Moving to a frame rotating at the drive frequency and applying the rotating-wave approximation gives

$$\hat{H}_{q,\text{rot}} = \frac{\hbar\Delta_q}{2} \hat{\sigma}_z + \frac{\hbar\Omega(t)}{2} (\cos\phi \hat{\sigma}_x + \sin\phi \hat{\sigma}_y),$$

where

$$\Delta_q = \omega_q - \omega_d$$

is the qubit-drive detuning. The pulse amplitude sets the rotation angle of the qubit state, while its phase selects the axis in the equatorial plane of the Bloch sphere. In particular, $\phi = 0$ corresponds to a rotation around the x axis, whereas $\phi = \pi/2$ corresponds to a rotation around the y axis.

1.3 Open Quantum Systems

The dispersive Hamiltonian introduced in the previous section describes the ideal coherent evolution of the coupled qubit-resonator system. A real superconducting qubit, however, cannot be treated as perfectly isolated: it interacts with uncontrolled environmental degrees of freedom and with the electromagnetic modes used for control and readout. These interactions cause the loss of energy and quantum coherence and therefore limit the time available for qubit manipulation and measurement.

1.3.1 Density Matrix Formalism

The state-vector formalism introduced in Section 1.1.1 is appropriate to describe a pure, perfectly isolated quantum state. Once the qubit interacts with an external environment, however, a more general description in terms of a density matrix is required.

For a statistical ensemble of states $|\psi_i\rangle$ occurring with probabilities p_i , the density matrix is defined as

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|, \quad p_i \geq 0, \quad \sum_i p_i = 1. \quad (1.17)$$

For a pure state, only one state occurs with unit probability. In this case, the density matrix reduces to

$$\rho = |\psi\rangle \langle \psi|. \quad (1.18)$$

For a pure state expressed in the Bloch sphere parametrization,

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle, \quad (1.19)$$

the corresponding density matrix can be written as

$$\rho = \frac{1}{2} (I + \vec{r} \cdot \vec{\sigma}) = \frac{1}{2} \begin{pmatrix} 1 + \cos\theta & e^{-i\phi} \sin\theta \\ e^{i\phi} \sin\theta & 1 - \cos\theta \end{pmatrix},$$

or equivalently,

$$\rho = \begin{pmatrix} \cos^2 \frac{\theta}{2} & e^{-i\phi} \cos \frac{\theta}{2} \sin \frac{\theta}{2} \\ e^{i\phi} \cos \frac{\theta}{2} \sin \frac{\theta}{2} & \sin^2 \frac{\theta}{2} \end{pmatrix} = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}.$$

Here, $\vec{r}$ is the Bloch vector and $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ denotes the vector of Pauli matrices. The diagonal elements of ρ quantify the populations of the two computational-basis states, while the off-diagonal elements encode the coherence and the relative phase between them. The length of the Bloch vector determines the purity of the state: pure states satisfy $|\vec{r}| = 1$ and lie on the surface of the Bloch sphere, while mixed states, for which $|\vec{r}| < 1$, lie inside it. The completely mixed state $\rho = I/2$ corresponds to $\vec{r} = 0$ and is located at the center of the sphere.

Pure states are characterized by

$$\text{Tr}(\rho^2) = 1. \quad (1.20)$$

Indeed, for a pure state,

$$\rho^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| = \rho,$$

so that $\text{Tr}(\rho^2) = \text{Tr}(\rho) = 1$.

Mixed states, which can arise due to coupling to uncontrolled environmental degrees of freedom, instead satisfy

$$\frac{1}{2} \leq \text{Tr}(\rho^2) < 1 \quad (1.21)$$

for a single qubit. The lower bound corresponds to the completely mixed state, while values approaching one describe states with a larger degree of purity.

A generic system-environment interaction for a single qubit can be modeled as

$$H_{\text{int}} = \hat{O}_q \otimes \hat{X}, \quad (1.22)$$

where $\hat{O}_q$ is an operator acting on the qubit Hilbert space and $\hat{X}$ is an operator acting on the environmental, or bath, degrees of freedom.

The specific form of $\hat{O}_q$ determines the physical nature of the induced noise. Noise components transverse to the qubit quantization axis, associated with the operators $\hat{\sigma}_x$ and $\hat{\sigma}_y$, induce transitions between the energy eigenstates. In contrast, longitudinal noise, associated with $\hat{\sigma}_z$, leads primarily to fluctuations of the qubit transition frequency and therefore causes dephasing.

1.3.2 Relaxation, dephasing and error sources

The interaction with the environment affects a superconducting qubit mainly through two processes: energy relaxation and loss of phase coherence. These effects are conventionally characterized by the times T_1 and T_2 , respectively. Energy relaxation refers to the transition from the excited state $|1\rangle$ to the ground state $|0\rangle$, which occurs at a characteristic decay rate

$$\Gamma_1 = \frac{1}{T_1}. \quad (1.23)$$

According to Fermi's Golden Rule, the relaxation rate is given by

$$\Gamma_1 \propto |\langle 0 | \hat{O}_q | 1 \rangle|^2 S(\omega_q), \quad (1.24)$$

where $S(\omega)$ denotes the noise power spectral density (PSD) evaluated at the qubit frequency¹.

In the low-temperature regime ($k_B T \ll \hbar\omega_q$), thermally activated excitation processes are exponentially suppressed:

$$\Gamma_{\uparrow} = e^{-\hbar\omega_q/k_B T} \Gamma_{\downarrow}, \quad (1.25)$$

such that the steady state is predominantly the ground state $|0\rangle$.

¹The PSD of a stationary noise process $\kappa(t)$ characterizes the distribution of its power over angular frequencies. It is defined as the Fourier transform of the autocorrelation function $C_\kappa(\tau) = \langle \kappa(t + \tau) \kappa(t) \rangle$:

$$S_\kappa(\omega) = \int_{-\infty}^{\infty} C_\kappa(\tau) e^{-i\omega\tau} d\tau.$$

The coherence of a superposition state decays due to both energy relaxation and pure dephasing, leading to

$$\frac{1}{T_2} = \frac{1}{2T_1} + \Gamma_\phi, \quad (1.26)$$

where Γ_ϕ is the pure dephasing rate. Pure dephasing manifests as fluctuations of the qubit transition frequency:

$$\omega_q \rightarrow \omega_q + \delta\omega(t). \quad (1.27)$$

Consequently, the accumulated phase can be written as

$$\phi(t) = \int_0^t \omega_q dt' + \delta\phi(t). \quad (1.28)$$

For Gaussian noise, the coherence decays as

$$\langle e^{i\delta\phi(t)} \rangle = e^{-\chi(t)}, \quad (1.29)$$

where $\chi(t)$ depends on the noise power spectrum.

Although pure dephasing does not involve an exchange of energy with the environment, it is generally irreversible in practice.

Within the Bloch–Redfield description of decoherence [11], [12] (see also Ref. [6, Eq. (43)]), the time evolution of the density matrix of a qubit initially prepared in the pure state $|\psi(0)\rangle = \alpha|0\rangle + \beta|1\rangle$ can be written as

$$\epsilon(\rho) = \begin{bmatrix} 1 + (|\alpha|^2 - 1)e^{-\Gamma_1 t} & \alpha\beta^* e^{i\delta\omega t} e^{-\Gamma_2 t} \\ \alpha^*\beta e^{-i\delta\omega t} e^{-\Gamma_2 t} & |\beta|^2 e^{-\Gamma_1 t} \end{bmatrix} \xrightarrow{t \gg (T_1, T_2)} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad (1.30)$$

where $\delta\omega = \omega_q - \omega_d$ denotes the detuning between the qubit frequency ω_q and the rotating-frame frequency ω_d , and $\Gamma_2 = 1/T_2$ is the total decoherence rate. In the limit $t \gg T_1, T_2$, all population relaxes into the ground state.

Physical sources of decoherence

In superconducting circuits, energy relaxation can arise from several mechanisms. A major source of decoherence is charge noise, originating from dielectric defects at material interfaces [13], which predominantly contributes to energy relaxation. In the transmon regime, sensitivity to charge noise is substantially reduced due to the large ratio of Josephson to charging energy. Pure dephasing is instead primarily attributed to magnetic flux noise [14]. Additional relaxation channels include photon-number fluctuations induced by residual microwave fields [15], quasi-particles (i.e., unpaired electrons) [16], and spontaneous emission through the readout resonator, known as Purcell decay [17].

Chapter 2

Semiclassical dynamics of the resonator

2.1 The resonator as an open quantum system

When the resonator is driven by a classical microwave field, its state is often well described by a coherent state $|\alpha\rangle$, defined by

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle.$$

In this regime the complex amplitude

$$\alpha = \langle \hat{a} \rangle$$

plays the role of the classical field amplitude, while the average intracavity photon number is

$$n = \langle \hat{n} \rangle = \langle \hat{a}^\dagger \hat{a} \rangle = |\alpha|^2.$$

The identification of α with a classical field amplitude motivates the semiclassical treatment used in the following sections. We first characterize the resonator as an open quantum system, introducing its linewidth and the distinction between internal loss and external coupling. We then derive the driven Langevin equation from input-output theory, following mainly Chapter 8 of Osada, Yamazaki and Noguchi [18], and we introduce the photon number and IQ-plane representation used to quantify the resulting state separation. Section 2.3 presents two concrete readout pulses, the rectangular and CLEAR pulses [1]. Finally, Section 2.4 uses the solutions derived earlier in the chapter to simulate and compare the two pulses numerically.

2.1.1 Resonator linewidth, lifetime and quality factor

Let ω_c denote the angular resonance frequency of a cavity mode, which can be modeled as a harmonic oscillator. If the resonator exhibits losses, a photon stored in the cavity possesses a finite lifetime. We denote by κ the total photon loss rate. The corresponding temporal evolution of the intracavity field follows the exponential decay law

$$n(t) = n(0) e^{-\kappa t}. \quad (2.1)$$

Since the field amplitude is proportional to the square root of the photon number, the complex field amplitude decays as

$$\alpha(t) = \alpha(0) e^{-\kappa t/2} e^{-i\omega_c t}. \quad (2.2)$$

Thus κ is the decay rate of the photon number, while $\kappa/2$ is the decay rate of the field amplitude.

In the frequency domain, the Fourier transform of an exponentially decaying oscillator gives a Lorentzian response. The cavity spectrum has the form

$$S_c(\omega) \propto \frac{1}{(\omega - \omega_c)^2 + (\kappa/2)^2}, \quad (2.3)$$

where the FWHM= κ .

A common way to characterize a resonator is through its quality factor

$$Q = \frac{\omega_c}{\kappa}. \quad (2.4)$$

The quantity Q compares the oscillation frequency of the field with its decay rate: a large Q means that the field oscillates many times before it leaks out of the resonator. If several independent loss channels are present, with decay rates κ_i , the total decay rate is the sum

$$\kappa = \sum_i \kappa_i, \quad (2.5)$$

which is equivalently expressed as

$$Q_{\text{tot}} = \left(\sum_i \frac{1}{Q_i} \right)^{-1}. \quad (2.6)$$

2.1.2 Internal loss and external coupling

A resonator can lose photons through uncontrolled internal mechanisms, such as dielectric loss or conductor loss, and through coupling to transmission lines used for measurement. We therefore separate the total linewidth into

$$\kappa = \kappa_{\text{in}} + \kappa_{\text{ex}}, \quad (2.7)$$

where κ_{in} is the intrinsic loss rate and κ_{ex} is the external coupling rate to the measurement line.

This distinction is physically important. Photons lost through the external measurement port can carry information to the detector, while photons lost internally are simply dissipated and do not contribute to the useful measurement signal. Therefore, in readout applications one often designs the resonator to be overcoupled, meaning that the external coupling dominates the internal loss:

$$\kappa_{\text{ex}} \gg \kappa_{\text{in}}. \quad (2.8)$$

2.2 The Langevin equation for the driven resonator

Section 2.1 characterized the resonator purely in terms of its linewidth κ , which describes how the field decays once it is left to evolve freely. To model an actual readout, however, we need to describe the resonator while it is being driven by an external microwave tone and dispersively coupled to the qubit. We first derive the equation of motion from input-output theory, treating the transmission line as a continuum of modes coupled to the resonator; we then specialize the resulting quantum Langevin equation to the driven, dispersively coupled qubit-resonator system; finally, we obtain the two classical, qubit-state-conditioned equations for the trajectories $\alpha_g(t)$ and $\alpha_e(t)$ used throughout the rest of this chapter.

2.2.1 Coupling between the resonator and the transmission line

The transmission line is treated as an open system supporting propagating microwave modes over a continuous range of frequencies. These modes can exchange photons with the resonator and are described by the operators $\hat{c}(\omega)$ and $\hat{c}^\dagger(\omega)$ that satisfy

$$[\hat{c}(\omega), \hat{c}^\dagger(\omega')] = 2\pi\delta(\omega - \omega'). \quad (2.9)$$

The resonator can be treated as the system of interest, and the transmission line as a thermal bath. The total Hamiltonian is written as

$$\hat{H}_{\text{tot}} = \hat{H}_s + \hat{H}_{\text{bath}} + \hat{H}_{\text{int}}.$$

The bath Hamiltonian is that of a continuum of independent harmonic modes,

$$\hat{H}_{\text{bath}} = \hbar \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} \omega \hat{c}^\dagger(\omega) \hat{c}(\omega).$$

The interaction Hamiltonian must describe the exchange of photons between the localized resonator mode $\hat{a}$ and the propagating modes $\hat{c}(\omega)$. It takes the following form

$$\hat{H}_{\text{int}} = -i\hbar\sqrt{\kappa} \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} [\hat{a}^\dagger \hat{c}(\omega) - \hat{c}^\dagger(\omega) \hat{a}].$$

The factor $-i$ is a phase convention and ensures that $\hat{H}_{\text{int}}$ is Hermitian, because the operator in square brackets is anti-Hermitian. The coupling amplitude is written as $\sqrt{\kappa}$ because κ is a decay rate, which is proportional to the squared coupling amplitudes for Fermi's Golden Rule. This leads to

$$\hat{H}_{\text{tot}} = \hat{H}_s + \hbar \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} \omega \hat{c}^\dagger(\omega) \hat{c}(\omega) - i\hbar\sqrt{\kappa} \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} [\hat{a}^\dagger \hat{c}(\omega) - \hat{c}^\dagger(\omega) \hat{a}]. \quad (2.10)$$

Here $\hat{H}_s$ is the system Hamiltonian of the resonator described in Eq. (1.16). The second term represents the freely propagating field in the transmission line, and the last term describes photon exchange between the resonator mode and the propagating continuum of propagating modes $\hat{c}(\omega)$.

The Heisenberg equation for the operator $\hat{c}(\omega)$ is

$$\frac{d\hat{c}(\omega, t)}{dt} = \frac{i}{\hbar} [\hat{H}_{\text{tot}}, \hat{c}(\omega, t)] = -i\omega \hat{c}(\omega, t) + \sqrt{\kappa} \hat{a}(t). \quad (2.11)$$

The cavity operator obeys

$$\frac{d\hat{a}(t)}{dt} = \frac{i}{\hbar} [\hat{H}_s, \hat{a}(t)] - \sqrt{\kappa} \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} \hat{c}(\omega, t). \quad (2.12)$$

We now solve Eq. (2.11). With an initial time t_0 , we obtain

$$\hat{c}(\omega, t) = e^{-i\omega(t-t_0)} \hat{c}(\omega, t_0) + \sqrt{\kappa} \int_{t_0}^t d\tau e^{-i\omega(t-\tau)} \hat{a}(\tau). \quad (2.13)$$

The first term is the freely propagating incoming field, while the second term is the field radiated into the line by the cavity. We define the input field operator

$$\hat{c}_{\text{in}}(t) \equiv \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} e^{-i\omega(t-t_0)} \hat{c}(\omega, t_0). \quad (2.14)$$

Substituting the solution for $\hat{c}(\omega, t)$ into Eq. (2.12), we obtain two contributions. The first one is proportional to the incoming field $\hat{c}_{\text{in}}(t)$. The second one contains the history of the cavity field,

$$\kappa \int_{t_0}^t d\tau \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} e^{-i\omega(t-\tau)} \hat{a}(\tau).$$

This term is a memory term: in principle, the present evolution of the cavity depends on its values at earlier times $\tau < t$. The Markov approximation consists in assuming that the transmission line has a broad and nearly flat spectral density around the resonator frequency, so that the bath correlation time is much shorter than the timescale over which $\hat{a}(t)$ changes. Physically, photons emitted into the line and returning to the resonator are negligible, so the environment has no memory of the system's past state. The integral is equal to:

$$\kappa \int_{t_0}^t d\tau \delta(t-\tau) \hat{a}(\tau) = \frac{\kappa}{2} \hat{a}(t).$$

This produces the damping term $-\kappa \hat{a}(t)/2$ in the Langevin equation.

We then obtain the quantum Langevin equation

$$\dot{\hat{a}}(t) = \frac{i}{\hbar} [\hat{H}_s, \hat{a}(t)] - \frac{\kappa}{2} \hat{a}(t) - \sqrt{\kappa} c_{\text{in}}(t). \quad (2.15)$$

Repeating the same construction with a final time $t_1 > t$ defines the output field $\hat{c}_{\text{out}}(t)$. Combining the input and output forms of the Langevin equation leads to the input-output relation

$$\hat{c}_{\text{out}}(t) = \hat{c}_{\text{in}}(t) + \sqrt{\kappa} \hat{a}(t). \quad (2.16)$$

This equation connects the intracavity field to the measurable outgoing microwave signal.

For a single-port measurement, namely when the cavity is coupled to just one propagating mode, Eq. (2.15), after replacing c_{in} by $-\hat{a}_{\text{in}}$ needed to match the sign convention of Ref. [9, Eq. (77)], can be expressed as

$$\dot{\hat{a}}(t) = \frac{i}{\hbar} [\hat{H}_s, \hat{a}(t)] - \frac{\kappa}{2} \hat{a}(t) + \sqrt{\kappa} \hat{a}_{\text{in}}(t). \quad (2.17)$$

This is the form that will be used below.

2.2.2 Derivation of the driven Langevin equation

Starting from Eq. (2.17), we now compute the commutator explicitly. The two identities needed are

$$[\hat{a}^\dagger \hat{a}, \hat{a}] = -\hat{a}, \quad [\hat{a}^\dagger + \hat{a}, \hat{a}] = -1. \quad (2.18)$$

Using Eq. (1.16), we obtain

$$\frac{i}{\hbar} [\hat{H}_s, \hat{a}] = i [(\delta_r + \chi \hat{\sigma}_z) \hat{a}^\dagger \hat{a}, \hat{a}] + i [\epsilon(t)(\hat{a}^\dagger + \hat{a}), \hat{a}] \quad (2.19)$$

$$= -i(\delta_r + \chi \hat{\sigma}_z) \hat{a} - i\epsilon(t). \quad (2.20)$$

Substituting this into Eq. (2.17) gives the quantum Langevin equation for the driven dispersive resonator:

$$\dot{\hat{a}}(t) = -i(\delta_r + \chi \hat{\sigma}_z) \hat{a}(t) - i\epsilon(t) - \frac{\kappa}{2} \hat{a}(t) + \sqrt{\kappa} \hat{a}_{\text{in}}(t). \quad (2.21)$$

where

- $\delta_r \equiv \omega_r - \omega_d$ is the detuning of the readout drive from the resonator frequency ω_r , which coincides with the midpoint of the qubit-state-dependent frequencies $\omega_r \pm \chi$ (see Sec. 2.2.3); it describes the phase rotation of the resonator field in the drive frame;
- $-\kappa \hat{a}/2$ is damping of the field amplitude;
- $\sqrt{\kappa} \hat{b}_{\text{in}}$ accounts for the incoming quantum field and vacuum noise;
- $-i\epsilon(t)$ is the readout drive.

2.2.3 State-dependent resonator dynamics

For readout simulations we are mainly interested in the average coherent field inside the resonator. We define the conditional amplitudes

$$\alpha_j(t) = \langle \hat{a}(t) \rangle_j, \quad j \in \{g, e\}, \quad (2.22)$$

where the expectation value is conditioned on the qubit state. In the dispersive regime, $\hat{\sigma}_z$ commutes with the effective Hamiltonian expressed in Eq. (1.16), so if the qubit starts in an eigenstate of $\hat{\sigma}_z$ it remains approximately in that eigenstate during an ideal quantum non-demolition measurement. We may therefore replace

$$\hat{\sigma}_z \rightarrow +1 \quad \text{for } |e\rangle, \quad \hat{\sigma}_z \rightarrow -1 \quad \text{for } |g\rangle. \quad (2.23)$$

Assuming that the mean of the vacuum noise vanishes,

$$\langle \hat{a}_{\text{in}}(t) \rangle = 0, \quad (2.24)$$

Eq. (2.21) becomes two classical differential equations:

$$\dot{\alpha}_e(t) = -\left[\frac{\kappa}{2} + i(\delta_r + \chi)\right] \alpha_e(t) - i\epsilon(t), \quad (2.25)$$

$$\dot{\alpha}_g(t) = -\left[\frac{\kappa}{2} + i(\delta_r - \chi)\right] \alpha_g(t) - i\epsilon(t). \quad (2.26)$$

These are the semiclassical Langevin equations used for dispersive readout. They show that the same drive pulse $\epsilon(t)$ produces two different resonator trajectories because the effective detuning depends on the qubit state:

$$\Delta_e = \delta_r + \chi, \quad \Delta_g = \delta_r - \chi. \quad (2.27)$$

Different sign conventions are possible depending on whether one defines the detuning as $\omega_r - \omega_d$ or $\omega_d - \omega_r$. The relative shift of the two resonator responses is 2χ . In a more compact form, Eq. (2.25) and Eq. (2.26), using Δ_e and Δ_g , can be written as (see Ref. [9, Eq. (111)])

$$\dot{\alpha}_j(t) = -\left(\frac{\kappa}{2} + i\Delta_j\right)\alpha_j(t) - i\epsilon(t), \quad j \in \{g, e\}. \quad (2.28)$$

2.2.4 Photon number and IQ plane representation

Having derived how the conditional amplitudes $\alpha_g(t)$ and $\alpha_e(t)$ evolve under the Langevin equation, we now turn to how this information is actually extracted from a measurement, and to the specific quantity that determines how well the two qubit states can be distinguished.

In physical implementations of superconducting qubits, the readout pulse is a classical microwave tone at frequency ω_{RO} sent to the resonator. After reflection or transmission, the signal can be written as (see Ref. [6, Eq. (147)])

$$s(t) = A_{\text{RO}} \cos(\omega_{\text{RO}}t + \theta_{\text{RO}}),$$

which in complex notation is the real part of

$$s(t) = \text{Re}\{A_{\text{RO}}e^{i\theta_{\text{RO}}}e^{i\omega_{\text{RO}}t}\}.$$

Defining the complex phasor $\alpha = A_{\text{RO}}e^{i\theta_{\text{RO}}}$, the signal quadratures are

$$I = \text{Re}(\alpha), \quad Q = \text{Im}(\alpha),$$

and the measurement outcome corresponds to a point (I, Q) in the I/Q plane. The resonator state under coherent drive is a coherent state $|\alpha\rangle$ obeying $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$ and the average photon number is

$$n = \langle \hat{a}^\dagger \hat{a} \rangle = |\alpha|^2 = I^2 + Q^2.$$

In the IQ plane, the two possible qubit states correspond to two different trajectories and, in steady state, to two different points. The relevant quantity for readout is the separation

$$\Delta\alpha(t) = \alpha_e(t) - \alpha_g(t). \quad (2.29)$$

A common choice is to drive near the midpoint of the two qubit-dependent resonances. In that case, the two responses can have approximately equal magnitude but different phase, so that the state information is encoded primarily as a rotation in the IQ plane.

2.3 Readout pulses

The previous section showed that the readout signal is encoded in the separation $\Delta\alpha(t) = \alpha_e(t) - \alpha_g(t)$ between the two conditional resonator trajectories in the IQ plane: since the qubit state is not measured directly, it must be inferred from the resonator response, which differs between $|g\rangle$ and $|e\rangle$ as a consequence of the dispersive shift. This separation, together with the intracavity photon population that produces it, is entirely determined by the shape of the drive $\epsilon(t)$. Different choices of $\epsilon(t)$ therefore lead to different trade-offs between measurement speed, state distinguishability, and residual photon population. This section identifies the physical requirements a good readout pulse should satisfy, and then presents concrete pulse shapes designed to meet them.

2.3.1 Objectives of a readout pulse

A readout pulse is therefore designed to make the two conditional responses $\alpha_g(t)$ and $\alpha_e(t)$ distinguishable while keeping the measurement fast, controlled and minimally invasive. The construction of an efficient readout pulse is a compromise between several physical requirements. The most important objectives are described below.

- Fast ring-up of the intracavity field: when the readout drive is switched on, the resonator does not reach its steady-state amplitude instantaneously. Reducing the ring-up time allows the measurement to reach the useful discrimination regime more rapidly.
- Large separation between $\alpha_g(t)$ and $\alpha_e(t)$ in the IQ plane: one of the main goals is to produce two distinguishable coherent states of the resonator field with respect to the noise of the measurement chain. The quantity that determines how distinguishable the two qubit states are is the separation $\Delta\alpha(t)$, which should be maximized during the measurement window.
- Short measurement time relative to the qubit relaxation time: the readout must be finished as quickly as possible, since a longer measurement duration increases the likelihood that the qubit will relax while the measurement is taking place.
- Controlled maximum photon number: the strength of the readout signal increases with the number of photons stored in the resonator,

$$n_j(t) = |\alpha_j(t)|^2.$$

This allows the signal to be better distinguished from background noise. However, the photon number cannot be increased arbitrarily. If an excessively large number of photons populates the resonator, several phenomena may arise, including nonlinear effects that can significantly distort the resonator's response and qubit ionization.

- Low residual photon population after the measurement: at the end of the measurement, the resonator should return as close as possible to the vacuum state. Residual photons are problematic because they continue to shift the qubit frequency through dispersive coupling. This can dephase the qubit and compromise operations performed shortly afterwards.

For a pulse, after the drive is switched off at time T , the field decays passively. However, this passive decay may be too slow. A shaped readout pulse can instead include an active depletion segment, designed to drive the resonator field back toward zero faster than natural decay alone.

- Minimal backaction on the qubit: dispersive readout is designed to be approximately quantum non-demolition, meaning that the measurement should reveal the qubit state without inducing unwanted transitions between $|g\rangle$ and $|e\rangle$.

In the following paragraphs, we introduce two types of pulses that will be employed later in this chapter to simulate the semiclassical reset dynamics of the resonator. The first, and conceptually simplest, is the rectangular pulse, which will serve as a fundamental reference. The second is the CLEAR pulse, which, while also relatively simple, is of particular interest because it enables the optimized rapid loading and unloading of photons in the resonator.

2.3.2 The rectangular pulse

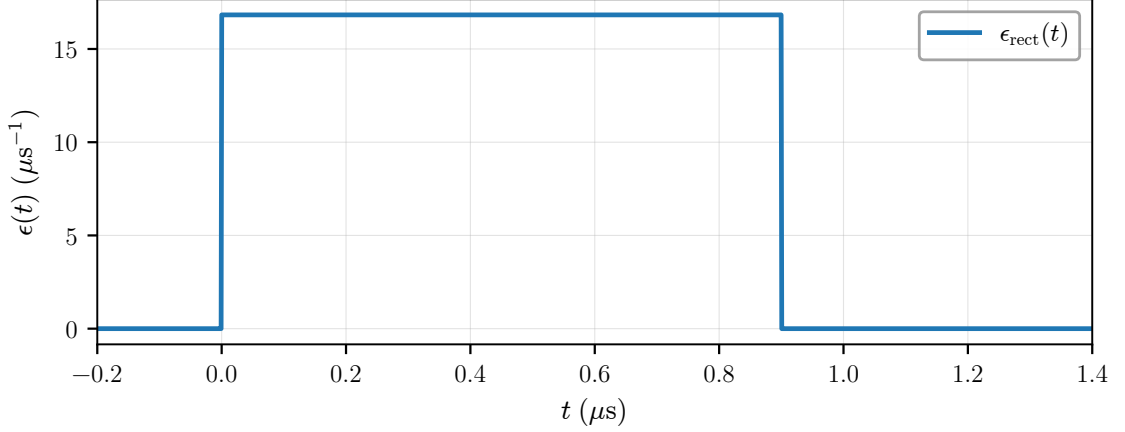


Figure 2.1: Shape of the rectangular readout pulse, $\epsilon(t) = A$ for $0 \leq t < T$.

The rectangular pulse consists of a constant drive $\epsilon(t) = A$. For this pulse, the conditional Langevin equation has the generic form

$$\dot{\alpha}_j(t) = -\left(\frac{\kappa}{2} + i\Delta_j\right)\alpha_j(t) - iA. \quad (2.30)$$

The steady-state amplitude is obtained by setting $\dot{\alpha}_j = 0$ (see Ref. [9, Eq. (112)]):

$$\alpha_j^{ss} = -\frac{iA}{\frac{\kappa}{2} + i\Delta_j}. \quad (2.31)$$

For the initial condition $\alpha_j(0) = 0$, which is equivalent to the vacuum, the full solution is

$$\alpha_j(t) = \alpha_j^{ss} \left[1 - e^{-\left(\frac{\kappa}{2} + i\Delta_j\right)t}\right]. \quad (2.32)$$

This expression contains both the ring-up and the phase rotation of the field. The real decay rate $\kappa/2$ determines how fast the amplitude approaches its steady-state value, whereas the imaginary part Δ_j determines the rotation in the IQ plane.

If the drive is switched off at time T , the subsequent evolution is

$$\alpha_j(t > T) = \alpha_j(T) e^{-\left(\frac{\kappa}{2} + i\Delta_j\right)(t-T)}. \quad (2.33)$$

Consequently, the photon number decays as

$$n_j(t > T) = |\alpha_j(T)|^2 e^{-\kappa(t-T)}. \quad (2.34)$$

This is the time-domain ring-down discussed in Section 2.1.1: the field amplitude decays with rate $\kappa/2$, while the stored energy or photon population decays with rate κ .

2.3.3 The CLEAR pulse

The CLEAR pulse is a widely recognized technique developed to achieve the steady state and reset the resonator more rapidly following measurement. The key idea is to use short additional drive segments, referred to as kicks, to speed up the resonator's transition between steady-state field amplitudes. The CLEAR pulse consists of five segments: two ring-up kicks, a central readout segment, and two ring-down kicks.

The presence of two kicks, rather than one, is important. Since the same drive pulse produces different trajectories depending on the qubit state, a single kick could bring one trajectory to the desired target point, but in general it would not bring both trajectories to the correct target. The two kicks make it possible to impose two constraints, one for each qubit state.

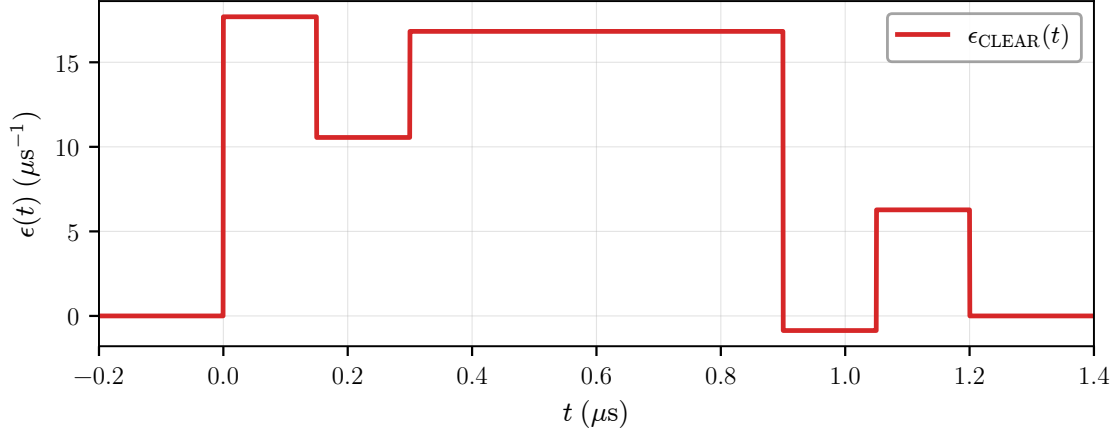


Figure 2.2: Example of a CLEAR pulse shape.

For a constant drive ϵ applied during a time interval τ , the solution of the Eq. (2.28) is

$$\alpha_j(t + \tau) = c_j \alpha_j(t) - i\epsilon \frac{1 - c_j}{\lambda_j}, \quad (2.35)$$

where

$$\lambda_j = \frac{\kappa}{2} + i\Delta_j, \quad c_j = e^{-\lambda_j \tau}.$$

Applying two constant kicks, ϵ_1 before and ϵ_2 after, each of duration t_{kick} , gives

$$\alpha_j^{(f)} = c_j^2 \alpha_j^{(i)} - i \frac{(1 - c_j)c_j}{\lambda_j} \epsilon_1 - i \frac{1 - c_j}{\lambda_j} \epsilon_2. \quad (2.36)$$

For the two qubit states, this becomes a 2×2 linear system:

$$M \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \end{pmatrix} = b, \quad (2.37)$$

with

$$M_{j1} = -i \frac{(1 - c_j)c_j}{\lambda_j}, \quad M_{j2} = -i \frac{1 - c_j}{\lambda_j},$$

and

$$b_j = \alpha_j^{(f)} - c_j^2 \alpha_j^{(i)}.$$

This construction is used twice. For the ring-up kicks the initial amplitude is $\alpha_j^{(i)} = 0$ and the target is the steady-state amplitude associated with the readout drive A , expressed in Eq. (2.31).

For the ring-down kicks the initial amplitude is $\alpha_{j,\text{ss}}$, while the target is $\alpha_j^{(f)} = 0$. Thus the ring-down part of the CLEAR pulse actively depletes the cavity field instead of waiting for passive exponential decay.

2.3.4 Overview of alternative pulse-shaping strategies

Besides piecewise-constant sequences, microwave pulses can be shaped using smoother envelopes. An abrupt change in the drive amplitude contains a broad range of frequency components and may excite unwanted modes or be distorted by the finite bandwidth of the control electronics. For this reason, the sharp edges of a rectangular pulse can be replaced by Gaussian, cosine-shaped, or other smooth ramps. This reduces the high-frequency content of the waveform and generally produces a more controlled transient response.

A more recent readout strategy is the active four-tone readout (A4R) protocol [19]. This pulse, similar to the CLEAR pulse, is designed to fill and empty the resonator rapidly. The waveform was first identified through model-free reinforcement learning and was subsequently generalized into a simple analytical sequence that can be calibrated directly on superconducting

hardware. A4R consists of four consecutive segments: ring-up, steady-state readout, photon depletion, and kickback. The first segment rapidly populates the resonator, while the second maintains the field during state discrimination. The depletion segment removes most of the stored photons after the measurement, and the final kickback corrects the small residual field that remains after the main reset segment.

The protocol is designed primarily for the regime $\frac{\kappa}{\chi} \gg 1$, where a single ring-up segment is sufficient because the phase accumulated through the dispersive interaction during this stage is relatively small. The two reset segments may have independently adjustable amplitudes and durations, providing a flexible active depletion of the resonator field. This differs from the standard CLEAR construction, in which the two segments associated with each ring-up or ring-down stage are commonly constrained to have equal duration.

Although the analytical A4R waveform is piecewise constant, its sharp transitions can be smoothed to account for the finite bandwidth of the control electronics. The resulting waveform preserves the four-segment structure while being more suitable for direct implementation on real hardware.

Another analytical strategy for dispersive readout with intrinsic resonator reset is Derivative Removal for Annulment of Cavity-Hybridized Measurement Adiabats (DRACHMA) [20]. Unlike the piecewise-constant CLEAR and A4R waveforms, DRACHMA starts from a smooth target intra-cavity response $a_T(t)$ that vanishes at the beginning and at the end of the measurement. The corresponding complex input pulse is then derived analytically from the state-dependent resonator response, so that the same waveform can return the cavity field to zero for both qubit states. The method can also be generalized to additional transmon states and extended to account for the resonator Kerr nonlinearity, providing a smooth alternative that combines qubit readout and active resonator reset.

The present work focuses on the CLEAR protocol because it was the first such protocol to be experimentally implemented and remains the simplest, with a piecewise-constant structure that follows directly from the linear semiclassical equations developed in this chapter.

2.4 Semiclassical simulation of resonator reset pulses

The rectangular and CLEAR envelopes constructed in Section 2.3.3 were obtained by solving the conditional Langevin equations of Section 2.2.3: first for a single constant segment, Eq. (2.32), and then for the two-kick sequence that constitutes the CLEAR pulse, Eq. (2.35). These expressions determine $\alpha_j(t)$ exactly for any choice of κ , χ , t_{kick} and drive amplitude, but on their own they don't show how the two pulses actually compare in practice: how large the transient overshoot in the photon number is, how compact the corresponding trajectory in the IQ plane becomes, and how much population is left in the resonator once the drive has been switched off. This section evaluates the chapter's solutions for a physically motivated set of parameters. The reference values are chosen from the parameter regime of the CLEAR reset experiment of McClure et al. [1], so that the comparison refers to a realistic implementation while remaining within the linear semiclassical model developed above.

2.4.1 Aim of the simulations

Three questions are addressed, first for a reference set of resonator parameters and then by varying two of them:

1. how the conditional photon population $n_j(t) = |\alpha_j(t)|^2$ evolves during and after the pulse, for both qubit states;
2. how the two conditional trajectories $\alpha_g(t)$ and $\alpha_e(t)$ separate in the IQ plane, since this separation is what determines how distinguishable the two qubit states are;
3. how much photon population is left in the resonator once the pulse has ended, since minimizing this residual population is the actual purpose of a reset protocol.

The comparison is restricted to the linear semiclassical model of this chapter and does not attempt to reproduce a full experimental characterization. The goal here is instead to isolate,

within a single self-consistent and analytically transparent model, the basic mechanism that makes active reset faster than passive decay, and to identify the trade-offs that a real implementation of the CLEAR pulse has to confront.

Working within a linear model has a limitation, which is worth stating explicitly before looking at the results. Changing the drive amplitude only rescales the resonator response in Eq. (2.32) and Eq. (2.35); it never introduces nonlinear effects, because the model contains no term coupling the photon number back to the resonator frequency. In a real device this coupling is present, in the form of a Kerr-like shift induced by the qubit itself, and becomes non-negligible at the photon numbers typically needed for fast, high-fidelity readout. If it were included, the effective detuning Δ_j would depend on $n_j(t)$, and a CLEAR pulse computed from the linear model would no longer cancel the residual field exactly. The simulations presented below should therefore be read as characterizing the best-case, low-power limit of the CLEAR protocol, in which the linear approximation is expected to hold.

2.4.2 Simulation method and parameters

For each qubit state $j \in \{g, e\}$, the resonator field is propagated using Eq. (2.28), with the two conditional detunings $\Delta_g = \delta_r - \chi$ and $\Delta_e = \delta_r + \chi$. Because the drive is piecewise constant on every segment of both pulses, each segment is evaluated directly with the closed-form single-segment solution, Eq. (2.32); the trajectory is then obtained by propagating the final amplitude of one segment into the initial condition of the next. The resonator is assumed to start empty, $\alpha_g(0) = \alpha_e(0) = 0$, and the two quantities used throughout this section to characterize the dynamics are the conditional photon population $n_j(t) = |\alpha_j(t)|^2$ and the state separation $S(t) = |\alpha_g(t) - \alpha_e(t)|$, already introduced when discussing the IQ-plane representation of the readout signal.

As in Section 2.3.3, the drive is applied at the midpoint between the two dispersively shifted resonator frequencies, $\delta_r = 0$, so that $|\Delta_g| = |\Delta_e| = \chi$. This choice is not simply a matter of convenience. McClure et al. note that driving at the midpoint frequency maximizes the readout signal, makes the steady-state photon number independent of the qubit state in the linear-response regime, and allows every segment of the CLEAR pulse to be applied along a single quadrature [1]. This last property can be verified directly from the two-kick construction of Eq. (2.35): although the conditional steady-state amplitudes $\alpha_{g,ss}$ and $\alpha_{e,ss}$ are individually complex, solving the 2×2 linear system for ϵ_1, ϵ_2 with $\Delta_g = -\Delta_e$ always yields kick amplitudes with a vanishing imaginary part, for any $\kappa, \chi, t_{\text{kick}}$ and A . This follows from the symmetry $\Delta_g = -\Delta_e$, which maps the linear system solved for the g trajectory onto the complex conjugate of the one solved for e , and forces the two solutions to combine into a real result; all five segments of the CLEAR pulse used below are therefore purely real, consistent with the experimental design of Ref. [1].

The parameters used in the simulations, listed in Table 2.1, are taken from the CLEAR pulse experiment of McClure et al. [1]. The resonator linewidth $\kappa/2\pi$ and dispersive shift $|\chi|/2\pi$ correspond to the values they extract from frequency-domain measurements, and the kick duration t_{kick} matches the value used in their experiment, chosen to be comparable to the cavity time constant $T_{\text{cav}} = 1/\kappa$. The target photon number n_{target} is chosen to match the normalized drive power $P_{\text{norm}} = 3.6$ used to produce the IQ-plane trajectories in Fig. 3 of Ref. [1], since in the linear-response regime the steady-state photon number coincides with P_{norm} by construction. These values are used here as realistic physical scales for the simplified linear model, and not as an attempt to reproduce the full experimental setup, which also includes a finite-bandwidth control chain, a thermal photon background, and the resonator nonlinearity discussed above.

The readout amplitude A is fixed by imposing the desired steady-state photon population during the central readout segment. For a constant drive, the steady-state amplitude is described by Eq. (2.31).

Since the drive is applied at the midpoint, $|\Delta_j| = |\chi|$, and therefore

$$n_{ss} = |\alpha_{j,ss}|^2 = \frac{|A|^2}{\left(\frac{\kappa}{2}\right)^2 + \chi^2}.$$

Imposing

$$n_{ss} = n_{\text{target}}$$

Parameter	Symbol	Value
Resonator linewidth	$\kappa/2\pi$	1.1 MHz
Angular linewidth	κ	$2\pi \times 1.1 = 6.91 \mu\text{s}^{-1}$
Cavity time constant	$T_{\text{cav}} = 1/\kappa$	0.145 μs
Base detuning	δ_r	0
Dispersive shift	$ \chi /2\pi$	1.3 MHz
Angular dispersive shift	$ \chi $	$2\pi \times 1.3 = 8.17 \mu\text{s}^{-1}$
Initial amplitude	α_0	0
Kick duration	t_{kick}	0.150 μs
Readout duration	t_{readout}	0.600 μs
Target photon number	n_{target}	3.6

Table 2.1: Parameters used in the simulations, taken from Ref. [1].

gives

$$A = \sqrt{n_{\text{target}} \left[\left(\frac{\kappa}{2} \right)^2 + \chi^2 \right]}.$$

Using the parameters in Table 2.1, this gives

$$A \simeq 16.83 \mu\text{s}^{-1}.$$

The rectangular pulse is kept on for the same duration as the ring-up plus readout part of the CLEAR pulse,

$$T_{\text{rect}} = 2t_{\text{kick}} + t_{\text{readout}},$$

and then it is switched off. The total simulation time is chosen equal to the full CLEAR duration,

$$T_{\text{CLEAR}} = 4t_{\text{kick}} + t_{\text{readout}}.$$

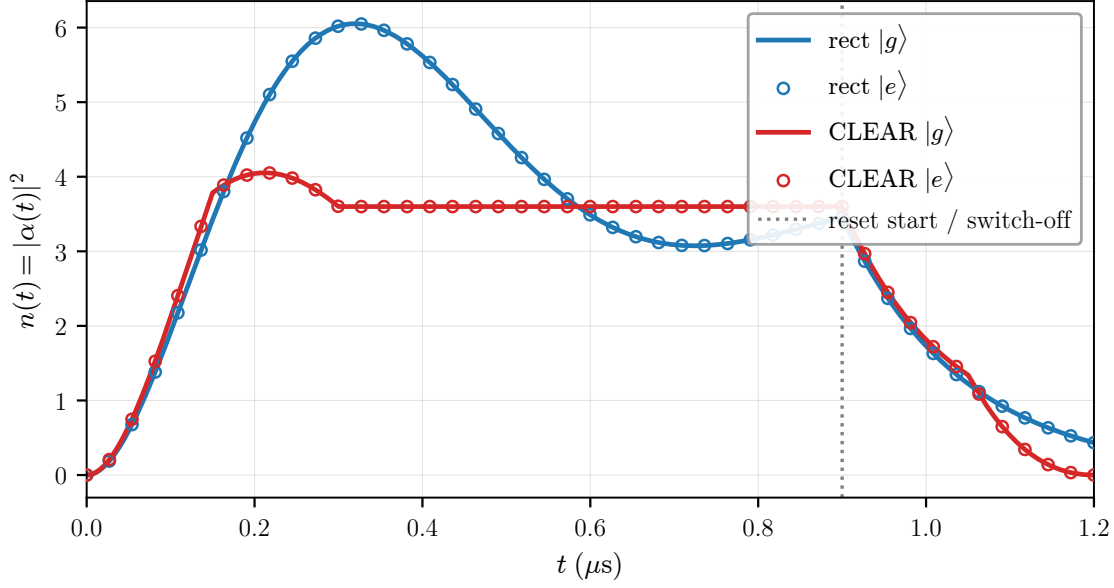
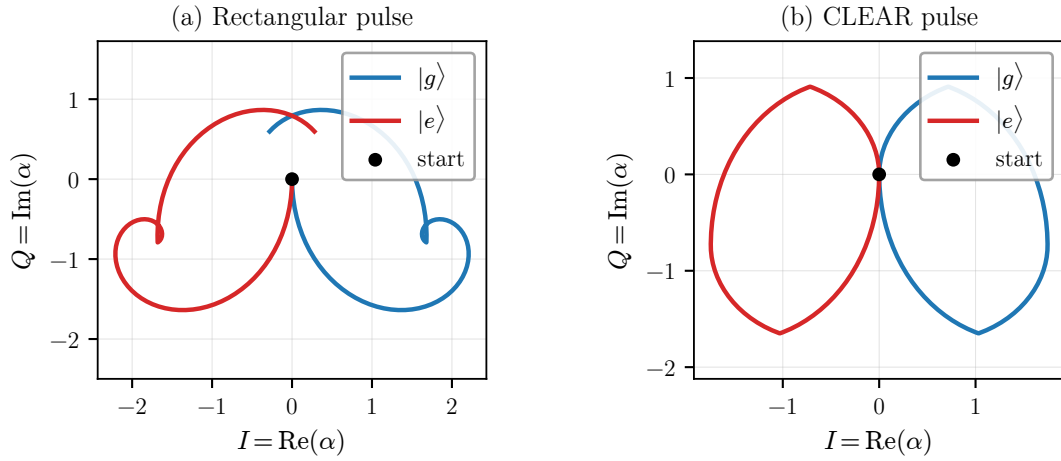
In this way, the rectangular pulse has a passive decay window of duration $2t_{\text{kick}}$, which can be directly compared with the two active CLEAR ring-down kicks: the two pulses are switched off and are allowed to reset over the same total time.

All simulations were implemented in Julia; the code returns, for each pulse and each qubit state, the photon population, the IQ-plane trajectory, the state separation, and the final residual photon population.

2.4.3 Results

Photon population and IQ-plane trajectories

Fig. 2.3 shows the conditional photon population $n_j(t) = |\alpha_j(t)|^2$ obtained for the reference parameters of Table 2.1. Since the drive is applied at the midpoint between the two dispersively shifted resonator frequencies, the two effective detunings have equal magnitude and opposite sign, $\Delta_g = -\chi$ and $\Delta_e = +\chi$, consistent with $\Delta_e = \delta_r + \chi$ and $\Delta_g = \delta_r - \chi$ for $\delta_r = 0$. As shown explicitly below, this symmetry makes the two conditional photon populations coincide exactly in the linear model, even though the underlying trajectories $\alpha_g(t)$ and $\alpha_e(t)$ are different complex functions of time; the difference between the two conditional responses only becomes visible once they are compared in the IQ plane, in Fig. 2.4.

Figure 2.3: Photon population $n_j(t)$ for the rectangular and CLEAR pulses.Figure 2.4: IQ-plane trajectories of $\alpha_g(t)$ and $\alpha_e(t)$.

For a constant drive, the dynamics is described by Eq. (2.32). It follows that

$$|\alpha_j(t) - \alpha_{j,ss}| = |\alpha_{j,ss}| e^{-\kappa t/2},$$

so the distance from the steady state decreases monotonically. However, the photon population depends on the distance from the origin,

$$n_j(t) = |\alpha_j(t)|^2.$$

Since

$$e^{-\lambda_j t} = e^{-\kappa t/2} e^{-i\Delta_j t},$$

the transient term both decays and rotates in the IQ plane. As a consequence, the trajectory can pass temporarily farther from the origin than the steady-state point, producing an overshoot in the photon population.

This can be seen explicitly by computing the modulus squared of Eq. (2.32):

$$n_j(t) = |\alpha_j(t)|^2 = \frac{|A|^2}{\left(\frac{\kappa}{2}\right)^2 + \Delta_j^2} \left[1 + e^{-\kappa t} - 2e^{-\kappa t/2} \cos(\Delta_j t) \right].$$

The oscillatory term $\cos(\Delta_j t)$ is responsible for the ringing observed in the rectangular pulse in Fig. 2.3, where it produces a pronounced overshoot above the target readout population before the photon number relaxes back toward its steady-state value. Because the cosine is an even function, $\cos(\Delta_j t)$ takes the same value for $\Delta_g = -\chi$ and $\Delta_e = +\chi$; the prefactor $|A|^2/[(\kappa/2)^2 + \Delta_j^2]$ is likewise unchanged under $\Delta_j \rightarrow -\Delta_j$, since it only depends on Δ_j^2 . The two conditional photon populations are therefore exactly equal in this symmetric configuration, $n_g(t) = n_e(t)$ for all t , which is why only two curves, rather than four, are visually distinguishable in Fig. 2.3: the rectangular and CLEAR traces plotted for $j = g$ and $j = e$ overlap pairwise.

This degeneracy in photon number does not mean that the qubit state is indistinguishable. Even though $|\alpha_g(t)| = |\alpha_e(t)|$ at every instant, the complex phases of $\alpha_g(t)$ and $\alpha_e(t)$ differ, since the field rotates at rate $-\Delta_j$ in the IQ plane while its magnitude decays or grows; this is exactly what Fig. 2.4 shows. For the rectangular pulse, the trajectory spirals outward past the target circle $|\alpha| = \sqrt{n_{\text{target}}}$ before settling, in line with the overshoot just described and the two conditional trajectories curve in opposite directions, clockwise for one qubit state and counterclockwise for the other, because Δ_g and Δ_e have opposite sign. For the CLEAR pulse, the two ring-up kicks steer both trajectories directly onto their steady-state points with essentially no overshoot, and the two ring-down kicks bring them back to the origin along a comparably direct path, producing the visibly more compact loops of Fig. 2.4(b). These pictures reproduce those obtained experimentally by McClure et al. for the same target photon number, $P_{\text{norm}} = 3.6$ (Fig. 3 of Ref. [1]).

At the vertical line in Fig. 2.3, the rectangular pulse is switched off. After this time the drive is zero, the field evolves freely, and the photon population decays according to Eq. (2.33): relying only on natural cavity decay, it is still far from zero at the end of the simulated window, leaving the finite residual population quantified below.

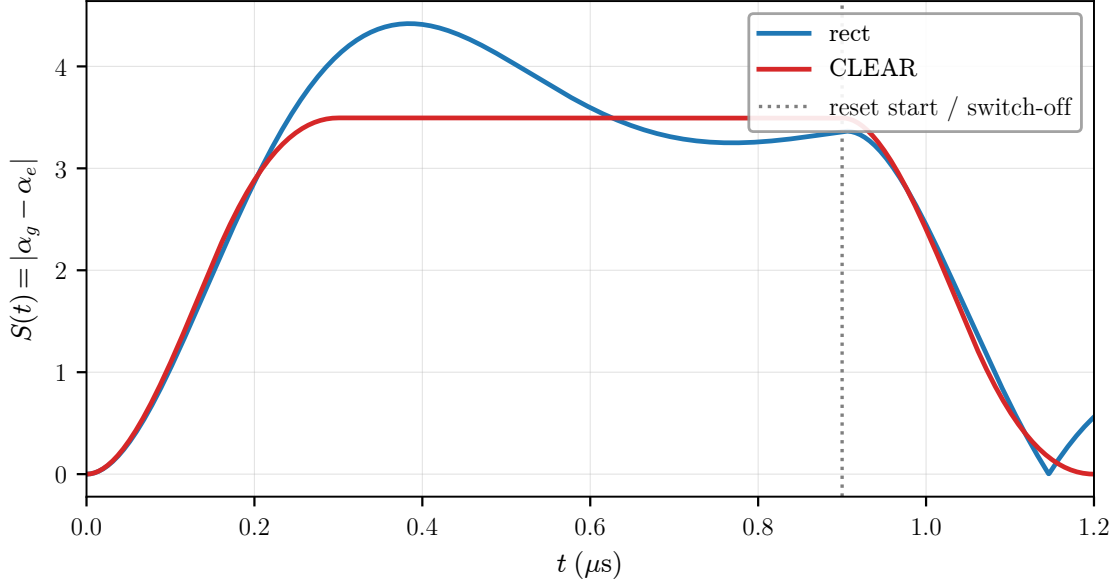
The CLEAR pulse is designed to avoid this passive behavior. During the ring-up stage, the first two kick segments are not simply equal to the readout drive amplitude A ; instead, they are computed so that both conditional trajectories reach the desired readout amplitudes at the end of the ring-up stage, which is why the photon population in Fig. 2.3 rises smoothly to its target value, without the overshoot seen for the rectangular pulse, and then stays flat during the central readout segment. In the final ring-down stage, the last two kicks actively drive the resonator field back to the origin of the IQ plane, so that the residual photon population is strongly suppressed compared with the rectangular pulse.

The reason why the CLEAR pulse produces this behavior is that the kick amplitudes are not arbitrary. As discussed in Section 2.3.3, they are obtained by solving a linear system that imposes boundary conditions on both conditional resonator trajectories. For the ring-up stage, the two kicks are chosen so that both trajectories reach their corresponding readout steady-state amplitudes,

$$\alpha_j^{(f)} = \alpha_{j,\text{ss}}, \quad j \in \{g, e\}.$$

For the ring-down stage, the same construction is applied with the opposite target:

$$\alpha_j^{(f)} = 0.$$

Figure 2.5: State separation $S(t)$ for the rectangular and CLEAR pulses.

quantity	rectangular pulse	CLEAR pulse
maximum residual population at the end	4.34×10^{-1}	5×10^{-32}
maximum population	6.05	4.05
maximum separation $S(t)$	4.42	3.49
time of maximum separation $S(t)$	$0.38 \mu s$	$0.68 \mu s$

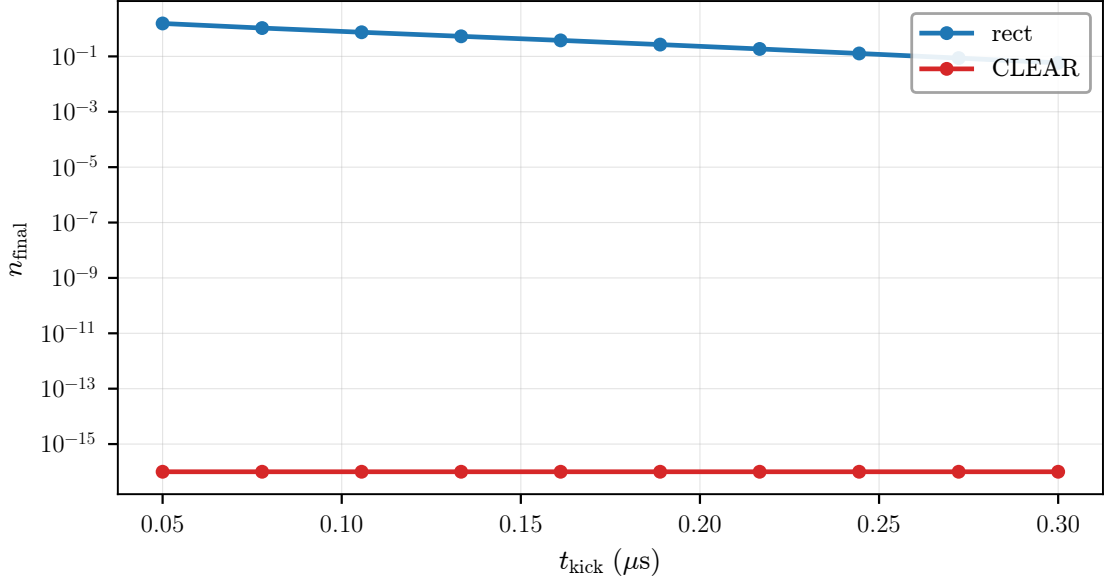
Table 2.2: Metrics extracted from the rectangular and CLEAR pulse simulations (parameters of Table 2.1).

Fig. 2.5 shows the corresponding state separation $S(t) = |\alpha_g(t) - \alpha_e(t)|$, and Table 3.7 summarizes the metrics extracted from both figures. For the rectangular pulse, an appreciable residual photon population, 4.34×10^{-1} , is still present at the end of the simulated window, consistent with the fact that its ring-down relies only on passive cavity decay. For the CLEAR pulse, the residual population is suppressed to the level of numerical noise (5×10^{-32}), confirming that the ring-down kicks return the field essentially to the vacuum state. This idealized, noiseless value should not be compared quantitatively with an experimentally measured residual population, which is bounded from below by thermal and technical noise; it should instead be read as confirmation that, within the linear model, the ring-down kicks cancel the field exactly rather than only approximately.

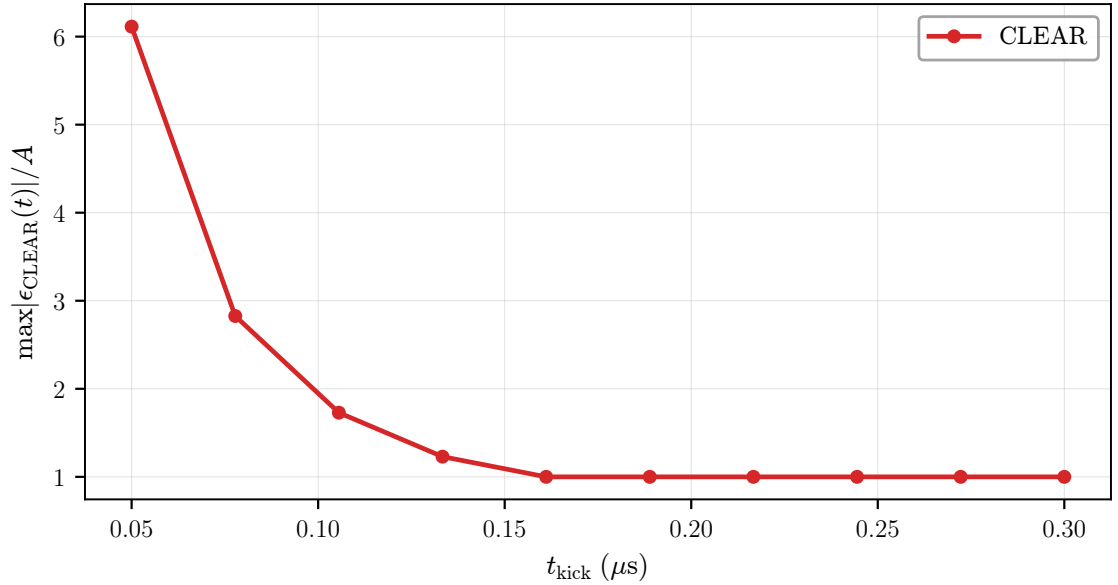
The maximum photon population reached during the CLEAR sequence (4.05) is also closer to the target value $n_{\text{target}} = 3.6$ than for the rectangular pulse (6.05), reflecting the overshoot discussed above. The rectangular pulse reaches a larger maximum separation S_{max} for the same reason, and it does so earlier ($t_{S_{\text{max}}} = 0.38 \mu s$) than the CLEAR pulse ($0.68 \mu s$), whose ring-up kicks are designed to bring the field directly to the steady state rather than to overshoot it.

Trade-off between reset speed and drive amplitude

After comparing the two pulse shapes for the reference set of parameters, the CLEAR kick duration is now varied in order to understand the trade-off between reset speed and pulse amplitude. In the ideal linear model, the CLEAR construction strongly suppresses the residual photon population for every kick duration considered, as shown in Fig. 2.6: unlike the rectangular pulse, whose residual population after the same window decreases only gradually as t_{kick} (and hence the total passive decay time) is increased, the CLEAR pulse leaves the resonator close to the numerical noise floor throughout the entire range.

Figure 2.6: Final residual photon population vs. kick duration t_{kick} .

This suppression is not free, however. Fig. 2.7 shows the maximum amplitude reached by the CLEAR pulse as a function of t_{kick} : for short kick durations the resonator field has to be displaced across the IQ plane in a very short time, so the construction of Eq. (2.35) requires correspondingly large kick amplitudes; as t_{kick} is increased, the same displacement can be achieved more gradually, and the required maximum amplitude decreases monotonically towards the central readout amplitude A . This is the main trade-off of the CLEAR protocol within the linear model: active reset can suppress the residual photon population almost arbitrarily well, but doing so faster requires a stronger drive.

Figure 2.7: Maximum CLEAR amplitude vs. kick duration t_{kick} .

In practice, this trade-off is more severe than the linear model alone suggests. Reducing t_{kick} increases the drive amplitude required, which for a fixed experimental setup eventually pushes the resonator response out of the linear regime assumed throughout this section; the CLEAR pulse computed from Eq. (2.35) then no longer cancels the residual field exactly, for the same reason discussed for high readout powers above. There is a second limitation that the present model cannot capture, because it treats every segment as instantaneously constant,

whereas in a real setup these transitions are rounded by the finite bandwidth and rise time of the control electronics. This rounding prevents the implemented pulse from being perfectly piecewise constant and can therefore reduce the cancellation of the residual field. The achievable kick duration in a real implementation is therefore bounded not only by the growing drive amplitude of Fig. 2.7, but also by how closely the control hardware can approximate a piecewise-constant pulse.

Effect of the ratio χ/κ on state separation

The comparison above uses a single value of the dispersive shift, $|\chi|/2\pi = 1.3$ MHz. We now vary this value while keeping the resonator linewidth κ fixed, to explore the effect of the ratio χ/κ , which controls how strongly the two qubit-state-dependent resonator responses separate in the IQ plane. For a drive applied at the midpoint between the two shifted resonator frequencies, the effective detunings remain $\Delta_g = -\chi$ and $\Delta_e = +\chi$, so larger values of χ increase the relative phase rotation between the two conditional fields; the photon decay rate after switch-off, however, is set by κ alone and is therefore unaffected. Changing χ/κ thus modifies the phase evolution of the field during ring-up and ring-down without changing how quickly a population, once created, decays passively. For consistency with the rest of this section, the readout amplitude A is recomputed at every value of χ so that the steady-state photon number is still $n_{\text{target}} = 3.6$, and t_{kick} and t_{readout} are kept fixed at the reference values of Table 2.1.

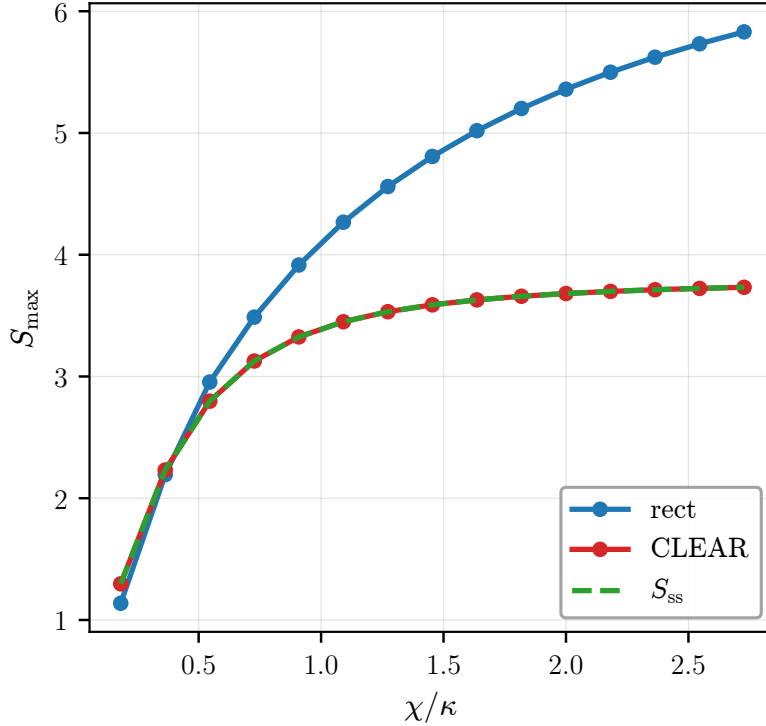


Figure 2.8: Maximum separation vs. χ/κ .

Fig. 2.8 shows the resulting maximum separation $S_{\text{max}} = |\alpha_e - \alpha_g|$ as a function of χ/κ , together with the steady-state prediction

$$S_{\text{ss}} = 2\sqrt{n_{\text{target}}} \frac{|\chi|}{\sqrt{(\kappa/2)^2 + \chi^2}},$$

obtained by evaluating $|\alpha_{e,\text{ss}} - \alpha_{g,\text{ss}}|$ at fixed $n_{\text{ss}} = n_{\text{target}}$. For the CLEAR pulse, S_{max} closely tracks S_{ss} at every value of χ/κ , confirming that the ring-up kicks drive the resonator directly to the target separation without overshooting it. The rectangular pulse reaches somewhat larger values of S_{max} , but, as discussed above, only because of the same transient overshoot that

inflates its maximum photon population, not because it approaches the target readout state more efficiently.

Better state distinguishability at larger χ/κ is not free either: Fig. 2.9 shows the maximum CLEAR drive amplitude in this same scan, normalized to the readout amplitude A . For $\chi/\kappa \gtrsim 1.2$ the ratio is essentially 1, meaning the kicks are barely stronger than the steady-state drive itself; but as χ/κ decreases below 1, the required kick amplitude grows quickly, reaching more than five times A at the smallest ratio considered. This trend has a simple interpretation: for small χ/κ the dispersive shift contributes little to steering the field away from the origin relative to the damping $\kappa/2$, so the two-kick construction of Eq. (2.35) must rely on brute-force amplitude, rather than on the detuning-driven phase rotation, to displace the resonator to its target in a fixed time t_{kick} . This mirrors the trade-off already found when varying t_{kick} at fixed χ : whenever the CLEAR pulse is asked to produce a given displacement in the IQ plane faster than the natural dynamics would allow, either by shortening the kick or by operating at a small dispersive shift, it can only do so by increasing the drive amplitude.

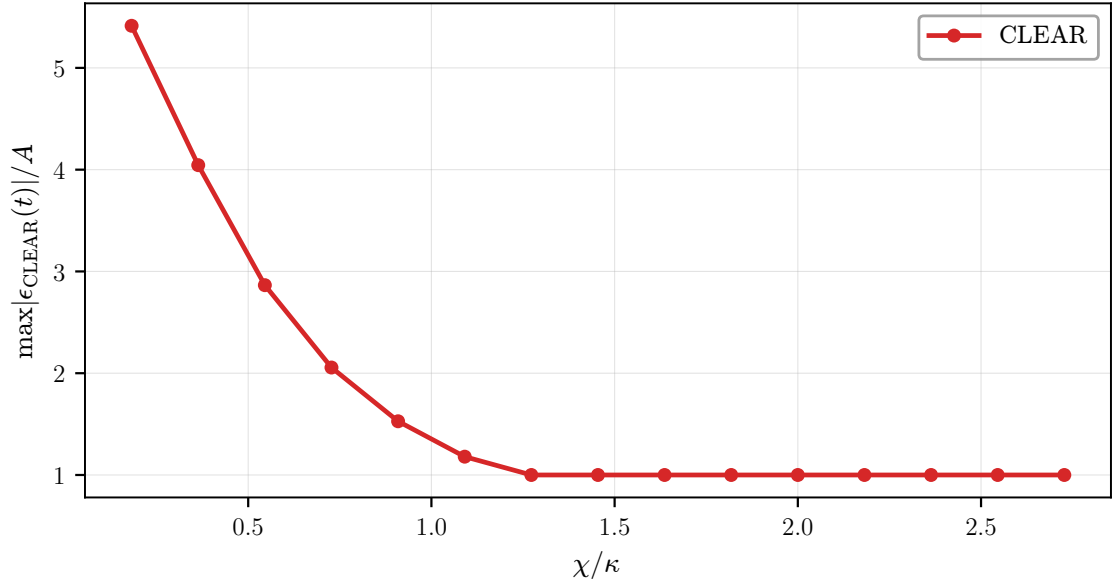


Figure 2.9: Maximum CLEAR amplitude, normalized to A , vs. χ/κ .

Taken together, these results show that χ/κ cannot be increased arbitrarily just because it improves state distinguishability and lowers the required drive amplitude. A larger χ brings the system closer to the limits of the dispersive approximation used to derive Eq. (2.28) in the first place; and if the ratio is instead increased by reducing κ , the resonator decay time $1/\kappa$ grows, which can slow down the reset even as it improves state distinguishability. As with the trade-off between kick duration and drive amplitude discussed above, this shows that the parameters of a CLEAR pulse must be chosen as a compromise between readout speed, state distinguishability, drive amplitude, and the validity of the underlying linear, dispersive description.

Chapter 3

Experimental implementation of CLEAR resonator reset

Chapters 1 and 2 introduced the theoretical basis of dispersive qubit readout and developed a semiclassical model for the resonator dynamics, which was then used to compare rectangular and CLEAR pulse shapes. The purpose of this chapter is to connect that model to the experimental workflow required to implement a CLEAR sequence on a superconducting qubit platform. The chapter is organized in three steps. Section 3.1 describes the experimental platform, from microwave signal propagation and cryogenic wiring to the RFSoc-based control electronics and the Qibo software stack. Section 3.2 presents the calibration of the qubit-resonator system, which fixes the operating frequencies and pulse parameters needed for readout. Section 3.3 then discusses the implementation of the CLEAR sequence on the calibrated setup and the measurements used to assess resonator depletion.

3.1 Experimental setup

The experimental setup must generate microwave pulses, deliver them to a millikelvin device with controlled attenuation and noise, and acquire the weak signals interacting with the resonator. This section first recalls the microwave concepts that determine how signals propagate through the setup, namely transmission lines, impedance matching and S-parameters. It then describes the cryogenic environment, the room-temperature RFSoc electronics used for pulse generation and acquisition, and finally the software layers that translate calibration routines into hardware instructions.

3.1.1 Microwave signal propagation and S-parameters

The hardware architecture of a superconducting quantum processor is built from components operating in the microwave regime, typically between 4 GHz and 8 GHz. At these frequencies the design of the experimental setup departs from low-frequency electronics, because the electrical wavelength becomes comparable to the physical size of the components.

While low-frequency circuits are well described by the lumped-element model, in which voltage and current are assumed uniform across each component, microwave engineering requires a distributed-parameter approach: the signal phase varies appreciably over the device's physical extent, so Kirchhoff's laws no longer apply. In this regime, circuits are instead analyzed through Maxwell's equations and transmission-line theory. Since in many cases, a description of microwave circuits in terms of voltages and current is cumbersome or even not possible, the natural observables become power and the scattering parameters (S parameters) [21].

A distributed-parameter description becomes possible by modeling each cable as a transmission line. A transmission line that supports a TEM (Transverse ElectroMagnetic) wave must consist of at least two conductors and is characterized by per-unit-length parameters: resistance (R), inductance (L), conductance (G), and capacitance (C). These parameters determine both the characteristic impedance and the propagation constant of the line.

Impedance matching and reflections

For a lossless transmission line the characteristic impedance is

$$Z_0 = \sqrt{\frac{L}{C}}, \quad (3.1)$$

where L and C are the inductance and capacitance per unit length, respectively. When the line is terminated on a load impedance Z_L , any mismatch between Z_0 and Z_L produces a partial reflection of the incident wave at the impedance discontinuity, quantified by the reflection coefficient (see Ref. [21, Eq. (2.35)])

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}. \quad (3.2)$$

If $\Gamma \neq 0$, part of the incident wave travels back toward the source. This reduces the efficiency of power transfer and can give rise to standing waves along the line; in addition, reflections may distort the transmitted signal (e.g. producing ripples), increase the risk of damaging circuit components, and degrade the signal-to-noise ratio. In quantum information experiments, this degradation can directly lower the readout fidelity and interfere with the coherent control of the qubit.

Impedance matching, implemented, for example, through single- or double-stub tuners and quarter-wave transformers, addresses this issue by inserting a lossless matching network between the load and the line, so as to realize an effective impedance transformation. Perfect matching ($\Gamma = 0$ over a finite bandwidth) is nonetheless fundamentally unattainable, as constrained by the Bode-Fano criterion (see Ref. [21, Ch. 5]).

Scattering parameters

At microwave frequencies, networks are conventionally characterized in terms of scattering parameters rather than voltages and currents directly. For a transmission line, the voltage at position z can be written as the superposition of a forward- and a backward-propagating wave (see Ref. [21, Eq. (2.14a)]),

$$V(z) = V_0^+ e^{-i\beta z} + V_0^- e^{i\beta z}, \quad (3.3)$$

where β is the propagation constant of the line.

For a general n -port microwave network, the scattering matrix S relates the amplitudes of the waves incident on each port, a_i , to the amplitudes of the waves reflected or transmitted from each port, b_i , through $b = S a$ (see Ref. [21, Eq. (4.40)]). For a two-port device, such as a single element of the readout chain, S_{11} quantifies the fraction of the incident wave reflected back at port 1 (and is directly related to the reflection coefficient Γ introduced above), while S_{21} quantifies the fraction of the signal transmitted from port 1 to port 2, i.e. the insertion gain or loss of the component. S-parameters are the natural experimental observables at microwave frequencies, since they can be measured directly with a vector network analyzer (VNA) without requiring access to the voltage or current at internal nodes of the circuit.

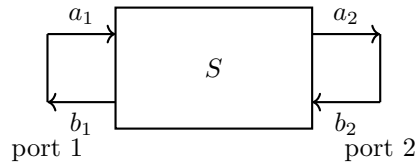


Figure 3.1: Schematic representation of a two-port microwave network (cf. the general n -port network of Ref. [21, Fig. 4.5]). The waves incident on port 1 and port 2 have amplitudes a_1 and a_2 , respectively, while the waves reflected or transmitted from each port have amplitudes b_1 and b_2 ; the scattering matrix relates them through $b = S a$ (Eq. (4.40) of Ref. [21]), i.e. $b_1 = S_{11}a_1 + S_{12}a_2$ and $b_2 = S_{21}a_1 + S_{22}a_2$. S_{11} and S_{22} are the reflection coefficients at port 1 and port 2, while S_{21} and S_{12} are the transmission coefficients from port 1 to port 2 and from port 2 to port 1, respectively.

3.1.2 Cryogenic environment and microwave lines

Superconducting qubits must be operated at temperatures low enough to suppress thermal excitations of the qubit transition and to keep the circuit in the superconducting regime. This is achieved with a dilution refrigerator, which cools the device from room temperature down to the millikelvin range. Fig. 3.2 shows the cryogenic wiring used to connect the room-temperature electronics to the quantum chip inside the refrigerator. The horizontal dashed lines represent the different temperature stages of the cryostat: room temperature (RT), the two pulse-tube stages, the still, the cold plate, and the mixing chamber (MC), where the sample is mounted.

The input microwave lines are used to send control and readout pulses towards the sample. As shown in Fig. 3.2, these lines include attenuators distributed across several temperature stages. The attenuators have two roles. First, they reduce the signal power to the level required at the device. Second, they thermalize the electromagnetic noise coming from warmer stages, preventing room-temperature thermal photons from reaching the qubit and degrading its coherence. Additional filters are placed close to the sample to suppress unwanted frequency components outside the relevant microwave band.

The output line has the opposite task: it must carry the weak signal emitted by the readout resonator back to room temperature while adding as little noise as possible. For this reason, the first amplification stages are placed at cryogenic temperatures. In Fig. 3.2 the readout signal passes through non-reciprocal components, such as isolators or circulators, before reaching the cold amplification chain¹. These components enforce a preferred direction of propagation and prevent noise generated by the amplifiers from being transmitted back towards the qubit. The signal is then amplified at low temperature and later further amplified at higher stages before being acquired by the room-temperature electronics.

The figure also shows auxiliary cryogenic lines and the shielding surrounding the chip. The sample is enclosed in nested shields, including a gold-plated copper shield, an aluminum shield and a μ -metal shield. These shields reduce electromagnetic and magnetic interference at the device position, while the copper enclosure also provides good thermalization of the sample holder. The overall purpose of the cryogenic wiring is therefore to deliver well-controlled microwave pulses to the qubit-resonator system and to extract the readout signal while preserving the low-noise environment required for coherent superconducting-qubit operation.

3.1.3 Room-temperature electronics

The cryogenic wiring connects the quantum chip to the outside of the dilution refrigerator, where the microwave pulses are generated and acquired. The role of the room-temperature electronics is therefore to synthesize the control and readout tones sent to the cryostat, and to digitize the amplified signal returning from the readout chain.

In this experiment, tasks are performed using an RFSoc board. RFSoc stands for Radio-Frequency System-on-Chip: it is an integrated platform that combines, on the same device, programmable logic, high-speed digital-to-analog converters, analog-to-digital converters, and processing units. A Field-Programmable Gate Array (FPGA) executes timing-critical instructions in real time, the Digital-to-Analog Converter (DACs) generate the analog microwave waveforms used for qubit control and resonator readout, and the Analog-to-Digital Converters (ADCs) digitize the signal returning from the cryostat. A processor handles configuration, data transfer, and communication with the control software.

¹The cold amplification chain typically starts with a Traveling Wave Parametric Amplifier (TWPA) as its first stage; however, in this experiment the TWPA was not employed, in order to simplify the operation of the setup.

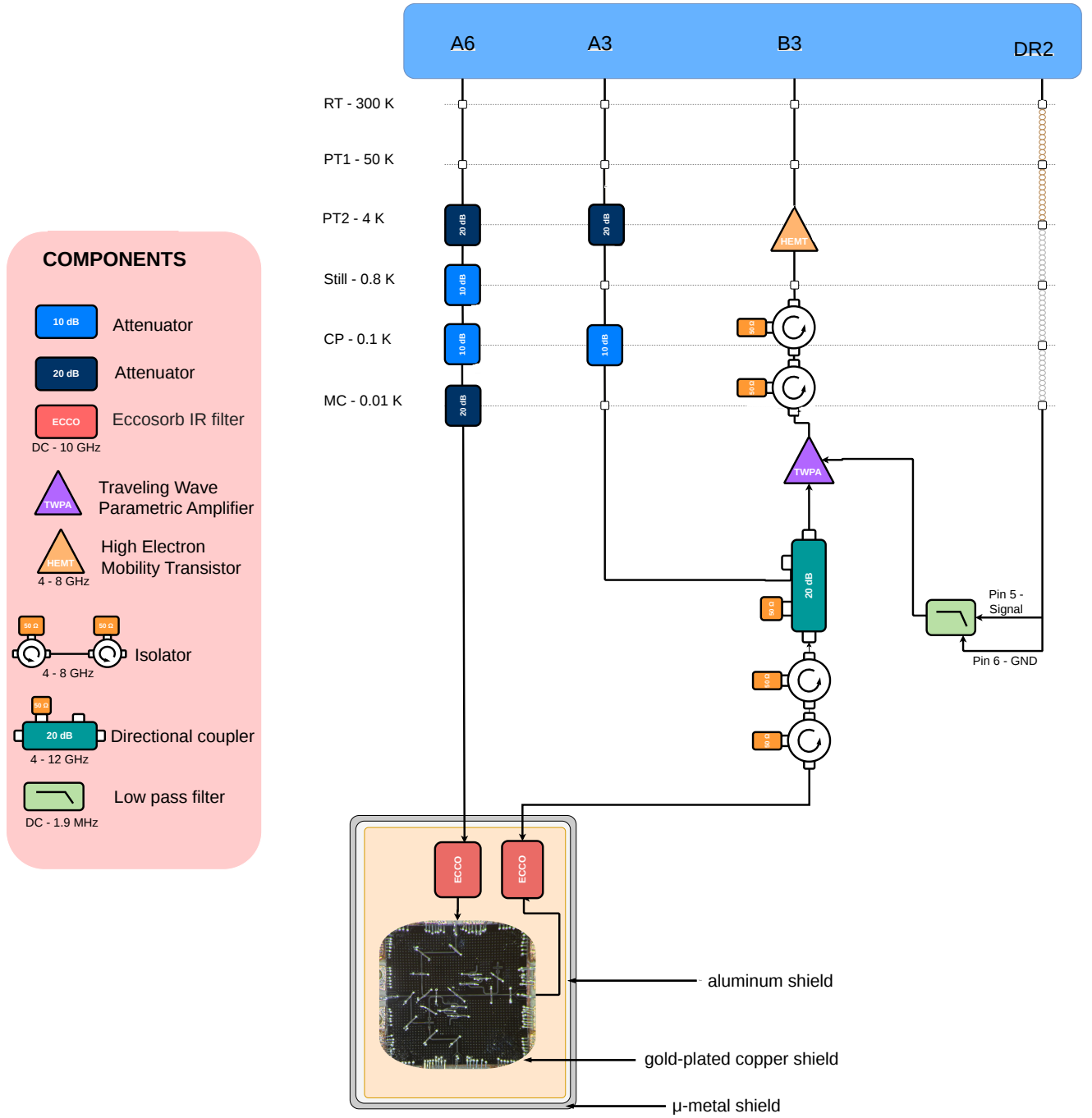


Figure 3.2: Cryogenic wiring of the experiment. The diagram shows the temperature stages of the dilution refrigerator, the microwave input lines with distributed attenuation and filtering, the output readout line with circulators and cryogenic amplification, and the shielded sample enclosure mounted at the mixing-chamber stage. The qubit used in the experiment is QB-1, shown on the chip in Fig. 1.2.

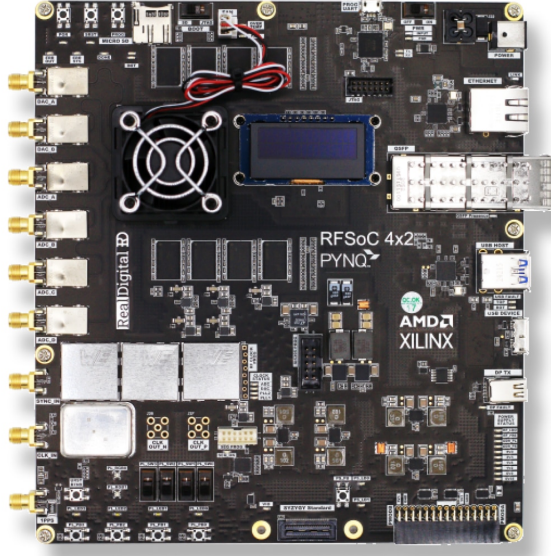


Figure 3.3: The RFSoc board

In our experimental platform, two DAC outputs are used to generate the microwave tones: DAC1 produces the readout probe tone, while DAC2 produces the qubit-drive tone. Each line passes through a variable attenuator, used to set the correct output power and to improve the impedance match of the DAC output, with an attenuation range of 7.5-7.9 dB for the DAC1 line and 5-5.9 dB for the DAC2 line. The two tones are then combined into a single coaxial line before entering the cryostat. Since the qubit drive and the readout probe are applied at different frequencies, they can be delivered through the same physical input line and later separated by their spectral content.

On the acquisition side, the signal returning from the cryostat has already been amplified by the cryogenic output chain. At room temperature, it is further amplified by a single stage with approximately +20 dB of gain. The conditioned signal is finally acquired by ADC1, and is later converted into digital data.

Outside the RFSoc rack, the same two lines are connected to the cryostat, sending the signals through A6 in Fig. 3.2 and reading out from B3.

3.1.4 The Qibo software stack

The experiments described in the following sections are implemented through the Qibo open-source software ecosystem [22]. Qibo provides a hardware-agnostic, high-level language through which quantum circuits, algorithms and calibration routines can be defined.

When the experiment is executed on a real, locally controlled quantum processor, the high-level instructions produced by Qibo are passed to a hardware backend. The connection between Qibo and the physical apparatus is provided by Qibolab [23], which acts as the hardware-control layer: it translates abstract quantum operations, such as quantum gates, into the corresponding control pulses. Each pulse is described by its carrier frequency, amplitude, phase, duration, and by which control or acquisition window it belongs to.

When the platform is based on an RFSoc board, Qibosoq [24] bridges the high-level pulse description produced by Qibolab to the low-level RFSoc layer. Calibration experiments are carried out using Qibocal [25]: experiments are specified through declarative YAML runcards, where a runcard defines the platform, the target qubits, and a list of actions, each corresponding to a calibration protocol together with its own parameters, without requiring the calibration code to be rewritten for each new measurement.

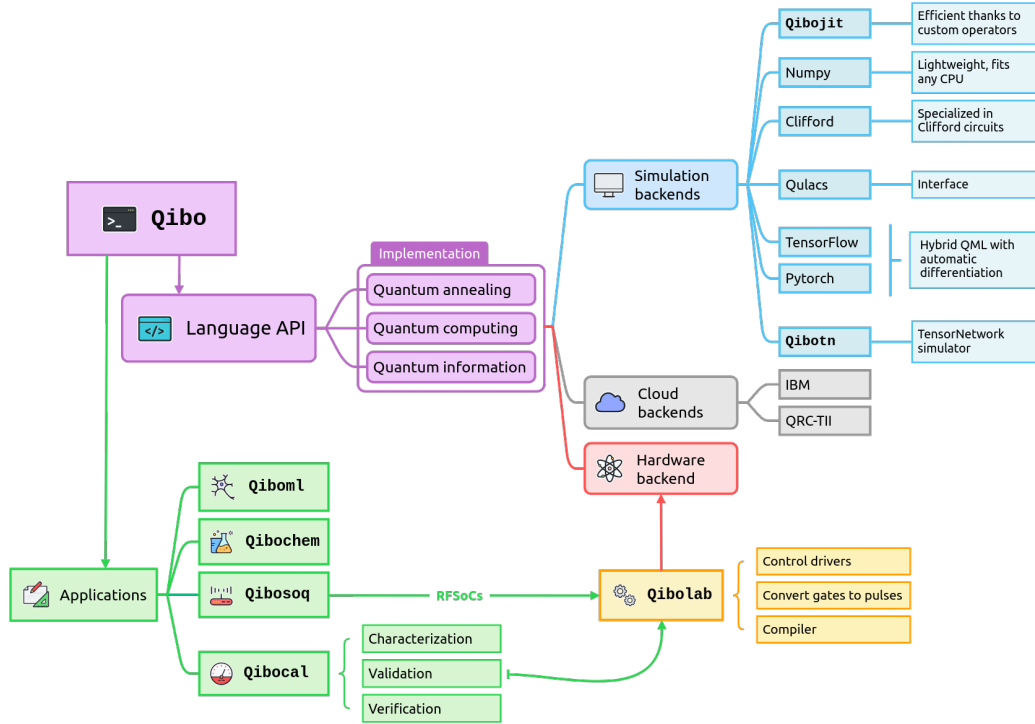


Figure 3.4: Overview of the Qibo software ecosystem. Qibo provides the language API; Qibolab translates quantum operations into control pulses and, through Qibosoq, into RFSoc-level instructions; Qibocal implements calibration and characterization routines. Reproduced from Ref. [22].

3.2 Calibration

Before the CLEAR pulse sequence can be implemented, the qubit-resonator system has to be characterized. This is not done in a single measurement: it is a chain of standard experiments, run in sequence, each one supplying a parameter that the next experiment needs as input. The main quantities determined along this chain are:

- the bare resonator frequency f_r , used as a first probe reference;
- the readout power operating point, identified through a resonator punchout;
- the qubit transition frequency f_{01} ;
- the drive amplitudes of the $\pi/2$ and π rotations used to control the qubit;
- the coherence times T_1 , T_2 and T_2^{echo} ;
- the dispersive shift χ between the state dependent resonator frequencies f_g and f_e .
- the mapping between the readout drive amplitude and the intracavity photon number n ;

All the experiments are run through Qibocal, as anticipated. For each experiments are both returned the raw acquired and fitted parameters in a file, and these are used to update the platform configuration before the next experiment is executed. The following section presents each experiment, together with the plot and the fitted value it produced.

3.2.1 Low-power resonator spectroscopy

The first protocol locates the bare readout resonance without any prior knowledge of the qubit-resonator system.

A single microwave tone is swept in frequency across the probe line, over a 4 MHz -wide range around the expected resonator frequency, while the resonator is driven at a low, fixed amplitude $\text{amp} = 0.06$, normalized to the full-scale DAC output (Table 3.1), and the transmitted amplitude is recorded after each shot. Each point of the trace is the average of a number of repeated single-shot acquisitions, which reduces the statistical noise on the estimated signal. Between repetitions, the sequence waits for a fixed relaxation time t_{relax} before probing again; here it is set to zero, since a low, continuous probe tone does not put the qubit in a well-defined excited state that needs to decay away. In the experiments that do excite the qubit, t_{relax} is instead set to several T_1 's, so that the qubit has returned to $|g\rangle$ before each repetition.

Working at low power is essential at this stage: for $n \ll n_{\text{crit}}$, the resonator response is described by the linear, semiclassical model of Chapter 2, and the transmitted signal traces out an approximately Lorentzian dip (or peak, depending on the type of resonator, which could be 2D or 3D),

$$S(f) = \frac{A}{1 + \left(\frac{f - f_r}{w}\right)^2} + c, \quad (3.4)$$

centered on the resonator frequency f_r , undistorted by the power-dependent frequency pulling that appears at higher drive amplitudes and that is exploited, instead, in the punchout measurement of Sec. 3.2.3.

Figure 3.5 shows the result of the scan, together with the center frequency returned by a fit of Eq. (3.4) to the trace. The fit locates the resonance at $f_r = 7583.54\text{ MHz}$ (Table 3.1). Since this measurement is performed below the punchout transition and at low readout power, with no drive preparing the qubit in $|e\rangle$, the qubit sits in its thermal ground state $|g\rangle$; the resonance therefore already corresponds to the state-dependent frequency f_g , which will be compared to f_e once the dispersive shift is measured directly (Sec. 3.2.8).

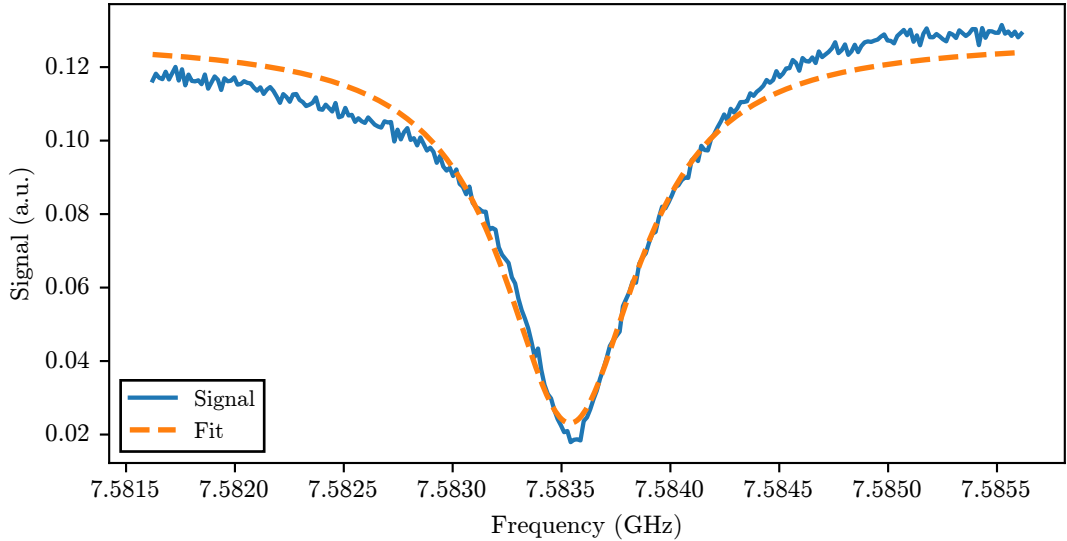


Figure 3.5: Low-power resonator spectroscopy.

3.2.2 Wide-range qubit power spectroscopy

Before the qubit transition frequency can be measured precisely, its approximate location has to be found without any prior knowledge. This is done with a two-dimensional scan of the

Parameters	Values
Readout Frequency [MHz]	7583.54
Amplitude	0.06
χ^2_{reduced}	2.57651×10^3

Table 3.1: Parameters extracted from low resonator spectroscopy.

drive frequency and drive amplitude: a short drive pulse ($1.0 \mu s$) is swept over a 400 MHz -wide frequency window while its amplitude is stepped from 0.001 to 0.046, and the readout signal at the resonator frequency of Sec. 3.2.1 is recorded for every (frequency, amplitude) pair.

Figure 3.6 shows the resulting map. Driving well below the qubit frequency has essentially no effect on the readout signal, while a faint, amplitude-dependent feature appears once the drive approaches the $0 \rightarrow 1$ transition. No automated fit is performed on this map, and it does not update the platform configuration: its only purpose is to provide a rough estimate of f_{01} , together with a drive amplitude at which the feature is visible, to be used as the starting point of the narrow, single-amplitude scan of Sec. 3.2.4.

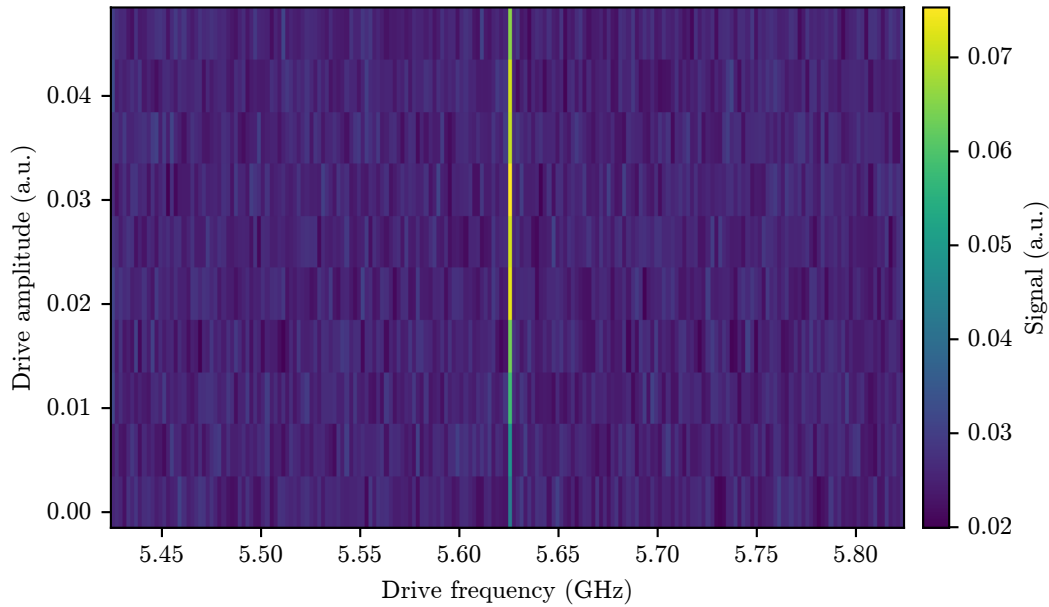


Figure 3.6: Wide-range qubit power spectroscopy: readout signal as a function of drive frequency and drive amplitude, used to roughly locate the qubit transition.

3.2.3 Resonator punchout

The resonator spectroscopy of Sec. 3.2.1 is performed at a fixed, low readout amplitude. In this regime the response is linear and the measured resonance corresponds to the dressed resonator frequency, namely the frequency of the resonator while it is dispersively coupled to the qubit. For readout, this is the relevant operating regime: the resonator frequency depends on the qubit state and can therefore be used to distinguish $|0\rangle$ from $|1\rangle$.

The resonator punchout experiment extends this measurement by repeating the resonator spectroscopy for different values of the readout-pulse amplitude. The result is a two-dimensional map as a function of probe frequency and amplitude. At low amplitude, the resonance remains at the dressed frequency. As the amplitude is increased, the resonator response enters a transition region: the qubit-resonator system departs from the simple dispersive picture; the underlying mechanism involves multiphoton resonances that can drive the qubit out of its computational subspace [26]. At high amplitude, the measured resonance approaches the bare cavity frequency, i.e. the frequency of the resonator when it is effectively decoupled from the qubit.

The purpose of the punchout measurement is to identify the range of readout amplitudes that can be safely used in the following calibrations: the amplitude should be large enough to provide a good signal-to-noise ratio, but still below the punchout transition, where the response becomes nonlinear and the readout is no longer performed in the dispersive linear-response regime.

Figure 3.7 shows the resonator punchout map obtained by repeating the resonator spectroscopy for different readout amplitudes. The horizontal axis is the probe frequency, while the vertical axis is the readout-pulse amplitude. The picture shows the normalized magnitude of the measured signal, and the dark feature marks the resonator response.

At low readout amplitude, the resonance occurs near f_{low} , which is reported in Table 3.2 as the low-power resonator frequency and corresponds to the dressed resonator frequency. At high amplitude, the resonance shifts toward f_{high} , also listed in Table 3.2 as the high-power resonator frequency, representing the bare resonator frequency.

The selected readout amplitude A_{RO} lies well below the transition region visible in Fig. 3.7. This confirms that the subsequent readout calibrations are performed in the low-power, linear dispersive regime assumed in the theoretical model.

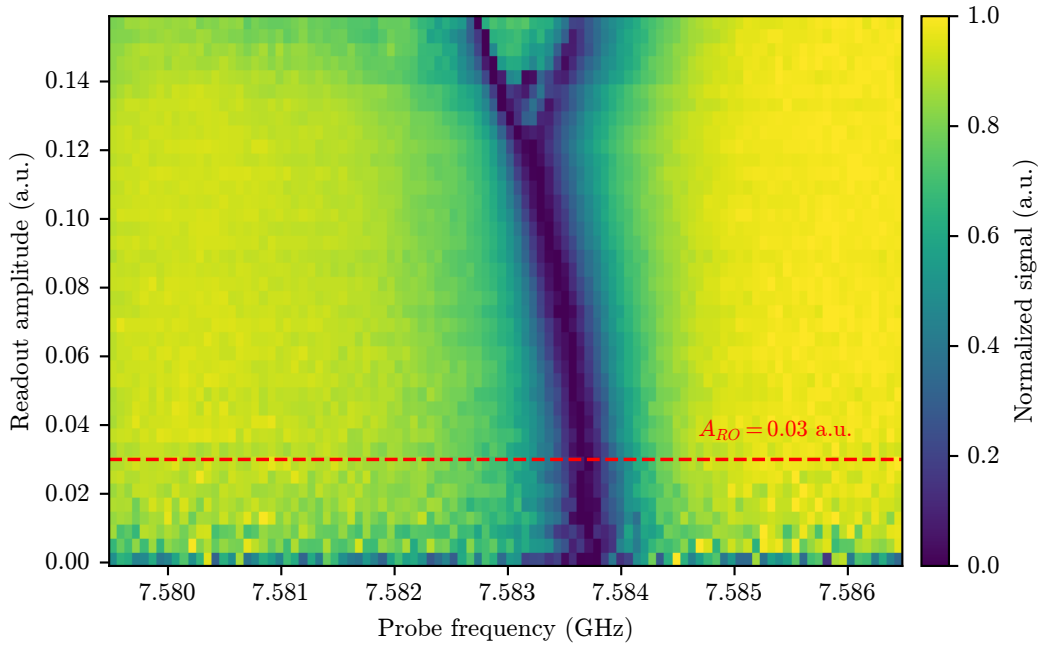


Figure 3.7: Plot for a resonator punchout experiment.

Parameters	Values
f_{low} [MHz]	7585.82
f_{high} [MHz]	7583.02
A_{RO} [a.u.]	0.03

Table 3.2: Parameters extracted from the resonator punchout experiment.

3.2.4 Qubit spectroscopy

With an approximate resonator frequency, readout amplitude, and an estimate of f_{01} from the wide-range scan of Sec. 3.2.2 available, the qubit transition can be located precisely. A low-power drive pulse is swept in frequency over a 10 MHz window centered on the expected qubit frequency, and is followed by a readout pulse at the resonator frequency of Sec. 3.2.1. When the drive is resonant with the $0 \rightarrow 1$ transition, the qubit is partially excited and the dispersive

shift of the resonator changes the measured signal accordingly, producing a Lorentzian feature analogous to the one used to locate the resonator frequency.

The readout amplitude used here is the same operating value A_{RO} adopted throughout the chapter (Sec. 3.2.3), low enough that it does not itself perturb the qubit frequency appreciably; the extracted center frequency is therefore a clean estimate of the bare f_{01} . Figure 3.8 shows the resulting trace: the fit locates the transition at $f_{01} = 5625.28 \text{ MHz}$, which replaces the generic default value used to configure the platform before this measurement.

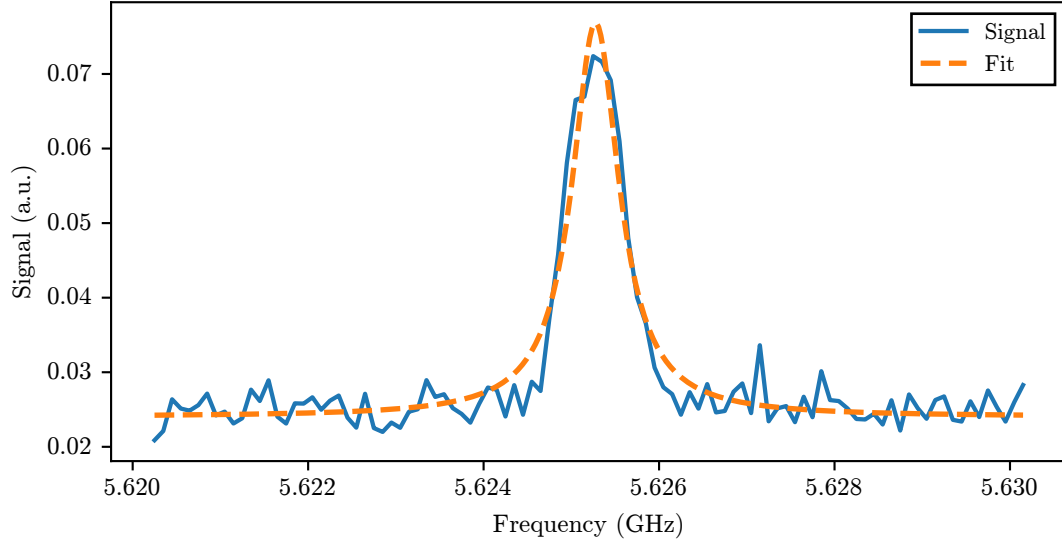


Figure 3.8: Qubit spectroscopy: readout signal as a function of the drive frequency, with a Lorentzian fit locating the $0 \rightarrow 1$ transition.

Parameters	Values
$f_{01} [\text{MHz}]$	5625.28

Table 3.3: Parameters extracted from qubit spectroscopy.

3.2.5 Rabi calibration of the π and $\pi/2$ pulse

With the qubit frequency fixed, the next step is to calibrate the drive pulse itself, so that it performs a definite rotation of the qubit state on the Bloch sphere. In particular the X gate, corresponding to the Pauli matrix

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad |0\rangle \rightarrow |1\rangle \quad \text{and} \quad |1\rangle \rightarrow |0\rangle,$$

and the $\pi/2$ pulse (RX90), corresponding to a rotation of angle $\pi/2$ about the x axis of the Bloch sphere,

$$R_X(\pi/2) = e^{-i\frac{\pi}{4}X} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}, \quad |0\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle).$$

A drive pulse resonant with f_{01} is determined, at fixed shape, by its duration and its amplitude, and the rotation angle it implements grows with their product (the pulse area); either one can, in principle, be swept at fixed value of the other to trace out a full oscillation.

Three variants of this measurement are therefore possible: sweeping the duration at fixed amplitude (Rabi length), sweeping the amplitude at fixed duration (Rabi amplitude), or sweeping both together on a two-dimensional grid. All three repeat the same sequence: prepare the qubit in $|0\rangle$, apply a drive pulse of given amplitude and/or length, measure, iterate. All three are expected to show the excited-state population P_e oscillating as

$$P_e \propto 1 - \cos(2\pi f_{\text{Rabi}} t), \quad (3.5)$$

either in t (Rabi length) or, since f_{Rabi} is itself approximately linear in the drive amplitude in the regime explored here, in the amplitude (Rabi amplitude). The two-dimensional version is the most complete but also the slowest, and is normally needed only when neither a reasonable pulse duration nor a reasonable amplitude range is known in advance; here a short duration of 30 ns was fixed a priori and only the drive amplitude was scanned.

In a Rabi experiment the physical quantity of interest is the excited-state population, which is encoded in the position of the measured IQ point along the readout axis connecting the responses associated with $|0\rangle$ and $|1\rangle$.

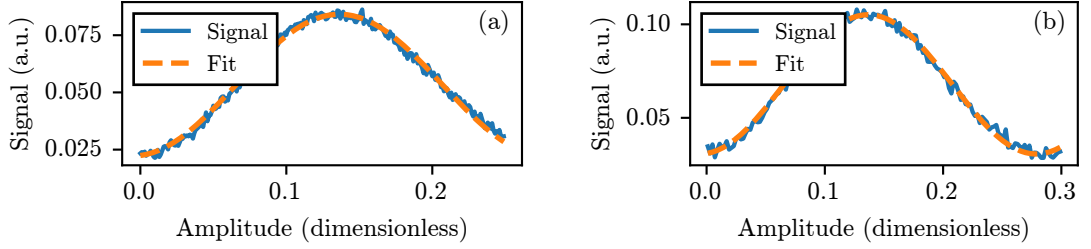


Figure 3.9: Rabi oscillation of the readout signal as a function of drive amplitude, at fixed pulse duration, used to calibrate (a) the π -pulse and (b) the RX90 ($\pi/2$) pulse. In panel (a), the amplitude corresponding to the first maximum identifies the amplitude required for a π rotation. In panel (b), two consecutive RX90 pulses with equal amplitude are applied. Their combined rotation is therefore π when each pulse implements a $\pi/2$ rotation, so that the amplitude corresponding to the maximum is taken as the RX90 amplitude.

Parameters	Values
π pulse amplitude [a.u.]	0.16
π pulse length [ns]	50.000
$\pi/2$ pulse amplitude [a.u.]	0.14
$\pi/2$ pulse length [ns]	30.000

Table 3.4: Parameters extracted from the Rabi experiment.

3.2.6 Ramsey calibration of the qubit transition frequency

The qubit transition frequency $f_{01} = 5625.28$ MHz, obtained from the qubit spectroscopy of Sec. 3.2.4, is limited by the frequency resolution of that scan. A more precise calibration can be obtained with a Ramsey experiment. The sequence consists of two $\pi/2$ pulses separated by a variable delay τ . The first pulse brings the qubit state onto the equatorial plane of the Bloch sphere, where it evolves freely during the waiting time. The second $\pi/2$ pulse converts the accumulated phase into a measurable population difference.

If the drive frequency is exactly resonant with the true qubit transition, the phase accumulated during the free evolution is stationary in the rotating frame. If instead there is a detuning between the drive and the qubit, the phase evolves during the delay τ , producing Ramsey fringes. The measured signal can be described by an exponentially decaying oscillation,

$$S(\tau) = S_0 + Ae^{-\tau/T_2} \cos(2\pi f_R \tau + \phi),$$

where f_R is the observed Ramsey oscillation frequency, ϕ is an additional phase offset, and T_2 is the unrefocused coherence time measured in a standard Ramsey experiment. It includes both irreversible decoherence and slow fluctuations of the qubit frequency that are not removed by an echo pulse.

Following the Qibocal Ramsey protocol, an artificial detuning

$$\Delta = 0.40 \text{ MHz}$$

is intentionally introduced between the drive and the approximate qubit frequency. This detuning is implemented as a phase ramp applied to the second $\pi/2$ pulse. If the frequency used by the control software were already equal to the true qubit frequency, the Ramsey fringe would oscillate at the applied detuning Δ . Any difference between the fitted oscillation frequency f_R and the applied detuning therefore reveals a residual error in the qubit frequency. This error is then used to update the value of f_{01} used in the following pulse sequences.

Figure 3.10 shows the measured Ramsey fringe together with the fitted decay. The fit gives the value of T_2 reported in Table 3.5. The same fit also corrects the qubit transition frequency, yielding f_{01} in Table 3.5. Although the correction is small, it is propagated to all subsequent pulse sequences. In particular, it fixes the qubit-drive frequency used in the final Rabi calibration of Sec. 3.2.10 and in the CLEAR characterization of Sec. 3.3.1.

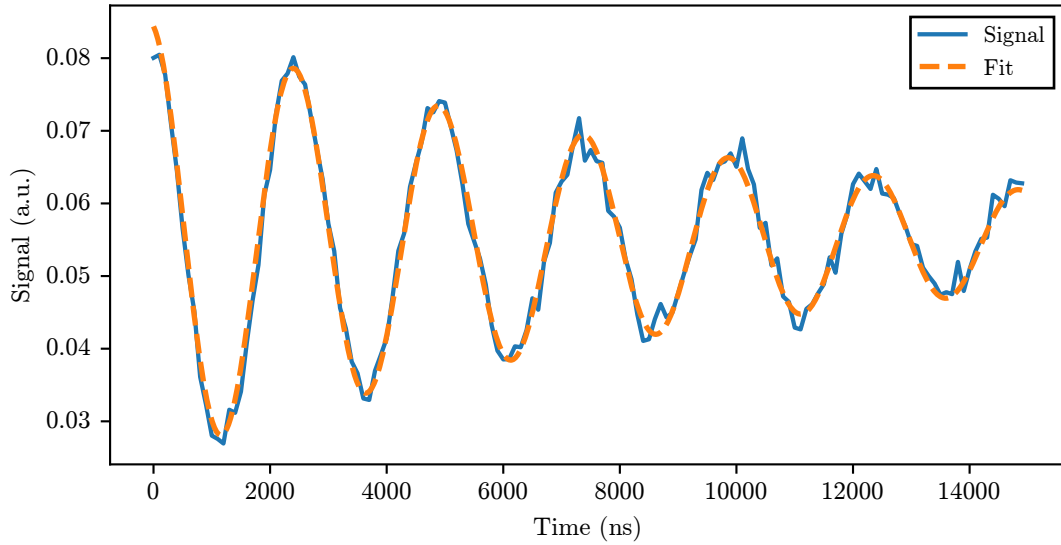


Figure 3.10: Ramsey fringe at a fixed applied detuning of 0.40 MHz , used to refine the qubit transition frequency.

Parameters	Values
$f_{01} [\text{MHz}]$	5625.27
$T_2 [\mu\text{s}]$	10.2

Table 3.5: Parameters extracted from the Ramsey experiment.

3.2.7 Coherence times: T_1 , T_2 , and Hahn-echo T_2

The next three protocols characterize the intrinsic coherence properties of the qubit introduced in Sec. 1.3.2. The corresponding decay rates are

$$\Gamma_1 = \frac{1}{T_1}, \quad \Gamma_2 = \frac{1}{T_2}.$$

These quantities are not control parameters to be optimized during the CLEAR calibration, but properties of the device itself. They are nevertheless relevant for the present work.

The relaxation time T_1 is measured by preparing the qubit in $|e\rangle$ with a π pulse, waiting for a variable delay t , and then measuring the remaining excited-state population. The measured signal is fitted to an exponential decay,

$$S(t) = S_0 e^{-t/T_1} + c, \quad (3.6)$$

where c accounts for a residual offset at long delay.

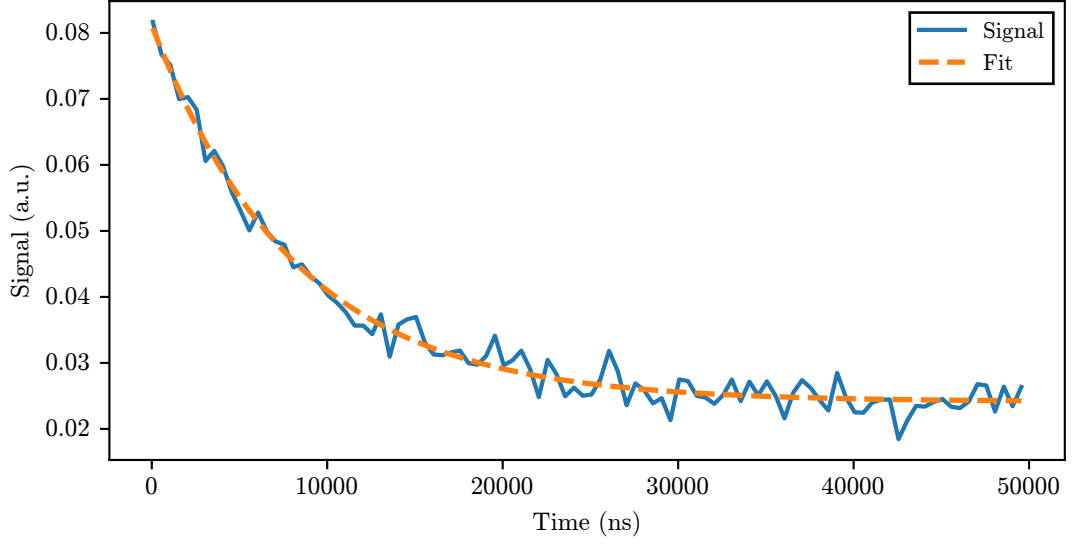


Figure 3.11: Coherence measurements: relaxation T_1 with fitted exponential.

The dephasing time T_2 is extracted from a Ramsey experiment. In this sequence, two $\pi/2$ pulses are separated by a variable delay. During this free-evolution time, the qubit phase evolves relative to the drive, producing Ramsey fringes whose contrast decays with a characteristic time T_2 . This decay contains two contributions: energy relaxation, already quantified by T_1 , and pure dephasing, described by T_ϕ . These contributions combine as

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}.$$

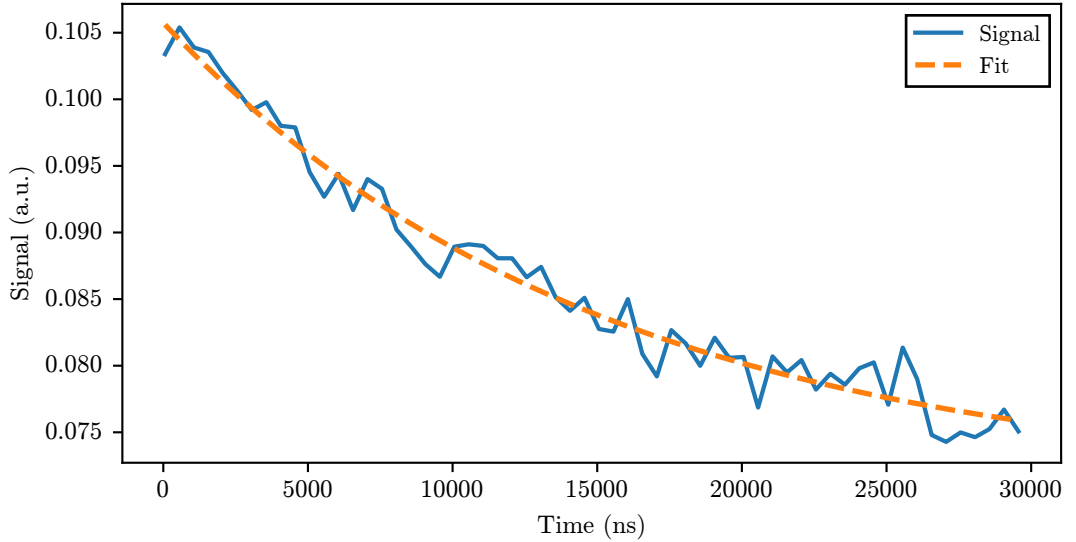


Figure 3.12: Coherence measurements: dephasing T_2 with fitted exponential.

A Hahn-echo sequence is used to distinguish slow dephasing from irreversible relaxation. It adds a π pulse halfway through the free-evolution interval, which reverses the phase accumulated

from slowly varying frequency shifts. The resulting echo decay time, denoted T_2^{echo} , is therefore typically longer than the unfocused Ramsey value T_2 .

Figs. 3.11, 3.12, 3.13 show the three coherence measurements performed consecutively on the same qubit. Each curve is fitted to an exponential decay of the form of Eq. (3.6), giving the values in Table 3.6.

The Ramsey value T_2 extracted from this first scan is slightly larger than the Hahn-echo value T_2^{echo} . This ordering is not the one usually expected: the echo sequence is designed to refocus slow frequency fluctuations and often gives a coherence time longer than, or at least comparable to, the Ramsey value. The inversion observed here should therefore not be interpreted as a physical effect. It is more likely a consequence of the short delay range swept compared to the coherence times themselves, which tends to bias an exponential fit toward overestimating the decay constant; an imperfectly calibrated π pulse in the echo sequence could also contribute to the inversion observed here.

The value of T_1 measured here is adopted in the rest of the chapter, and for the dephasing rate we will use the coherence T_2 , and define the corresponding effective rate as

$$\Gamma_2 = \frac{1}{T_2}.$$

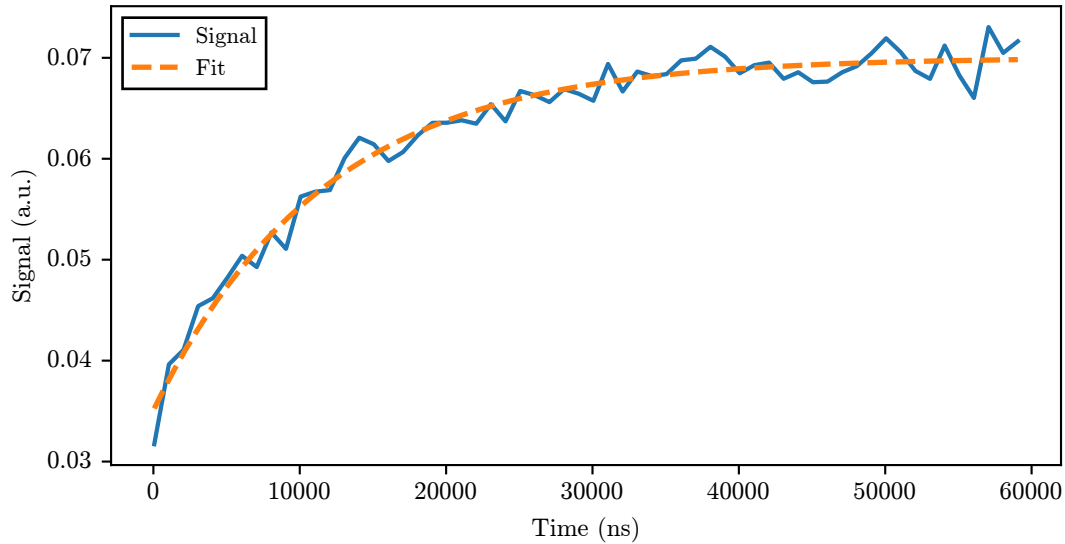


Figure 3.13: Coherence measurements: dephasing T_2^{echo} with fitted exponential.

Parameters	Values
T_1 [μs]	8.2 ± 0.3
T_2 [μs]	15.0 ± 1.5
T_2^{echo} [μs]	11.7 ± 0.6

Table 3.6: Parameters extracted from the coherence experiments.

3.2.8 Dispersive shift

The dispersive shift χ is the single most important parameter for the CLEAR construction of Sec. 3.3.1: it fixes both member of the effective-detuning pair $\Delta_g = \delta_r - \chi$, $\Delta_e = \delta_r + \chi$ that enters Eq. (2.35), and, through Sec. 3.3.4, the probe frequency at which the CLEAR pulse is actually driven. It is measured directly, rather than inferred from f_r and f_{01} , by repeating the resonator spectroscopy of Sec. 3.2.1 twice in the same run, once with the qubit left in $|g\rangle$ and

once after a π pulse prepares it in $|e\rangle$, and locating the two resulting resonance frequencies f_g and f_e independently; χ then follows from

$$\chi/2\pi = \frac{f_g - f_e}{2}. \quad (3.7)$$

Because the readout tone itself populates the resonator and can Stark-shift the very lineshapes being fitted, the measurement is performed at low readout amplitude, with a fine frequency step and a large number of shots to compensate for the reduced signal-to-noise ratio (Fig. 3.14). This gives $f_g = 7583.63 \text{ MHz}$ and $f_e = 7582.51 \text{ MHz}$, corresponding to

$$\chi/2\pi = \frac{7583.63 - 7582.51}{2} \text{ MHz} \simeq 0.56 \text{ MHz}. \quad (3.8)$$

The two state-dependent frequencies measured at low power also fix a first estimate of the probe midpoint frequency, $(f_g + f_e)/2 \simeq 7583.07 \text{ MHz}$; as discussed in Sec. 3.3.4, however, this naive average is not accurate enough to be used directly.

The resonator linewidth κ adopted in the rest of the chapter is also refined here, using the half-width of the $|g\rangle$ -state Lorentzian dip of Fig. 3.14: unlike the $|e\rangle$ -state trace, which is broadened by the mixture of the two qubit-state responses caused by T_1 decay during the measurement, the $|g\rangle$ -state trace gives a clean, single-Lorentzian estimate of κ .

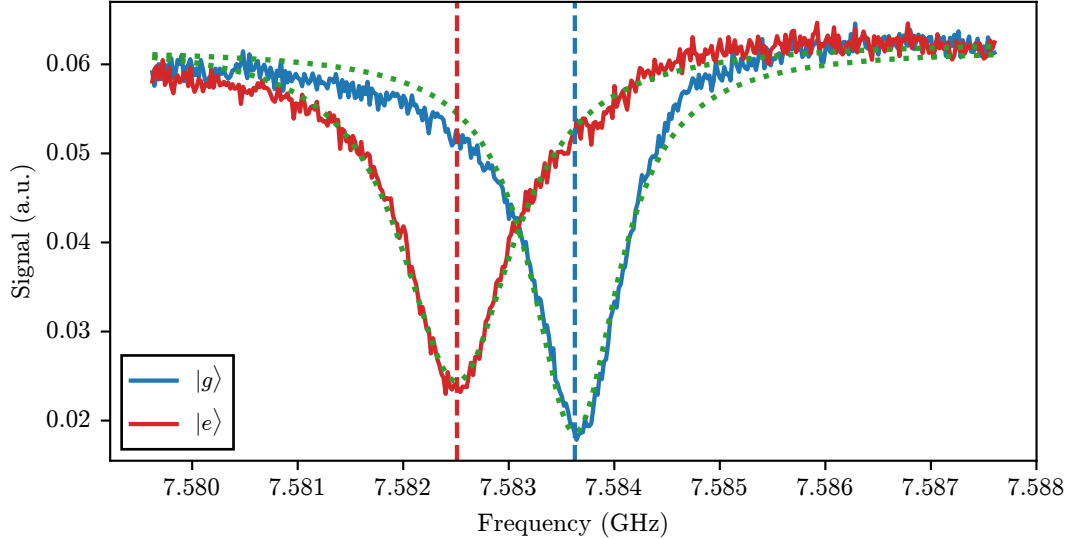


Figure 3.14: Dispersive-shift measurement: resonator spectroscopy with the qubit prepared in $|g\rangle$ (blue) and $|e\rangle$ (red), at low readout power. The dashed vertical lines mark the two fitted resonance frequencies f_g and f_e .

Parameters	Values
f_g [MHz]	7583.63
f_e [MHz]	7582.51
$\chi/2\pi$ [MHz]	0.56
$\kappa/2\pi$ [MHz]	1.00

Table 3.7: Parameters extracted from the dispersive shift experiment at low power.

3.2.9 AC-Stark-shift calibration of the intracavity photon number

Before implementing the CLEAR pulse, several system parameters must be calibrated on the physical device. This section describes the AC Stark shift measurement used to translate the

readout pulse amplitude, which is only defined in the arbitrary DAC units of the RFSoc, into the corresponding mean photon number $\bar{n}$ in the resonator. This is achieved by the fact that the same dispersive qubit-resonator coupling that, in the regime where the qubit-resonator detuning greatly exceeds the coupling strength, produces the qubit-state-dependent pull of the resonator frequency used for dispersive readout (Chapter 2), also produces, symmetrically, a photon-number-dependent shift of the qubit transition frequency,

$$f_q(n) = f_q(0) - 2(\chi/2\pi)n, \quad (3.9)$$

where $\chi/2\pi$ is the same dispersive shift that sets the resonator pull. This effect, derived from the dispersive Jaynes-Cummings Hamiltonian, can be found in, e.g., in Ref. [6]. Measuring the qubit frequency as a function of readout amplitude, and hence of the number of photons it injects into the resonator, therefore provides the desired calibration.

The measurement uses a two-tone pulse sequence. A first tone, with variable amplitude, populates the resonator with photons, playing the same role as the readout pulse itself. Its frequency is fixed at their midpoint,

$$f_{\text{probe}} = \frac{f_{r,g} + f_{r,e}}{2}, \quad (3.10)$$

where $f_{r,g}$ and $f_{r,e}$ are the resonator frequencies measured with the qubit in $|g\rangle$ and $|e\rangle$, respectively. Only at this midpoint is the steady-state intracavity photon number, to leading order, independent of the qubit state, so that the single calibration obtained here remains valid later, when the CLEAR pulse is applied irrespective of which state the qubit happens to be in. While this tone is on, a second, weak tone sweeps its frequency across the expected qubit transition, probing the qubit spectroscopically under the AC Stark shift induced by the photons in the resonator. The qubit state is then read out with a standard dispersive readout pulse. Repeating this sequence for a grid of probe amplitudes and drive frequencies yields the map shown in Fig. 3.15, where the diagonal feature traces the Stark-shifted qubit resonance: as the probe amplitude increases, and with it the number of photons in the resonator, the qubit resonance moves to lower frequency.

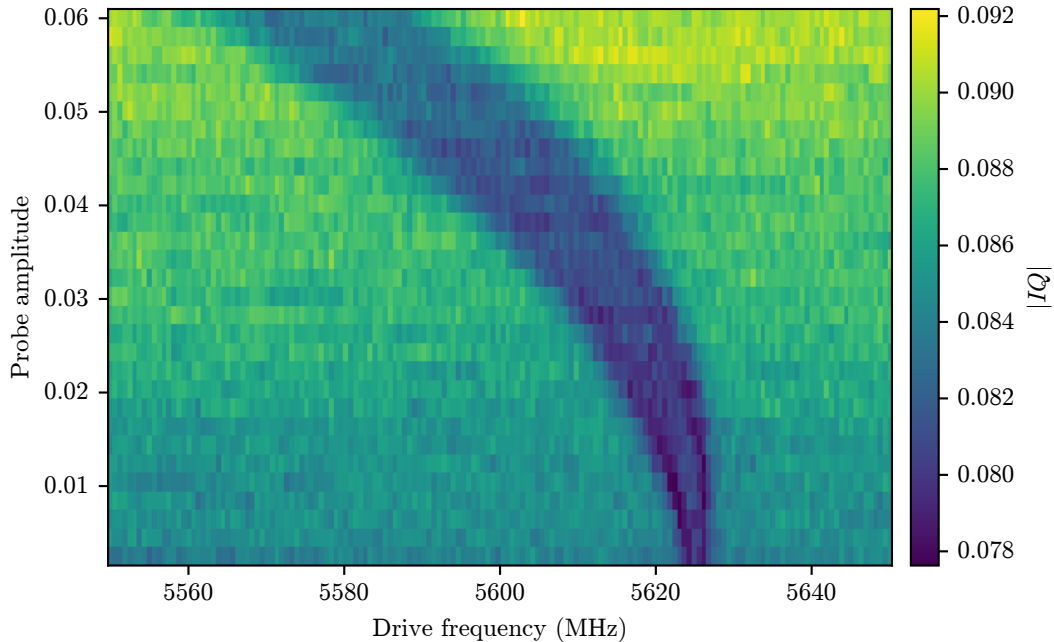


Figure 3.15: Two-tone spectroscopy map used to extract the AC Stark shift. For each probe (readout) amplitude, the drive frequency is swept across the qubit transition and the readout signal $|IQ|$ is recorded.

For each probe amplitude, the qubit resonance frequency is extracted by fitting the corresponding row of the map with a Lorentzian line shape. The resulting resonance frequency as a function of probe amplitude is shown in Fig. 3.16: it decreases monotonically at low amplitude, while at higher amplitude the trend becomes noisier, as the system departs from the dispersive regime and the Lorentzian fit becomes less reliable.

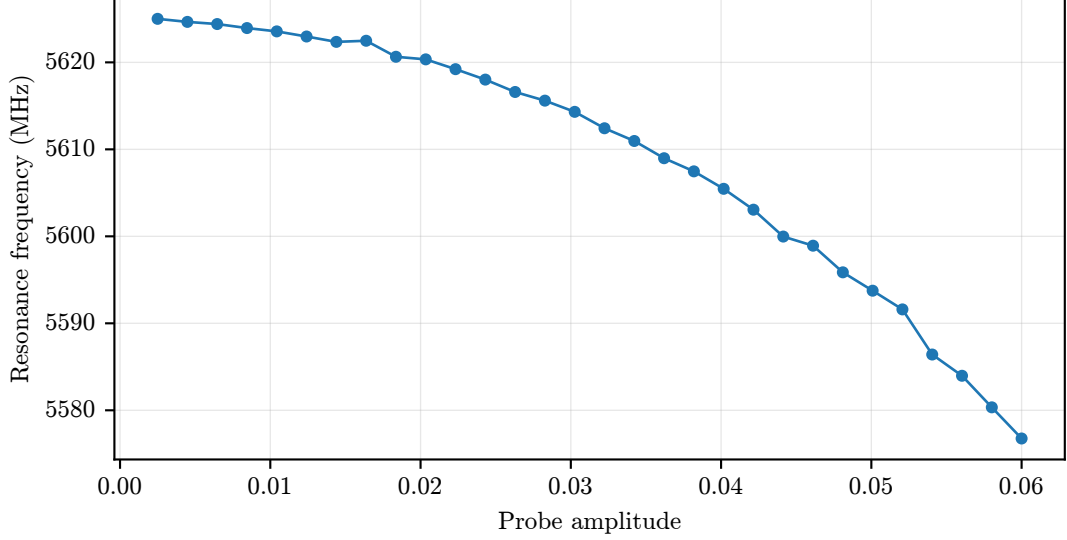


Figure 3.16: Qubit resonance frequency, extracted from a Lorentzian fit of each row of Fig. 3.15, as a function of the probe amplitude.

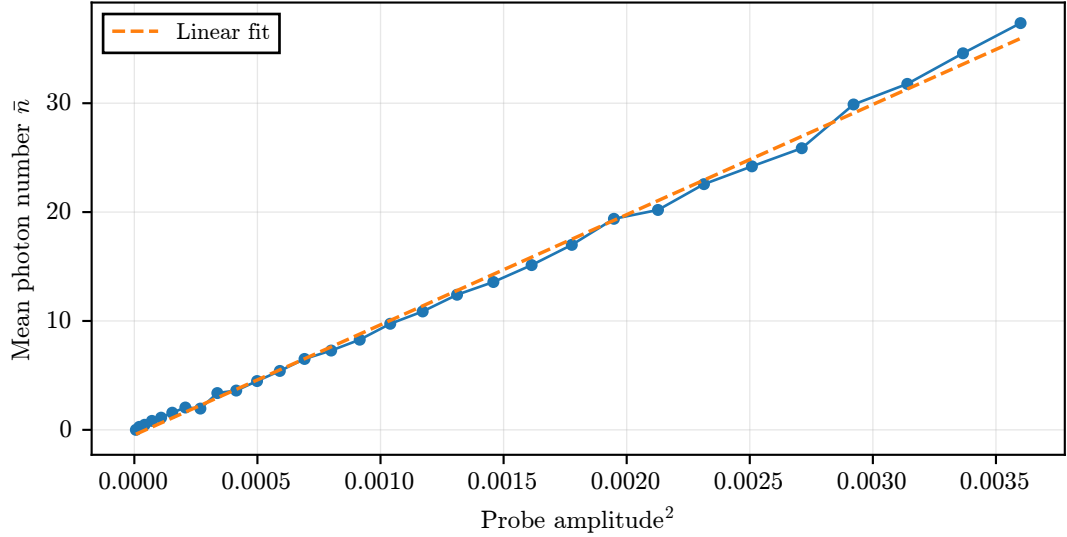


Figure 3.17: Mean intracavity photon number $\bar{n}$, obtained from the Stark shift of Fig. 3.16 via $\bar{n} = -\Delta f / (2\chi/2\pi)$, as a function of the squared probe amplitude A^2 . The dashed line is a linear fit $\bar{n} = kA^2$, with $k \approx 10117$ used to convert readout amplitude into photon number.

The frequency shift Δf measured with respect to the unshifted qubit frequency (probe amplitude close to zero) is converted into a mean photon number through $\bar{n} = -\Delta f / (2\chi/2\pi)$, where $\chi/2\pi = (f_{r,g} - f_{r,e})/2$ is the dispersive shift, with $f_{r,g}$ and $f_{r,e}$ the same state-dependent resonator frequencies used above to fix the probe tone, measured independently with a dispersive-shift experiment. Plotting $\bar{n}$ against the square of the probe amplitude, as shown in Fig. 3.17, confirms

the expected linear scaling of the intracavity photon number with the readout power. A linear fit to the 30 measured points gives

$$\bar{n} = kA^2 \quad (3.11)$$

with $k \approx 10117$.

3.2.10 Final calibrated parameters

The previous experiments listed can be repeated, as the parameters may change over time. Here we present the recalibrated parameters and report all the final results that we will use in the next section.

With the qubit frequency corrected in Sec. 3.2.6, the first experiment that is repeated is the Rabi calibration of Sec. 3.2.5 for both the π (RX) and $\pi/2$ (RX90) pulses, using the corrected frequency so that the drive is no longer slightly off-resonant. This second pass also uses a finer amplitude step and more shots than the exploratory calibration of Sec. 3.2.5, since accurate π and $\pi/2$ amplitudes are needed both for the state-preparation pulse of the CLEAR-characterization sequence (Sec. 3.3.2) and for every Ramsey-type measurement performed afterwards. The fits give a π pulse amplitude of 0.1627, for a pulse of length 50 ns, and a $\pi/2$ pulse amplitude of 0.1363, for a pulse of length 30 ns; the latter differs by less than one percent from the value of Sec. 3.2.5.

Table 3.8 collects the qubit and resonator parameters extracted in this section. Together with the resonator linewidth κ discussed in Sec. 3.2.8, they form the operating point used to configure the CLEAR pulse in Sec. 3.3.1.

It should be noted that the value of $\chi/2\pi$ reported in Table 3.8 differs from the one directly measured in the dispersive-shift experiment of Sec. 3.2.8. The discrepancy arises because $f_{r,g}$ was adjusted, rather than re-measured, to make the state-dependent resonator frequencies consistent with the probe midpoint determined in Sec. 3.3.4, while $f_{r,e}$ was left at its measured value. This value of χ is used for the AC Stark Shift in Sec. 3.2.9, which already reports the final calibrated parameter of k .

Quantity	Symbol	Value
Resonator linewidth	$\kappa/2\pi$	1.00 MHz
Dispersive shift	$\chi/2\pi$	0.65 MHz
Qubit transition frequency	f_{01}	5625.27 MHz
Relaxation time	T_1	8.2 μ s
Ramsey dephasing time	T_2	15.0 μ s
Hahn-echo dephasing time	T_2^{echo}	11.7 μ s
π -pulse amplitude	-	0.16 (0.05 μ s)
$\pi/2$ -pulse amplitude	-	0.14 (0.03 μ s)
Stark-shift coefficient	k	10117

Table 3.8: Summary of the device parameters obtained from the calibration campaign of Sec. 3.2.1-3.2.10. These values configure the CLEAR pulse implementation of Sec. 3.3.1.

3.3 CLEAR pulse implementation

The previous sections fixed, one at a time, every parameter that the CLEAR pulse construction of Chapter 2 requires as input: the resonator linewidth, the dispersive shift, and the relation between drive amplitude and intracavity photon number. This section starts from those numbers and addresses two distinct but intertwined problems. The first is purely practical: translating the analytical shape of the pulse, so far defined only in terms of ratios between segments, into a sequence of amplitudes that can actually be loaded onto the RFSoc and compared, under matched conditions, with a conventional readout pulse. The second is more subtle, and is the real experimental crux of the chapter: unlike the qubit state, which the platform returns as a direct signal, the number of photons left in the resonator at the end of a readout pulse is not something that can be measured directly, and has to be reconstructed from how those leftover photons perturb something else. Establishing a reliable estimator for this quantity, before it can

even be used, is therefore just as necessary a step as building the pulse itself. The rest of the section follows exactly this logic, from the hardware sequence to its validation, before finally using the resulting estimator to put the central claim of this thesis to the test on real hardware, comparing the CLEAR pulse against a rectangular one under conditions that, until now, had only been explored theoretically.

3.3.1 From calibrated parameters to hardware pulse amplitudes

In Sec. 2.3.3 we obtain the shape of the CLEAR pulse, but this does not by itself say what voltage the RFSoc should actually output. In order to achieve that we need four more pieces of information: how quickly the resonator relaxes (κ), how far apart the two qubit-state-dependent resonator frequencies are (χ), how long a ring-up or ring-down segment should last (t_{kick}), and how strongly the resonator is driven during the readout window itself (A). The first three are properties of the device and of a chosen pulse timing; the fourth is set by how many photons we want in the resonator during readout, which is the quantity calibrated in Sec. 3.2.9.

Using the resonator linewidth of Table 3.8, a kick duration $t_{\text{kick}} = 0.16 \mu\text{s}$, comparable to the cavity time constant $1/\kappa$, and a steady-state readout duration $t_{\text{readout}} = 1.0 \mu\text{s}$, the linear system of Eq. (2.37) is solved twice, once for the ring-up pair and once for the ring-down pair, as described in Sec. 2.3.3, but now for the measured device parameters. The resulting ratios are listed in Table 3.9.

Segment	Ratio to A
Ring-up 1	2.4866
Ring-up 2	0.0901
Steady (readout)	1.0000
Ring-down 1	-1.4866
Ring-down 2	0.9099

Table 3.9: CLEAR segment amplitudes, relative to the steady-state readout drive A , obtained by solving Eq.(2.37) for $\kappa/2\pi = 1.00 \text{ MHz}$, $\chi/2\pi = 0.65 \text{ MHz}$ and $t_{\text{kick}} = 0.16 \mu\text{s}$.

As anticipated in Sec. 2.4.2, driving exactly at the midpoint between the two dispersively shifted resonator frequencies, $\delta_r = 0$, guarantees that all five ratios of Table 3.9 are purely real. This has a direct, practical consequence for the hardware implementation: a negative ratio such as that of the first ring-down segment does not need to be synthesized as an arbitrary complex (in-phase/quadrature) waveform, only as a pulse of amplitude, and a relative phase of 180° with respect to the readout segment.

The overall scale A is finally set by the target photon number, through Eq. (3.11), where $n_{\text{target}} = 2$, for which the steady-state amplitude obtained is $A \simeq 0.0141$; multiplying it by the ratios of Table 3.9 gives the five physical amplitudes actually loaded onto the RFSoc.

The rectangular reference pulse used for comparison throughout this section is defined, as in Sec. 2.4.2, to have the same steady-state amplitude A and to last $T_{\text{rect}} = 2t_{\text{kick}} + t_{\text{readout}}$, i.e. the same duration as the ring-up and readout part of the CLEAR pulse. The two pulse shapes therefore only differ in how the resonator is actively driven back to vacuum once the readout window has ended, which is precisely the effect this chapter sets out to isolate and measure.

3.3.2 Pulse-sequence implementation in Qibolab

In order to compare the rectangular and the CLEAR pulse, we need to measure the photon population left in the resonator once the readout pulse has ended. The strategy adopted here, following Ref. [1], is to use the qubit itself as the detector, through a Ramsey experiment performed immediately after the pulse under test. Therefore, the pulse sequence has to prepare a known qubit state, apply the pulse being characterized, and then run a Ramsey sequence that is sensitive to whatever is left over in the cavity, before finally reading out the qubit in the usual way.

The five segments of Table 3.9 (or, for the rectangular reference, the single steady segment of duration T_{rect}) are compiled into a hardware pulse sequence through Qibolab [23] and, on the

RFSoc platform used here, through Qibosoq [24], as introduced in Sec. 3.1.4. The sequence used to characterize both pulse shapes reproduces Fig. 1(b) of Ref. [1]:

$$(RX) \rightarrow M_1 \rightarrow t_{\text{relax}} \rightarrow RX90 \rightarrow t_R \rightarrow RX90 \rightarrow t_{\text{buffer}} \rightarrow M_2, \quad (3.12)$$

where M_1 is the measurement pulse under test (rectangular or CLEAR), t_{relax} is an optional wait time inserted after M_1 , $RX90 \rightarrow t_R \rightarrow RX90$ is the Ramsey pair used to probe the residual cavity population (Sec. 3.3.3), t_{buffer} is a short, fixed delay, and M_2 is the platform's calibrated readout pulse used to measure the qubit state at the end of the sequence. The optional initial RX pulse prepares the qubit in $|e\rangle$ before M_1 ; omitting it leaves the qubit in $|g\rangle$, so that both preparations can be probed with the same sequence. The Ramsey detuning is implemented as a phase ramp applied to the second RX90 pulse, exactly as in the calibration of Sec. 3.2.6, rather than as a shift of the drive frequency.

For the rectangular pulse, t_{relax} is set equal to the duration of the two CLEAR ring-down segments, $2t_{\text{kick}} = 0.32 \mu\text{s}$, so that the two pulse shapes expose the qubit to M_1 for the same total time and therefore lose the same fraction of an excited preparation to T_1 decay by passive means; for the CLEAR pulse, $t_{\text{relax}} = 0$, since the active ring-down segments already play this role. This choice is what makes the comparison in Sec. 3.3.5 a fair one: any difference in the residual photon population reflects how the two pulses handle the cavity, not how long the qubit was left sitting in M_1 .

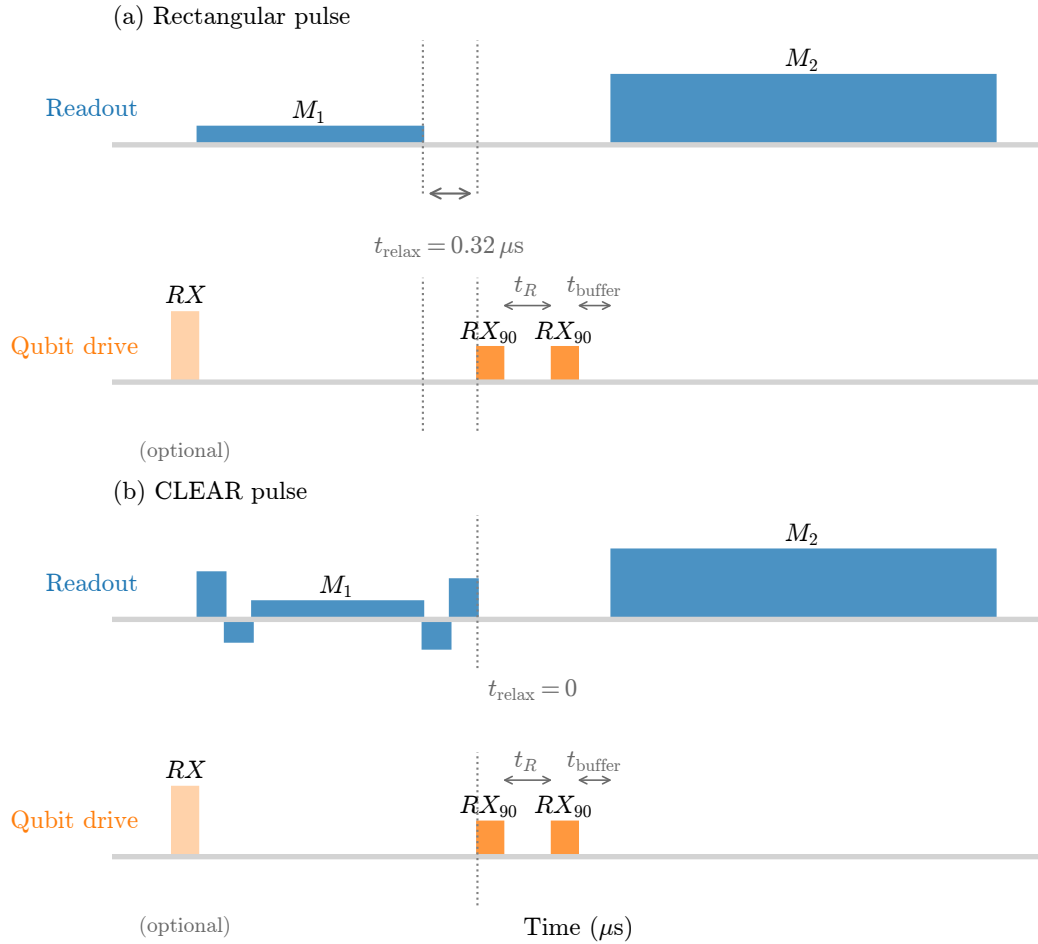


Figure 3.18: Pulse sequence (3.12), shown for (a) the rectangular M_1 and (b) the CLEAR M_1 . The optional initial RX pulse prepares the qubit in $|e\rangle$; omitting it leaves it in $|g\rangle$. The Ramsey pair and M_2 start at the same instant in both panels, since t_{relax} is chosen to equalize the total time the qubit is exposed to M_1 .

A second, shorter version of the pulse sequence (3.12), with the Ramsey pair and t_R removed,

$$(RX) \rightarrow M_1 \rightarrow t_{\text{relax}} \rightarrow M_2,$$

is used to measure directly the fraction p of the ensemble still in $|e\rangle$ at the moment the Ramsey pair would have started. This is needed because M_1 itself takes an appreciable amount of time: with M_1 lasting $4 t_{\text{kick}} + t_{\text{readout}} = 1.64 \mu\text{s}$ for the CLEAR pulse, i.e. about 20% of the relaxation time $T_1 = 8.2 \mu\text{s}$, a bare estimate

$$p \simeq e^{-1.64/8.2} \simeq 0.82$$

already predicts an appreciable loss of the excited preparation before the Ramsey even begins. This bare exponential is only a lower bound on the real effect, however, since the readout drive itself, especially at the higher powers explored in Sec. 3.3.5, can induce additional transitions out of $|e\rangle$ that a simple T_1 decay does not capture; the measured p is therefore used wherever an excited-state correction is needed (Sec. 3.3.3).

3.3.3 Extracting the residual photon population from a Ramsey trace

The quantity of interest in this section is the residual intracavity photon population n_0 left after M_1 , for a qubit prepared either in $|g\rangle$ or in $|e\rangle$. As mentioned above, this will be achieved by following the pulse sequence described in Ref. [1].

The principle is simple. Photons left in the cavity after M_1 shift the qubit frequency through the AC Stark effect, the same effect used for the calibration in Sec. 3.2.9. During the Ramsey delay t_R , the cavity field decays, and the corresponding Stark shift decays with it. This time-dependent frequency shift distorts what would otherwise be an ordinary Ramsey fringe. Extracting n_0 is therefore equivalent to fitting this transient distortion.

Explicitly, the excited-state signal as a function of the Ramsey delay is modeled as

$$S(t_R) = \frac{1}{2} [1 - \text{Im} \{ \exp [-(\Gamma_2 + i\Delta)t_R + i(\varphi_0 - 2n_0\chi\tau(t_R))] \}], \quad (3.13)$$

with

$$\tau(t_R) = \frac{1 - e^{-(\kappa + 2i\chi)t_R}}{\kappa + 2i\chi}. \quad (3.14)$$

This is Eq. (1) of Ref. [1]. Here Δ is the Ramsey detuning introduced in Sec. 3.3.2, $\Gamma_2 = 1/T_2$ is the decoherence rate, φ_0 is the initial Ramsey phase, and n_0 is the residual photon population at the beginning of the Ramsey pair.

The factor $\tau(t_R)$ turns a fixed initial number of leftover photons into a time-dependent phase shift. It accounts both for the decay of the cavity field at rate κ and for the fact that, in the two Ramsey branches, the residual cavity field evolves at two qubit-state-dependent frequencies separated by 2χ . For long delays, $t_R \rightarrow \infty$, we have

$$\tau(t_R) \rightarrow 1/(\kappa + 2i\chi).$$

At this point the cavity has emptied, the additional phase has saturated to a constant, and $S(t_R)$ reduces to an ordinary exponentially damped Ramsey fringe oscillating at the applied detuning Δ .

At short delays, by contrast, the cavity is still populated. The term $2n_0\chi\tau(t_R)$ then actively pulls the fringe frequency away from Δ . This early-time, non-sinusoidal distortion is the feature that makes n_0 recoverable from a fit to the Ramsey trace.

Fitting Eq. (3.14) to a measured trace requires two practical modifications with respect to a literal reading of Ref. [1].

First, $S(t_R) \in [0, 1]$ is a probability, whereas the digitized quadrature recorded by the acquisition chain has an arbitrary scale and offset. These are set by the readout fidelity and by the projection axis used to convert raw IQ points into an excited-state probability. A free multiplicative scale and a free additive offset are therefore applied before fitting.

Second, for a qubit prepared in $|e\rangle$, only a fraction p of the ensemble is still excited when the first $RX90$ pulse is applied. The remaining fraction $1 - p$ has already decayed to $|g\rangle$, as described

in Sec. 3.3.2. What matters for the sign of the Ramsey fringe is the qubit state at the first $RX90$, not the state at preparation. Moreover, because the cavity re-equilibrates on the timescale $1/\kappa$, much shorter than M_1 , a trajectory that decays during M_1 carries the ground-state residual photon population $n_{0,g}$, not the excited-state value $n_{0,e}$. Its fringe is also shifted by π .

The excited-state trace is therefore fitted to the two-component mixture

$$S_e(t_R) = p S(t_R; n_{0,e}, \varphi_0) + (1 - p) S(t_R; n_{0,g}, \varphi_0 + \pi). \quad (3.15)$$

Here $n_{0,g}$ is fixed to the value obtained independently from the corresponding ground-state trace, which is fitted first using the plain form of Eq. (3.14).

3.3.4 Probe-frequency midpoint calibration

The construction of Sec. 3.3.1 assumes that the readout tone is applied exactly at the midpoint between the two dispersively shifted resonator frequencies. This corresponds to $\delta_r = 0$.

A first estimate of the midpoint is obtained from the low-power dispersive-shift measurement of Sec. 3.2.8:

$$\frac{f_g + f_e}{2} \simeq 7583.07 \text{ MHz.}$$

This value, however, is not accurate enough to be used directly. For this reason, the midpoint is calibrated with a more direct observable:

$$\Delta n_0(f) = n_{0,e}(f) - n_{0,g}(f),$$

where $n_{0,e}$ and $n_{0,g}$ are the residual photon populations extracted with the Ramsey estimator of Sec. 3.3.3. The measurement is performed using a rectangular M_1 pulse, not a CLEAR pulse, with $t_{\text{relax}} = 0$. Physically, at the true midpoint, the rectangular pulse drives the two qubit-state-dependent resonator responses symmetrically. It should therefore leave the same residual photon population for a qubit prepared in $|g\rangle$ and in $|e\rangle$, so that

$$\Delta n_0(f_{\text{probe}}) = 0.$$

The measurement is performed with $t_{\text{relax}} = 0$ and repeated for a grid of probe frequencies around the naive midpoint.

The choice of $\Delta n_0(f)$ is also useful because it is a difference measured at fixed probe frequency. Therefore, it does not depend on the Stark-shift coefficient k of Sec. 3.2.9 or on the readout contrast: both quantities cancel between the two qubit-state branches. This makes the midpoint calibration reliable even before the photon-number axis has been fully trusted. Moreover, $\Delta n_0(f)$ is expected to change sign close to the true midpoint, since the two branches correspond to Lorentzian responses centered on opposite sides of that frequency. The zero crossing can therefore be located precisely.

In practice, the crossing is found by fitting $n_{0,g}(f)$ and $n_{0,e}(f)$ independently with straight lines in a local window around the sign change of $\Delta n_0(f)$. The midpoint is then taken as the intersection of the two fitted lines.

The result is shown in Fig. 3.19. The two fitted segments cross at

$$f_{\text{probe}} = 7583.16 \pm 0.01 \text{ MHz.}$$

The measured crossing frequency is adopted as the operating probe frequency for the rest of this section.

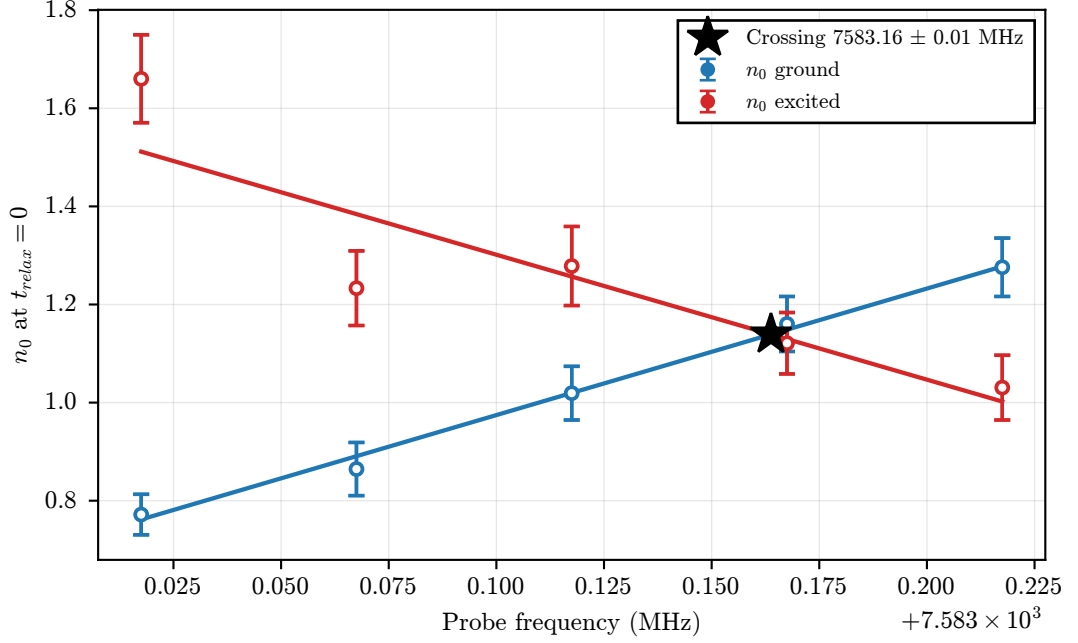


Figure 3.19: Residual photon population n_0 at $t_{\text{relax}} = 0$ for the qubit prepared in $|g\rangle$ (blue) and $|e\rangle$ (red), as a function of the probe frequency, used to calibrate the probe-frequency midpoint. The star marks the crossing of the local linear fits to the two branches, and the dashed line marks the frequency previously stored in the configuration.

3.3.5 Residual population versus drive power

The central comparison of this chapter, reproducing Fig. 3(c) of Ref. [1], is the residual photon population n_0 as a function of the readout drive power, for the rectangular and the CLEAR pulse, with the qubit prepared in either $|g\rangle$ or $|e\rangle$. This measurement converts the qualitative claim of Sec. 2.3.3, that the CLEAR pulse should deplete the resonator more rapidly than passive decay into a quantitative characterization across the full range of drive powers relevant to practical readout conditions. The steady-state segment amplitude is swept over a normalized drive power $P_{\text{norm}} = P/P_{1\text{ photon}}$, where $P_{1\text{ photon}}$ is the power corresponding to $n_{\text{target}} = 1$ through Eq. (3.11); in the linear-response regime P_{norm} coincides with the steady-state photon number itself, so the horizontal axis of the comparison below can be read directly as an intended photon number, independently of the absolute calibration of the readout chain. For every value of P_{norm} and for both pulse shapes, the CLEAR ratios of Table 3.9 are rescaled by the corresponding amplitude $A(P_{\text{norm}})$ following Sec. 3.3.1, and n_0 is extracted with the estimator of Sec. 3.3.3 from the sequence of Eq. (3.12), using, for each pulse shape, the excited fraction p measured once at the largest power of the scan, where the drive-induced contribution to qubit decay is largest and therefore easiest to resolve above the statistical noise.

Figure 3.20 shows the resulting comparison, for P_{norm} ranging from 0.5 to 50. For the rectangular pulse, n_0 grows approximately linearly with P_{norm} , reaching about 5 residual photons at $P_{\text{norm}} = 50$, essentially indistinguishable between the two qubit preparations: with no active mechanism to remove them, leftover photons simply accumulate in proportion to how many were driven in. For the CLEAR pulse, n_0 instead remains below about 0.5 photons over the entire range explored, more than an order of magnitude smaller than the rectangular case at the largest powers, confirming that the active ring-down mechanism of Sec. 2.3.3, rather than only the passive decay available to the rectangular pulse in the matched window T_{rect} of Sec. 3.3.1, is responsible for the improvement. This qualitative behavior matches the idealized simulation of Sec. 2.4.3 and the original measurement of Ref. [1]; quantitatively, however, the residual population measured here for the CLEAR pulse does not reach the numerical noise floor found in the noiseless linear simulation of Table 3.7, remaining instead at the level of a few tenths of a photon and growing slowly with power. This is consistent with the limitations already

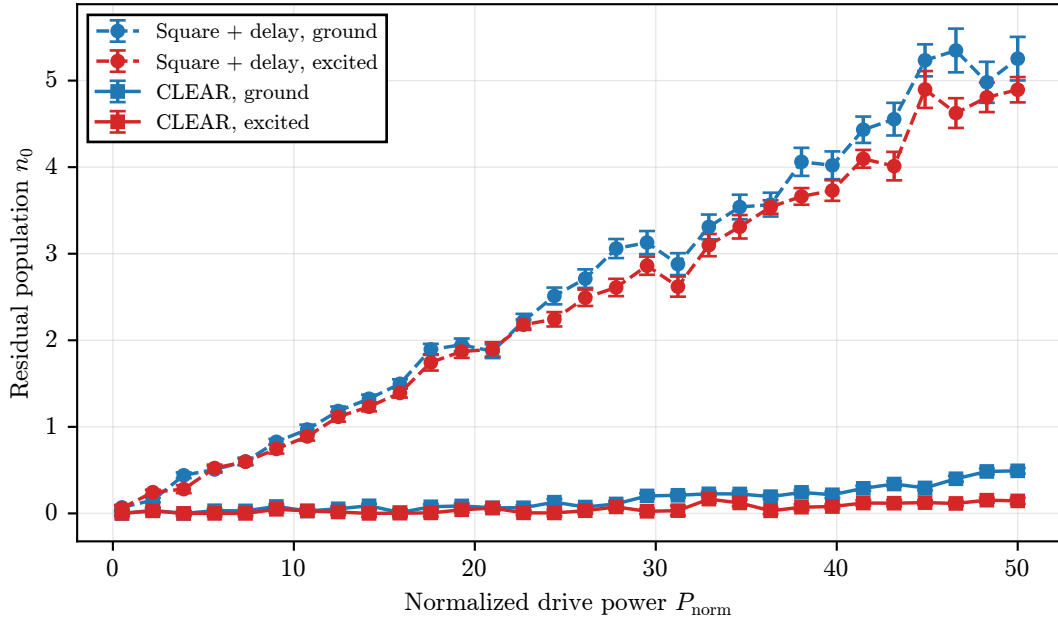


Figure 3.20: Residual photon population n_0 as a function of the normalized drive power P_{norm} , for the rectangular (“square + delay”) and CLEAR pulse, with the qubit prepared in $|g\rangle$ or $|e\rangle$.

anticipated in Sec. 2.4.1.

3.3.6 Cross-check: residual population versus relaxation delay

The power scan of Sec. 3.3.5 demonstrates that CLEAR leaves fewer photons behind than the rectangular pulse once both have finished, but it does not by itself show how that difference comes about. As an independent cross-check, reproducing the style of Fig. 2(b) of Ref. [1], the residual population is measured instead as a function of the wait time t_{relax} inserted between the end of M_1 and the start of the Ramsey pair, at fixed drive power ($n_{\text{target}} = 2$). This changes the observable from power to time, and directly tests the mechanism rather than only its end result. For the rectangular pulse, in the absence of an active ring-down segment, $n_0(t_{\text{relax}})$ is expected to decay passively as $e^{-\kappa t_{\text{relax}}}$. For the CLEAR pulse, by contrast, the ring-down segments already return the cavity close to vacuum by the end of M_1 itself, so n_0 is expected to be small already at $t_{\text{relax}} = 0$ and to stay approximately constant as t_{relax} is increased further - there is little left for passive decay to remove.

Figure 3.21 confirms this picture. For the rectangular pulse (Fig. 3.21a), n_0 decays from about 1.1 photons at $t_{\text{relax}} = 0$ to a value compatible with zero within a few hundred nanoseconds, broadly consistent with the passive time constant $1/\kappa \simeq 0.2 \mu\text{s}$ expected from the linewidth $\kappa/2\pi = 1.00 \text{ MHz}$ of Table 3.8, to within the precision of this exploratory scan. For the CLEAR pulse (Fig. 3.21b), n_0 is already compatible with zero, within its statistical uncertainty, at $t_{\text{relax}} = 0$, and remains so throughout the scan: the active ring-down has already completed, within M_1 , the same task that the rectangular pulse only accomplishes gradually and passively over the following few hundred nanoseconds of t_{relax} .

Taken together with the drive-power comparison of Sec. 3.3.5, this closes the experimental demonstration set out at the beginning of this chapter: the CLEAR pulse, constructed analytically in Sec. 2.3.3, simulated under idealized conditions in Sec. 2.4, and calibrated on real hardware in Sec. 3.2 and in this section, empties the readout resonator on a timescale set by the chosen kick duration t_{kick} rather than by the passive cavity lifetime $1/\kappa$, over the whole range of drive powers accessible in this measurement campaign.

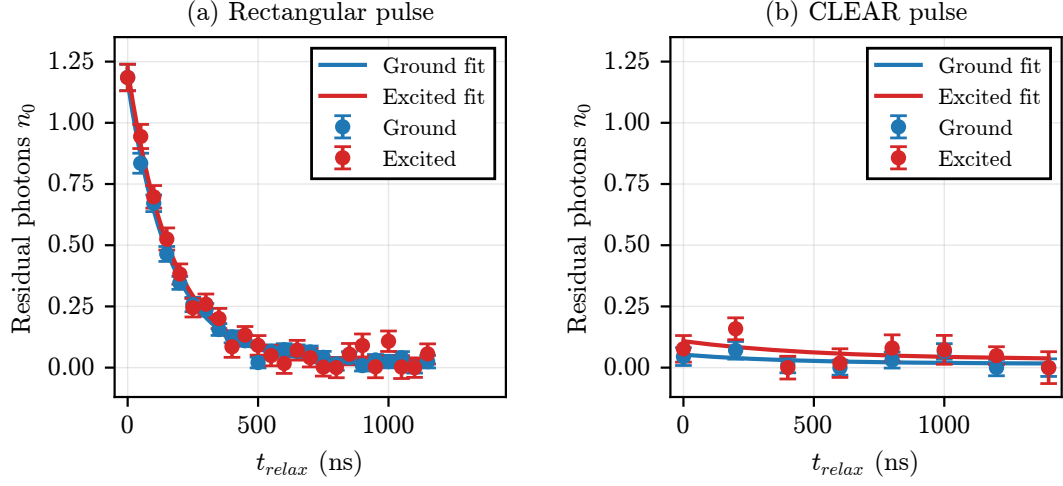


Figure 3.21: Residual photon population n_0 as a function of the relaxation delay t_{relax} after M_1 , for the rectangular (a) and CLEAR (b) pulse, with the qubit prepared in $|g\rangle$ or $|e\rangle$. The dashed curve is the passive decay $e^{-\kappa t_{\text{relax}}}$ expected for the rectangular pulse.

Conclusions

Fast, high-fidelity readout is not a marginal detail of a superconducting-qubit experiment: it is one of its most frequently repeated operations, and whenever it leaves the resonator populated with residual photons, it returns part of that cost as measurement-induced dephasing on every operation that follows. In this thesis I studied the dispersive readout of superconducting qubits, focusing on the optimization of the readout pulse shape used to suppress the intracavity photon population left behind after each measurement. The work developed on three levels, moving from the theoretical framework describing the qubit-resonator system, i.e. the transmon qubit, the Jaynes-Cummings and dispersive Hamiltonians, and the density-matrix description of an open quantum system, to a semiclassical model of the readout resonator dynamics derived from input-output theory, and finally to a full experimental implementation on a real transmon device. Within this semiclassical model I compared, both analytically and through numerical simulation, a conventional rectangular readout pulse against the CLEAR pulse, obtaining the amplitudes of its four kick segments by solving the linear system that forces the qubit-state-dependent field trajectories to reach the desired steady-state amplitude at the start of the readout and to vanish at its end; the simulations confirmed that, while the rectangular pulse leaves about 0.43 residual photons in the resonator once it is switched off, the CLEAR pulse suppresses this residual population down to the level of numerical noise. Bringing this comparison onto real hardware required setting up and executing a full calibration chain on an RFSoc-based control system, driven through the Qibo software stack (Qibo, Qibolab, Qibosoq and Qibocal), from resonator and qubit spectroscopy through Rabi, Ramsey and coherence-time measurements to the direct measurement of the dispersive shift and the AC-Stark-shift calibration that converts the readout amplitude, defined only in arbitrary DAC units, into an actual intracavity photon number; with these parameters calibrated, we implemented the CLEAR pulse and developed a Ramsey-based estimator, adapted from the protocol of Ref. [1], capable of extracting the residual photon population left in the resonator after a generic readout pulse. The resulting measurement, repeated over a wide range of normalized drive powers, showed that the residual population of the rectangular pulse grows approximately linearly with power, reaching about five residual photons at the highest power explored, while that of the CLEAR pulse remains below roughly half a photon over the entire range, more than an order of magnitude smaller at the largest powers; a complementary measurement of the residual population as a function of the relaxation delay after the readout pulse confirmed that this improvement stems from the active ring-down segments of the CLEAR pulse, rather than from the passive cavity decay alone on which the rectangular pulse depends.

Beyond the specific numbers, the resulting calibration routines, pulse-generation code and photon-population estimator are directly available as a tested building block for any future experiment in the group that needs a fast, low-dephasing qubit reset, rather than a one-off result confined to this thesis. Taken together, these results provide, on a single real device, a coherent path from the theoretical description of dispersive readout to a quantitative experimental demonstration that pulse shaping can suppress measurement-induced dephasing and shorten the effective readout time, a result directly relevant to any experiment in which qubits must be read out rapidly and repeatedly, and one that could be extended by combining the CLEAR strategy with complementary active-reset or machine-learning-assisted readout schemes, or by applying the same calibration and estimation pipeline developed here to multi-qubit and higher-photon-number readout scenarios.

Bibliography

- [1] D. T. McClure et al. “Rapid Driven Reset of a Qubit Readout Resonator”. In: *Physical Review Applied* 5 (2016), p. 011001. DOI: 10.1103/PhysRevApplied.5.011001.
- [2] A. G. J. MacFarlane, Jonathan P. Dowling, and Gerard J. Milburn. “Quantum technology: the second quantum revolution”. In: *Philosophical Transactions of the Royal Society of London, Series A: Physical and Engineering Sciences* 361.1809 (June 2003), pp. 1655–1674. ISSN: 0962-8428. DOI: 10.1098/rsta.2003.1227. eprint: <https://royalsocietypublishing.org/rsta/article-pdf/361/1809/1655/323777/rsta.2003.1227.pdf>. URL: <https://doi.org/10.1098/rsta.2003.1227>.
- [3] B.D. Josephson. “Possible new effects in superconductive tunnelling”. In: *Physics Letters* 1.7 (1962), pp. 251–253. ISSN: 0031-9163. DOI: [https://doi.org/10.1016/0031-9163\(62\)91369-0](https://doi.org/10.1016/0031-9163(62)91369-0). URL: <https://www.sciencedirect.com/science/article/pii/0031916362913690>.
- [4] B. D. JOSEPHSON. “Coupled Superconductors”. In: *Rev. Mod. Phys.* 36 (1 Jan. 1964), pp. 216–220. DOI: 10.1103/RevModPhys.36.216. URL: <https://link.aps.org/doi/10.1103/RevModPhys.36.216>.
- [5] Jens Koch et al. “Charge-insensitive qubit design derived from the Cooper pair box”. In: *Physical Review A* 76.4 (Oct. 2007). ISSN: 1094-1622. DOI: 10.1103/physreva.76.042319. URL: <http://dx.doi.org/10.1103/PhysRevA.76.042319>.
- [6] P. Krantz et al. “A quantum engineer’s guide to superconducting qubits”. In: *Applied Physics Reviews* 6 (2019), p. 021318. DOI: 10.1063/1.5089550.
- [7] R. Barends et al. “Coherent Josephson Qubit Suitable for Scalable Quantum Integrated Circuits”. In: *Phys. Rev. Lett.* 111 (8 Aug. 2013), p. 080502. DOI: 10.1103/PhysRevLett.111.080502. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.111.080502>.
- [8] Roberto Moretti et al. “Transmon Qubit Modeling and Characterization for Dark Matter Search”. In: *IEEE Transactions on Quantum Engineering* 7 (2026), pp. 1–8. DOI: 10.1109/TQE.2025.3633176.
- [9] Alexandre Blais et al. “Circuit quantum electrodynamics”. In: *Rev. Mod. Phys.* 93 (2 May 2021), p. 025005. DOI: 10.1103/RevModPhys.93.025005. URL: <https://link.aps.org/doi/10.1103/RevModPhys.93.025005>.
- [10] Alexandre Blais et al. “Cavity quantum electrodynamics for superconducting electrical circuits: An architecture for quantum computation”. In: *Phys. Rev. A* 69 (6 June 2004), p. 062320. DOI: 10.1103/PhysRevA.69.062320. URL: <https://link.aps.org/doi/10.1103/PhysRevA.69.062320>.
- [11] F. Bloch. “Generalized Theory of Relaxation”. In: *Physical Review* 105.4 (1957), p. 1206. DOI: 10.1103/PhysRev.105.1206.
- [12] A. G. Redfield. “On the Theory of Relaxation Processes”. In: *IBM Journal of Research and Development* 1.1 (1957), pp. 19–31. DOI: 10.1147/rd.11.0019.
- [13] C. Wang et al. “Surface participation and dielectric loss in superconducting qubits”. In: *Applied Physics Letters* 107.16 (Oct. 2015), p. 162601. ISSN: 0003-6951. DOI: 10.1063/1.4934486. eprint: https://pubs.aip.org/aip/apl/article-pdf/doi/10.1063/1.4934486/14470271/162601_1_online.pdf. URL: <https://doi.org/10.1063/1.4934486>.

- [14] Roger H. Koch, David P. DiVincenzo, and John Clarke. “Model for $1/f$ Flux Noise in SQUIDS and Qubits”. In: *Phys. Rev. Lett.* 98 (26 June 2007), p. 267003. DOI: 10.1103/PhysRevLett.98.267003. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.98.267003>.
- [15] Jay Gambetta et al. “Qubit-photon interactions in a cavity: Measurement-induced dephasing and number splitting”. In: *Phys. Rev. A* 74 (4 Oct. 2006), p. 042318. DOI: 10.1103/PhysRevA.74.042318. URL: <https://link.aps.org/doi/10.1103/PhysRevA.74.042318>.
- [16] G. Catelani et al. “Decoherence of superconducting qubits caused by quasiparticle tunneling”. In: *Phys. Rev. B* 86 (18 Nov. 2012), p. 184514. DOI: 10.1103/PhysRevB.86.184514. URL: <https://link.aps.org/doi/10.1103/PhysRevB.86.184514>.
- [17] A. A. Houck et al. “Controlling the Spontaneous Emission of a Superconducting Transmon Qubit”. In: *Phys. Rev. Lett.* 101 (8 Aug. 2008), p. 080502. DOI: 10.1103/PhysRevLett.101.080502. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.101.080502>.
- [18] Alto Osada, Rekishu Yamazaki, and Atsushi Noguchi. “Electromagnetic Cavities and Cavity Quantum Electrodynamics”. In: *Introduction to Quantum Technologies*. Vol. 1004. Lecture Notes in Physics. Springer Singapore, 2022, pp. 159–184. DOI: 10.1007/978-981-19-4641-7_8.
- [19] Aniket Chatterjee, Jonathan Schwinger, and Yvonne Y. Gao. “Enhanced Qubit Readout via Reinforcement Learning”. In: *Physical Review Applied* 23.5 (May 2025), p. 054057. DOI: 10.1103/PhysRevApplied.23.054057.
- [20] M. Jerger et al. *Dispersive Qubit Readout with Intrinsic Resonator Reset*. 2024. DOI: 10.48550/arXiv.2406.04891. arXiv: 2406.04891 [quant-ph].
- [21] David M. Pozar. *Microwave Engineering*. 4th. Wiley, 2012.
- [22] The Qibo team. *Qibo documentation*. <https://qibo.science/qibo/stable/>. Accessed: 2026-08-24.
- [23] The Qibo team. *Qibolab documentation*. <https://qibo.science/qibolab/stable/>. Accessed: 2026-08-24.
- [24] The Qibo team. *Qibosoq documentation*. <https://qibo.science/qibosoq/stable/>. Accessed: 2026-08-24.
- [25] The Qibo team. *Qibocal documentation*. <https://qibo.science/qibocal/stable/>. Accessed: 2026-08-24.
- [26] Marie F. Dumas et al. “Measurement-Induced Transmon Ionization”. In: *Physical Review X* 14 (2024), p. 041023.